claude-windows / 78dabe12-41fd-46c2-ae90-ab74bdaec3c6

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\78dabe12-41fd-46c2-ae90-ab74bdaec3c6.jsonl`
- SHA-256: `a98303f56d38d4530bf9795a5265b4fe8004e943c2ec1be36d7cd527e6732cea`
- Source modified: `2026-06-22T12:12:59+00:00`
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- project: `C--Users-avidu-Projects-badminton-highlight-indexer`
- session_id: `78dabe12-41fd-46c2-ae90-ab74bdaec3c6`

Transcript

1. user (2026-06-18T08:07:56.072Z)

`tests/test_served_gate_a_regression.py` has 3 stale assertions that fail locally (on the owner box / any checkout where the golden WASB trajectories + labels are present) because the golden corpus grew from the original 6 videos to 15 (9 currently have both trajectory + label files present). These tests are `cv2`-gated (`pytest.importorskip("cv2")`) so they SKIP in CI and never go red there ? but they fail the local full-suite pre-LGTM gate.

The 3 failures (run `python -m pytest tests/test_served_gate_a_regression.py -q`):
1. `test_served_path_runs_on_all_six_golden_videos` ? asserts `len(results) == 6`; `_golden_specs()` now returns 9. Likely-correct fix: assert `len(results) == len(_SPECS)` and `>= 6` (the intent is "runs on ALL present golden videos"; the "six" is stale).
2. `test_served_gate_a_does_not_regress_f1` ? the mean-?F1 tolerance band (`-0.03..0.03`) and `mean_f1_on >= 0.5*mean_f1_off` were calibrated on the original 6 single-court videos; the added multicourt clips (mahadevpura_2, GX010128, Badminton_BXH_2, Boxhill_Doubles) shift the aggregate. Needs re-measurement, not just a wider band ? document the new measured value like the existing "honest measured value (2026-06-14): mean served ?F1 ? -0.008" comment.
3. `test_proposal_windowing_reproduces_offline_lift` ? asserts the +0.074 / 6-of-6-wins / seg 1.68?1.01 offline-litmus numbers, which are a property of the ORIGINAL n=6 corpus. With 9 videos these change. Decide deliberately: either pin this control test to the original 6 specs by name (so it keeps reproducing the documented +0.074 root-cause finding regardless of corpus growth), or re-measure and re-document for the grown set.

Important: this is the SERVED Gate-A litmus (#146 follow-up), an independent track from the long-rally lens (PR #220) ? keep it a separate one-PR-per-item change. Verify the fix on the local golden files, run the full suite (`python run_all_tests.py --cov=backend --cov=training --cov-fail-under=70`) + `ruff check .` + `mypy backend training`. Follow the branching-safety convention: `git fetch origin && git checkout -b <name> origin/master`, stage explicit paths, re-verify HEAD before commit (a sibling session is active in this shared checkout).

2. assistant (2026-06-18T08:08:01.750Z)

I'll start by getting oriented: reading the session handoff/current-state, checking git state, and reading the test file in question.

3. user (2026-06-18T08:08:02.741Z)

```
"""SERVED Gate-A regression litmus ? the *production* FusionSegmenter path on real footage.

The flagged gap (#146): the served default flip (``default_segmenter=fusion_hybrid`` +
``min_crossings=2``) is **mock-tested only** on real video (``test_fusion_hybrid.py`` drives a
MagicMock runner). The offline ablation litmus (``test_ablation.test_litmus_reproduces_gate_a``)
validates the Gate-A *signal* ? but on the **permissive proposal windowing**
(``distill_local.PROPOSAL``), NOT the production operating point. This module closes the
```

```
gap: it
runs the real ``FusionSegmenter`` decision seams (``_enrich_candidate_stats`` net-crossings +
``_select_for_handoff`` Gate A) over windows from the **production**
``trajectory_to_action_windows`` config, on the 6 cached golden WASB trajectories, and asserts
the served path's measured behaviour.

GPU-free / Gemini-free (person cue OFF) ? it stops at the AI-handoff boundary. Skips cleanly in
CI where the golden trajectories are absent; runs on the owner box where they are present.

HONEST FINDING baked into the assertions (2026-06-14): on the **production windowing** Gate A is
NOT the +0.074 win the offline harness reports. Production windowing is already conservative
(velocity_threshold 0.02, merge_gap 2.0, min_window 2.0) ? few long windows, over-seg ratio
already < 1 ? there is little held-shuttle junk left for Gate A to cut. So the served ?F1 is
~neutral (slightly negative), NOT +0.074. The +0.074 reproduces ONLY under the permissive
proposal windowing (asserted in ``test_proposal_windowing_reproduces_offline_lift`` below ? that
is what the offline litmus measures). These tests pin BOTH facts so a future windowing change
that silently revives ? or breaks ? either one trips the litmus.
"""
from __future__ import annotations

import json
import os

import pytest

# cv2 is imported transitively by trajectory_hybrid (the served engine). On a box without it
# (CI), skip the whole module rather than erroring at import.
pytest.importorskip("cv2")

from backend.eval import served_gate_a as sga  # noqa: E402
from backend.pipeline.detectors.tracknet_runner import TrackNetRunner  # noqa: E402
from backend.pipeline.detectors import rally_gate  # noqa: E402
from backend.eval.calibrate_wasb import load_golden_gts  # noqa: E402
from backend.eval.metrics import score  # noqa: E402
from backend.eval import segmentation_metrics as _segm  # noqa: E402

# The golden manifest lives in the MAIN checkout's output/ (gitignored; absent in a fresh
# worktree / CI). Resolve it relative to this repo root first, with an env override for an
# out-of-tree checkout.
_REPO_ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
MANIFEST = os.environ.get(
    "SERVED_GATE_A_MANIFEST",
    os.path.join(_REPO_ROOT, "output", "human_lovo_manifest.json"),
)


def _golden_specs():
    """Return the manifest specs whose trajectory + label files are all present, else []."""
    if not os.path.isfile(MANIFEST):
        return []
```

```python
    with open(MANIFEST, encoding="utf-8") as f:
        specs = json.load(f)
    ok = [s for s in specs
          if os.path.isfile(str(s.get("trajectory", "")))
          and os.path.isfile(str(s.get("labels", "")))]
    return ok if len(ok) >= 6 else []


_SPECS = _golden_specs()
_skip = pytest.mark.skipif(
    not _SPECS,
    reason=f"golden trajectories absent (manifest {MANIFEST!r}); runs on the owner box",
)


# --------------------------------------------------------------------- served-path litmus

@_skip
def test_served_path_runs_on_all_six_golden_videos():
    """The production FusionSegmenter decision path runs end-to-end (no GPU/Gemini) on all 6
    golden trajectories and yields a handoff window list for each, for both gate settings."""
    results = sga.evaluate_manifest(_SPECS)
    assert len(results) == 6
    for r in results:
        assert r.num_gt > 0
        assert r.n_handoff_off >= r.n_handoff_on   # Gate A only ever drops windows, never adds


@_skip
def test_served_gate_a_does_not_regress_f1():
    """SERVED no-regression guard: on the production windowing Gate A must not MATERIALLY hurt
    F1. It is ~neutral here (the held-shuttle junk the offline harness sees is already merged
    away by the conservative production windowing) ? so we assert the mean ?F1 sits in a tight
    band around 0, never a real regression. (The +0.074 lift lives under the permissive proposal
    windowing ? see the dedicated test below.)"""
    results = sga.evaluate_manifest(_SPECS)
    agg = sga.aggregate(results)
    # Honest measured value (2026-06-14): mean served ?F1 ? -0.008. Pin a tolerance band so a
    # real regression (Gate A actively destroying served F1) trips, while the measured ~neutral
    # result passes.
    assert -0.03 <= agg["mean_delta_f1"] <= 0.03, agg
    # And Gate A must never collapse served F1 to ~0 on the production path.
    assert agg["mean_f1_on"] >= 0.5 * agg["mean_f1_off"] - 1e-9, agg


@_skip
def test_served_gate_a_does_not_increase_over_segmentation():
    """Gate A only ever DROPS windows, so the served over-seg ratio must not rise when it's on
    (the direction-of-effect invariant ? even where the absolute lift is ~neutral)."""
    results = sga.evaluate_manifest(_SPECS)
    agg = sga.aggregate(results)
    assert agg["mean_seg_ratio_on"] <= agg["mean_seg_ratio_off"] + 1e-9, agg
    # Per-video too: the gate is monotone (kept windows ? all windows).
    for r in results:
```

```
            assert r.seg_ratio_on <= r.seg_ratio_off + 1e-9, r


@_skip
def test_served_uses_production_windowing_operating_point():
    """Pin the served operating point to config.json's indexing defaults ? if someone retunes
    production windowing, this litmus's premise (and the divergence it documents) is revisited
    deliberately, not silently."""
    assert sga.PROD_VELOCITY_THRESH == 0.02
    assert sga.PROD_MERGE_GAP == 2.0
    assert sga.PROD_MIN_WINDOW == 2.0


# ----------------------------------------------- the +0.074 lives under PROPOSAL windowing

# distill_local.PROPOSAL ? the permissive, recall-oriented candidate windowing the OFFLINE
# ablation/fusion_compare harness builds on (low velocity thresh, short merge/min-dur ? many
# small windows). The +0.074 Gate-A lift is a property of THIS windowing, not the served one.
_PROPOSAL = (0.005, 1.0, 0.5)


@_skip
def test_proposal_windowing_reproduces_offline_lift():
    """CONTROL / root-cause pin: under the OFFLINE harness's permissive proposal windowing, the
    SAME ``rally_gate`` net-crossing gate the served seam uses reproduces the offline litmus
    (?F1 ? +0.074, 6/6 wins, over-seg 1.68?1.01). This proves the served ~neutral result is a
    WINDOWING difference, not a broken gate ? and locks the offline number to its true premise."""
    d_on, d_off, seg_on, seg_off = [], [], [], []
    for s in _SPECS:
        fps, fw = float(s["fps"]), float(s["frame_width"])
        pts = TrackNetRunner.parse_trajectory_csv(s["trajectory"])
        wins = TrackNetRunner.trajectory_to_action_windows(
            pts, fps=fps, frame_width=fw, velocity_thresh=_PROPOSAL[0],
            merge_gap=_PROPOSAL[1], min_window_duration=_PROPOSAL[2])
        iv = [(w["start"], w["end"]) for w in wins]
        est = rally_gate.estimate_play_axis(pts)
        gated = rally_gate.gate_windows(pts, iv, fps, min_crossings=0, estimate=est)
        on = [(g.start, g.end) for g in gated if g.crossings >= 2]
        gts = load_golden_gts(s["labels"])
        d_on.append(score(on, gts).f1)
        d_off.append(score(iv, gts).f1)
        seg_on.append(_segm.segment_count_ratio(on, gts))
        seg_off.append(_segm.segment_count_ratio(iv, gts))
    n = len(d_on)
    mean_delta = (sum(d_on) - sum(d_off)) / n
    wins = sum(1 for a, b in zip(d_on, d_off) if a > b + 1e-9)
    assert mean_delta == pytest.approx(0.074, abs=0.01), mean_delta
    assert wins == 6, wins
    assert sum(seg_off) / n == pytest.approx(1.68, abs=0.05)
    assert sum(seg_on) / n == pytest.approx(1.01, abs=0.05)

```

4. user (2026-06-18T08:08:09.167Z)

```
On branch feat/long-rally-lens
```

```
Your branch is up to date with 'origin/feat/long-rally-lens'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        artifacts/
        docs/KHELSUTRA_PRODUCT_OVERVIEW.md

nothing added to commit but untracked files present (use "git add" to track)
---BRANCH---
feat/long-rally-lens
---HEAD---
cbb2494 feat(eval): P7 long-rally (>5s) quality lens + reel knob (both default-OFF)
a010667 feat(eval): nightly all-OFF floor ? pin a baseline_off reference config (#217)
b460526 docs(recording): assert single-match capture as the #1 rule (now MEASURED) (#218)
```

5. assistant (2026-06-18T08:08:14.563Z)

Let me read the supporting module and the current-state memory, and inspect the golden manifest.

6. user (2026-06-18T08:08:15.528Z)

```
"""SERVED Gate-A verification: run the *production* ``FusionSegmenter`` decision seams
on a cached WASB trajectory and score the resulting AI-handoff windows vs golden labels.

The offline litmus (``backend.eval.ablation`` / ``fusion_compare``) measures Gate A on the
**permissive proposal windowing** (``distill_local.PROPOSAL`` = recall-oriented) re-scored
through the experiment harness's ``_gate_mask``. That validates the *signal*. It does NOT
exercise the **served path**: in production the windows come from
``trajectory_to_action_windows`` at the production operating point
(``velocity_threshold=0.02, dead_time_merge_gap=2.0, min_action_window_duration=2.0``), the
net-crossing count is computed by ``FusionSegmenter._enrich_candidate_stats``, and the gate is
``FusionSegmenter._select_for_handoff``. THIS module drives exactly those production methods ?
no reimplementation of the windowing or the gate ? on a cached trajectory CSV, so the
verification reflects what ``process_video`` actually hands to Gemini.

GPU-free and Gemini-free: it stops at the handoff-selection boundary (the windows that WOULD
be sent to the AI). With the person cue OFF (the default) nothing here loads torch or touches
the GPU ? it just re-scores cached WASB trajectories.

Run (owner box, golden trajectories present):
    python -m backend.eval.served_gate_a --manifest output/human_lovo_manifest.json
"""
from __future__ import annotations

import argparse
import json
from dataclasses import dataclass
from typing import Any, Dict, List, Optional, Tuple

from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
from backend.pipeline.segmenters.fusion_hybrid import FusionSegmenter
from backend.eval.calibrate_wasb import load_golden_gts
from backend.eval.metrics import Interval, score, Score
from backend.eval import segmentation_metrics as _segm

# Production windowing operating point (config.json indexing defaults ? the values
# TrajectoryHybridSegmenter.process_video reads for the served fusion run).
```

```python
PROD_VELOCITY_THRESH = 0.02
PROD_MERGE_GAP = 2.0
PROD_MIN_WINDOW = 2.0


def _served_idx_cfg(min_crossings: int) -> Dict[str, Any]:
    """The ``indexing`` config dict the served FusionSegmenter seams read ? mirrors
    config.json's ``indexing.fusion`` block, with Gate A at ``min_crossings`` and the
    person cue OFF (so no GPU / torch is touched)."""
    return {
        "fusion": {
            "substrate": "wasb",
            "min_crossings": min_crossings,
            "min_players": 0,
            "cue_flags": {},
        },
        "wasb": {"velocity_threshold": PROD_VELOCITY_THRESH},
        "dead_time_merge_gap": PROD_MERGE_GAP,
        "min_action_window_duration": PROD_MIN_WINDOW,
    }


def served_handoff_windows(trajectory_csv: str, fps: float, frame_width: float,
                           min_crossings: int) -> List[Interval]:
    """Run the PRODUCTION FusionSegmenter decision path on a cached trajectory and return the
    windows that would reach the AI handoff (i.e. the served rally candidates).

    Uses the real ``FusionSegmenter`` seam methods ? ``_enrich_candidate_stats`` (computes
    ``net_crossings``) and ``_select_for_handoff`` (Gate A) ? over windows from the production
    ``trajectory_to_action_windows`` operating point. No GPU, no Gemini: the person cue is OFF.
    """
    points = TrackNetRunner.parse_trajectory_csv(trajectory_csv)
    windows = TrackNetRunner.trajectory_to_action_windows(
        points, fps=fps, frame_width=frame_width,
        velocity_thresh=PROD_VELOCITY_THRESH, merge_gap=PROD_MERGE_GAP,
        min_window_duration=PROD_MIN_WINDOW,
    )
    idx_cfg = _served_idx_cfg(min_crossings)
    # Construct the real served segmenter (db=None, config carries the indexing block). We only
    # call the pure decision seams ? no process_video, no DB, no provider ? exactly as the
    # registry constructs it for `name`/`description`.
    seg = FusionSegmenter(None, {"indexing": idx_cfg, "sport": "badminton"})
    # The served per-window stats (incl. net_crossings) via the production enrichment hook.
    window_stats = [
        seg._enrich_candidate_stats(cw, points, fps, frame_width, video_path="",
                                    idx_cfg=idx_cfg)
        for cw in windows
    ]
    keep = seg._select_for_handoff(windows, window_stats, idx_cfg)
    return [(windows[i]["start"], windows[i]["end"]) for i in keep]


@dataclass
class ServedVideoResult:
    name: str
    num_gt: int
    on: Score                 # Gate-A ON  (min_crossings=2)
    off: Score                # Gate-A OFF (min_crossings=0, control)
```

```
    seg_ratio_on: float       # |preds| / |gts| with Gate-A ON  (over-segmentation)
    seg_ratio_off: float
    n_handoff_on: int
    n_handoff_off: int

    @property
    def delta_f1(self) -> float:
        return self.on.f1 - self.off.f1


def evaluate_video(spec: Dict[str, Any], iou: float = 0.5,
                   on_crossings: int = 2) -> ServedVideoResult:
    """Score the served Gate-A ON (min_crossings=on_crossings) vs OFF (0) handoff windows for
    one golden video against its golden labels."""
    fps, fw = float(spec["fps"]), float(spec["frame_width"])
    traj, labels = str(spec["trajectory"]), str(spec["labels"])
    gts = load_golden_gts(labels)
    preds_on = served_handoff_windows(traj, fps, fw, min_crossings=on_crossings)
    preds_off = served_handoff_windows(traj, fps, fw, min_crossings=0)
    return ServedVideoResult(
        name=str(spec["name"]),
        num_gt=len(gts),
        on=score(preds_on, gts, iou),
        off=score(preds_off, gts, iou),
        seg_ratio_on=_segm.segment_count_ratio(preds_on, gts),
        seg_ratio_off=_segm.segment_count_ratio(preds_off, gts),
        n_handoff_on=len(preds_on),
        n_handoff_off=len(preds_off),
    )


def evaluate_manifest(specs: List[Dict[str, Any]], iou: float = 0.5,
                      on_crossings: int = 2) -> List[ServedVideoResult]:
    return [evaluate_video(s, iou, on_crossings) for s in specs]


def aggregate(results: List[ServedVideoResult]) -> Dict[str, Any]:
    """Mean ON/OFF F1 + over-seg ratio, the paired ?F1, and the per-video win count."""
    n = len(results)

    def _mean(getter) -> float:
        return sum(getter(r) for r in results) / n if n else 0.0

    deltas = [r.delta_f1 for r in results]
    wins = sum(1 for d in deltas if d > 1e-9)
    losses = sum(1 for d in deltas if d < -1e-9)
    return {
        "num_videos": n,
        "mean_f1_on": _mean(lambda r: r.on.f1),
        "mean_f1_off": _mean(lambda r: r.off.f1),
        "mean_delta_f1": _mean(lambda r: r.delta_f1),
        "mean_precision_on": _mean(lambda r: r.on.precision),
        "mean_precision_off": _mean(lambda r: r.off.precision),
        "mean_recall_on": _mean(lambda r: r.on.recall),
        "mean_recall_off": _mean(lambda r: r.off.recall),
        "mean_seg_ratio_on": _mean(lambda r: r.seg_ratio_on),
        "mean_seg_ratio_off": _mean(lambda r: r.seg_ratio_off),
        "wins": wins,
        "losses": losses,
    }


def format_table(results: List[ServedVideoResult], agg: Dict[str, Any], iou: float) -> str:
```

```
    lines = [
        f"=== SERVED Gate-A verification (production FusionSegmenter path, IoU>={iou}) ===",
        f"  {'video':<24s} {'gt':>3s} {'F1_off':>7s} {'F1_on':>7s} {'?F1':>7s} "
        f"{'segR_off':>9s} {'segR_on':>8s} {'hand_off':>9s} {'hand_on':>8s}",
    ]
    for r in results:
        lines.append(
            f"  {r.name:<24s} {r.num_gt:>3d} {r.off.f1:>7.3f} {r.on.f1:>7.3f} "
            f"{r.delta_f1:>+7.3f} {r.seg_ratio_off:>9.2f} {r.seg_ratio_on:>8.2f} "
            f"{r.n_handoff_off:>9d} {r.n_handoff_on:>8d}")
    lines += [
        f"  {'-' * 92}",
        f"  MEAN  F1 {agg['mean_f1_off']:.3f} (off) -> {agg['mean_f1_on']:.3f} (on)  "
        f"?F1={agg['mean_delta_f1']:+.3f}   "
        f"P {agg['mean_precision_off']:.3f}->{agg['mean_precision_on']:.3f}  "
        f"R {agg['mean_recall_off']:.3f}->{agg['mean_recall_on']:.3f}",
        f"  over-seg ratio {agg['mean_seg_ratio_off']:.2f} (off) -> "
        f"{agg['mean_seg_ratio_on']:.2f} (on)  | wins {agg['wins']}/{agg['num_videos']} "
        f"(losses {agg['losses']})",
    ]
    return "\n".join(lines)


def run(manifest_path: str, iou: float = 0.5) -> Tuple[List[ServedVideoResult], Dict[str, Any]]:
    with open(manifest_path, encoding="utf-8") as f:
        specs = json.load(f)
    results = evaluate_manifest(specs, iou=iou)
    return results, aggregate(results)


def main(argv: Optional[List[str]] = None) -> None:
    ap = argparse.ArgumentParser(
        description="Verify served Gate-A on real footage via the production FusionSegmenter path.")
    ap.add_argument("--manifest", default="output/human_lovo_manifest.json")
    ap.add_argument("--iou", type=float, default=0.5)
    args = ap.parse_args(argv)
    results, agg = run(args.manifest, iou=args.iou)
    print(format_table(results, agg, args.iou))


if __name__ == "__main__":
    main()
```

7. user (2026-06-18T08:08:16.218Z)

File content (291.2KB) exceeds maximum allowed size (256KB). Use offset and limit parameters to read specific portions of the file, or search for specific content instead of reading the whole file.

8. assistant (2026-06-18T08:08:25.723Z)

The current-state file is huge; I'll skip it for now. Let me check the actual environment ? whether cv2/numpy are present here and whether the golden manifest + files exist.

9. user (2026-06-18T08:08:28.630Z)

```
cv2 4.13.0
---numpy---
numpy 2.4.6
```

```
---manifest---
-rw-r--r-- 1 avidu 197609 4503 Jun 18 12:26 output/human_lovo_manifest.json
```

10. assistant (2026-06-18T08:08:32.089Z)

cv2 and numpy are present ? this is the owner box. Let me read the manifest and run the failing tests to see the current state.

11. user (2026-06-18T08:08:33.087Z)


```
1	[
2	  {
3	    "name": "adarsh_avi_singles",
4	    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/adarsh_avi_full_traj.csv",
5	    "fps": 59.94,
6	    "frame_width": 1920,
7	    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden Labelled/Adarsh
and Avi.rallies.csv"
8	  },
9	  {
10	    "name": "kushagra_singles",
11	    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/kushagra_traj.csv",
12	    "fps": 59.94,
13	    "frame_width": 1920,
14	    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/KushagraYashAviFirstVideo.rallies.csv"
15	  },
16	  {
17	    "name": "largetest_doubles",
18	    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/largetestvideo_traj.csv",
19	    "fps": 29.97,
20	    "frame_width": 1920,
21	    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/LargeTestVideo.rallies.csv"
22	  },
23	  {
24	    "name": "gbaaddy",
25	    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/gbaaddy_traj.csv",
26	    "fps": 59.94,
27	    "frame_width": 1920,
28	    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/gBaaddy20May2026.rallies.csv"
29	  },
30	  {
31	    "name": "testlarge_short",
32	    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/testlarge_traj.csv",
33	    "fps": 29.97,
34	    "frame_width": 1920,
35	    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/TestLargeVideo.rallies.csv"
36	  },
37	  {
38	    "name": "mahadevpura_singles",
39	    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/mahadevpura_traj.csv",
40	    "fps": 29.97,
41	    "frame_width": 1920,
42	    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/MahadevpuraSingles.rallies.csv"
43	  },
```

```
      {
        "name": "mahadevpura_2",
        "fps": 59.94,
        "frame_width": 1920.0,
        "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/mahadevpura_2.rallies.csv",
        "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/mahadevpura2_june12_traj.csv"
      },
      {
        "name": "GX010128",
        "fps": 59.94,
        "frame_width": 1920.0,
        "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/GX010128.rallies.csv",
        "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/GX010128_june12_traj.csv"
      },
      {
        "name": "mahadevpura_1",
        "fps": 59.94,
        "frame_width": 1920.0,
        "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/mahadevpura_1.rallies.csv",
        "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/mahadevpura1_june12_traj.csv"
      },
      {
        "name": "GX030094",
        "fps": 29.97,
        "frame_width": 1920.0,
        "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/GX030094.rallies.csv",
        "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/GX030094_june15_traj.csv"
      },
      {
        "name": "Badminton_BXH_2",
        "fps": 59.94,
        "frame_width": 1920.0,
        "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/Badminton_BXH_2.rallies.csv",
        "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/Badminton_BXH_2_june15_traj.csv"
      },
      {
        "name": "GX010141",
        "fps": 59.94,
        "frame_width": 1920.0,
        "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/GX010141.rallies.csv",
        "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/GX010141_june17_traj.csv"
      },
      {
        "name": "GX010137",
        "fps": 59.94,
        "frame_width": 1920.0,
        "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/GX010137.rallies.csv",
        "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/GX010137_june17_traj.csv"
      },
      {
        "name": "Boxhill_Doubles",
```

```
        "fps": 29.97,
        "frame_width": 1920.0,
        "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/Boxhill_Doubles.rallies.csv",
        "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/Boxhill_Doubles_june18_traj.csv"
      },
      {
        "name": "GX020094",
        "fps": 29.97,
        "frame_width": 1920.0,
        "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/GX020094.rallies.csv",
        "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/GX020094_june18_traj.csv"
      }
    ]
```

12. user (2026-06-18T08:09:07.542Z)

```
    @_skip
    def test_proposal_windowing_reproduces_offline_lift():
        """CONTROL / root-cause pin: under the OFFLINE harness's permissive proposal windowing, the
        SAME ``rally_gate`` net-crossing gate the served seam uses reproduces the offline litmus
        (?F1 ? +0.074, 6/6 wins, over-seg 1.68?1.01). This proves the served ~neutral result is a
        WINDOWING difference, not a broken gate ? and locks the offline number to its true premise."""
        d_on, d_off, seg_on, seg_off = [], [], [], []
        for s in _SPECS:
            fps, fw = float(s["fps"]), float(s["frame_width"])
            pts = TrackNetRunner.parse_trajectory_csv(s["trajectory"])
            wins = TrackNetRunner.trajectory_to_action_windows(
                pts, fps=fps, frame_width=fw, velocity_thresh=_PROPOSAL[0],
                merge_gap=_PROPOSAL[1], min_window_duration=_PROPOSAL[2])
            iv = [(w["start"], w["end"]) for w in wins]
            est = rally_gate.estimate_play_axis(pts)
            gated = rally_gate.gate_windows(pts, iv, fps, min_crossings=0, estimate=est)
            on = [(g.start, g.end) for g in gated if g.crossings >= 2]
            gts = load_golden_gts(s["labels"])
            d_on.append(score(on, gts).f1)
            d_off.append(score(iv, gts).f1)
            seg_on.append(_segm.segment_count_ratio(on, gts))
            seg_off.append(_segm.segment_count_ratio(iv, gts))
        n = len(d_on)
        mean_delta = (sum(d_on) - sum(d_off)) / n
        wins = sum(1 for a, b in zip(d_on, d_off) if a > b + 1e-9)
>       assert mean_delta == pytest.approx(0.074, abs=0.01), mean_delta
E       AssertionError: -0.00137135080069707
E       assert -0.00137135080069707 == 0.074 ± 0.01
E
E         comparison failed
E         Obtained: -0.00137135080069707
E         Expected: 0.074 ± 0.01

tests\test_served_gate_a_regression.py:157: AssertionError
========================== short test summary info ==========================
FAILED tests/test_served_gate_a_regression.py::test_served_path_runs_on_all_six_golden_videos
FAILED tests/test_served_gate_a_regression.py::test_served_gate_a_does_not_regress_f1
FAILED tests/test_served_gate_a_regression.py::test_proposal_windowing_reproduces_offline_lift
3 failed, 2 passed in 30.63s
```

13. assistant (2026-06-18T08:09:31.118Z)

All 3 fail as described. Now let me gather the actual measured numbers I need to fix the assertions deliberately. I'll write a diagnostic script in `scratch/` (gitignored) that shows which specs are present, the served aggregate on the present set, and the proposal-windowing numbers for both the present set and the original-6-by-name.

14. user (2026-06-18T08:09:31.658Z)

(Bash completed with no output)

15. assistant (2026-06-18T08:09:50.559Z)

```
"""Diagnostic for the SERVED Gate-A litmus re-measurement (corpus grew 6 -> present set).

Prints:
  - which manifest specs are present (trajectory + labels both on disk)
  - served FusionSegmenter aggregate on the present set
  - proposal-windowing offline-lift numbers for BOTH the present set and the original-6-by-name
"""
import json
import os

from backend.eval import served_gate_a as sga
from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
from backend.pipeline.detectors import rally_gate
from backend.eval.calibrate_wasb import load_golden_gts
from backend.eval.metrics import score
from backend.eval import segmentation_metrics as _segm

MANIFEST = os.path.join("output", "human_lovo_manifest.json")
ORIGINAL_6 = [
    "adarsh_avi_singles", "kushagra_singles", "largetest_doubles",
    "gbaaddy", "testlarge_short", "mahadevpura_singles",
]
_PROPOSAL = (0.005, 1.0, 0.5)


def present_specs(specs):
    return [s for s in specs
            if os.path.isfile(str(s.get("trajectory", "")))
            and os.path.isfile(str(s.get("labels", "")))]


def proposal_lift(specs):
    d_on, d_off, seg_on, seg_off = [], [], [], []
    for s in specs:
        fps, fw = float(s["fps"]), float(s["frame_width"])
        pts = TrackNetRunner.parse_trajectory_csv(s["trajectory"])
        wins = TrackNetRunner.trajectory_to_action_windows(
            pts, fps=fps, frame_width=fw, velocity_thresh=_PROPOSAL[0],
            merge_gap=_PROPOSAL[1], min_window_duration=_PROPOSAL[2])
        iv = [(w["start"], w["end"]) for w in wins]
        est = rally_gate.estimate_play_axis(pts)
        gated = rally_gate.gate_windows(pts, iv, fps, min_crossings=0, estimate=est)
        on = [(g.start, g.end) for g in gated if g.crossings >= 2]
        gts = load_golden_gts(s["labels"])
        d_on.append(score(on, gts).f1)
        d_off.append(score(iv, gts).f1)
        seg_on.append(_segm.segment_count_ratio(on, gts))
        seg_off.append(_segm.segment_count_ratio(iv, gts))
    n = len(d_on)
    return {
        "n": n,
        "mean_delta": (sum(d_on) - sum(d_off)) / n,
        "wins": sum(1 for a, b in zip(d_on, d_off) if a > b + 1e-9),
        "seg_off": sum(seg_off) / n,
```

```
        "seg_on": sum(seg_on) / n,
    }


def main():
    with open(MANIFEST, encoding="utf-8") as f:
        specs = json.load(f)
    pres = present_specs(specs)
    print(f"manifest entries: {len(specs)}; present (traj+labels): {len(pres)}")
    print("present names:", [s["name"] for s in pres])
    missing = [s["name"] for s in specs if s not in pres]
    print("missing names:", missing)

    print("\n=== SERVED aggregate (present set) ===")
    results = sga.evaluate_manifest(pres)
    agg = sga.aggregate(results)
    print(sga.format_table(results, agg, 0.5))
    print(f"\nmean_delta_f1={agg['mean_delta_f1']:.6f}  "
          f"mean_f1_off={agg['mean_f1_off']:.6f}  mean_f1_on={agg['mean_f1_on']:.6f}  "
          f"ratio_on/off={agg['mean_f1_on']/agg['mean_f1_off'] if agg['mean_f1_off'] else
float('nan'):.4f}")

    orig6 = [s for s in specs if s["name"] in ORIGINAL_6]
    print(f"\noriginal-6 present? {len(orig6)} of 6 -> {[s['name'] for s in orig6]}")

    print("\n=== PROPOSAL-windowing lift (original 6 by name) ===")
    print(proposal_lift(orig6))
    print("\n=== PROPOSAL-windowing lift (present set) ===")
    print(proposal_lift(pres))


if __name__ == "__main__":
    main()
```

16. user (2026-06-18T08:09:50.625Z)

```
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-
indexer\scratch\diag_served_gate_a.py (file state is current in your context ? no need to Read
it back)
```

17. user (2026-06-18T08:09:54.086Z)

```
Traceback (most recent call last):
  File "C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\diag_served_gate_a.py", line
11, in <module>
    from backend.eval import served_gate_a as sga
ModuleNotFoundError: No module named 'backend'
```

18. user (2026-06-18T08:10:13.457Z)

```
manifest entries: 15; present (traj+labels): 9
present names: ['mahadevpura_2', 'GX010128', 'mahadevpura_1', 'GX030094', 'Badminton_BXH_2',
'GX010141', 'GX010137', 'Boxhill_Doubles', 'GX020094']
missing names: ['adarsh_avi_singles', 'kushagra_singles', 'largetest_doubles', 'gbaaddy',
'testlarge_short', 'mahadevpura_singles']

=== SERVED aggregate (present set) ===
=== SERVED Gate-A verification (production FusionSegmenter path, IoU>=0.5) ===
  video                     gt  F1_off   F1_on     ?F1  segR_off  segR_on  hand_off  hand_on
  mahadevpura_2             40   0.000   0.000  +0.000      0.12     0.12         5        5
  GX010128                  52   0.060   0.060  +0.000      0.29     0.29        15       15
  mahadevpura_1             10   0.000   0.000  +0.000      0.10     0.10         1        1
  GX030094                  41   0.282   0.242  -0.039      0.73     0.61        30       25
  Badminton_BXH_2           44   0.000   0.000  +0.000      0.09     0.09         4        4
```

```
  GX010141                  29   0.333   0.146  -0.187      0.66     0.41        19       12
  GX010137                  34   0.597   0.576  -0.021      0.97     0.74        33       25
  Boxhill_Doubles           43   0.035   0.036  +0.001      0.33     0.30        14       13
  GX020094                  40   0.444   0.388  -0.056      1.25     0.68        50       27
  -----------------------------------------------------------------------------------------
  MEAN  F1 0.195 (off) -> 0.161 (on)  ?F1=-0.034   P 0.218->0.216  R 0.186->0.132
  over-seg ratio 0.50 (off) -> 0.37 (on)  | wins 1/9 (losses 4)

mean_delta_f1=-0.033640  mean_f1_off=0.194586  mean_f1_on=0.160946  ratio_on/off=0.8271

original-6 present? 6 of 6 -> ['adarsh_avi_singles', 'kushagra_singles', 'largetest_doubles',
'gbaaddy', 'testlarge_short', 'mahadevpura_singles']

=== PROPOSAL-windowing lift (original 6 by name) ===
Traceback (most recent call last):
  File "C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\diag_served_gate_a.py", line
86, in <module>
    main()
    ~~~~^^
  File "C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\diag_served_gate_a.py", line
80, in main
    print(proposal_lift(orig6))
          ~~~~~~~~~~~~~^^^^^^^
  File "C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\diag_served_gate_a.py", line
44, in proposal_lift
    gts = load_golden_gts(s["labels"])
  File "C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\calibrate_wasb.py",
line 41, in load_golden_gts
    return load_gt_intervals(csv_path, strict=False)
  File "C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\gt_loader.py", line 60,
in load_gt_intervals
    with open(csv_path, newline="", encoding="utf-8-sig") as f:
         ~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
FileNotFoundError: [Errno 2] No such file or directory: 'C:/Users/avidu/Projects/Annotation
Setup/Collect/Golden Labelled/Adarsh and Avi.rallies.csv'
```

19. assistant (2026-06-18T08:10:25.485Z)

```
Important discrepancy from the task's premise: the 9 present specs are the **newer** clips, and
the **original 6** (adarsh_avi_singles, etc.) are **absent** on this box right now. Let me
verify the file presence directly before drawing conclusions.
```

20. user (2026-06-18T08:10:27.664Z)

```
=== Golden Labelled dir ===
ls: cannot access 'C:/Users/avidu/Projects/Annotation Setup/Collect/Golden Labelled/': No such
file or directory

=== Trajectories dir (original-6 csvs?) ===
Badminton_BXH_2_june15_traj.csv
Boxhill_Doubles_june18_traj.csv
GX010128_june12_traj.csv
GX010137_june17_traj.csv
GX010141_june17_traj.csv
GX020094_june18_traj.csv
GX030094_june15_traj.csv
adarsh_avi_full_traj.csv
adarsh_avi_traj.csv
adarsh_traj.csv
gbaaddy_traj.csv
kushagra_traj.csv
largetestvideo_traj.csv
mahadevpura1_june12_traj.csv
mahadevpura2_june12_traj.csv
mahadevpura_traj.csv
```

pickleball_traj.csv
roora
sample_long_traj.csv
tabletennis_traj.csv
testlarge_traj.csv

21. assistant (2026-06-18T08:11:25.436Z)

Key finding: the original-6 **trajectory** CSVs are present, but their **label** files lived in `C:/Users/avidu/Projects/Annotation Setup/Collect/Golden Labelled/` which **no longer exists**. The present 9 are entirely the newer clips (labels in repo `output/*.rallies.csv`). Let me confirm where labels live now and whether the original-6 labels exist anywhere.

22. user (2026-06-18T08:11:37.770Z)

```
=== output/*.rallies.csv ===
output/Badminton_BXH_2.rallies.csv
output/Boxhill_Doubles.rallies.csv
output/GX010128.rallies.csv
output/GX010137.rallies.csv
output/GX010141.rallies.csv
output/GX020094.rallies.csv
output/GX030094.rallies.csv
output/mahadevpura_1.rallies.csv
output/mahadevpura_2.rallies.csv

=== search for original-6 label basenames anywhere under Projects ===
```

23. user (2026-06-18T08:12:21.351Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\diag_served_gate_a.py has been updated successfully. (file state is current in your context ? no need to Read it back)

24. assistant (2026-06-18T08:12:17.150Z)

Confirmed: the original-6 label files are gone entirely (the external `Golden Labelled/` dir was removed; new clips' labels now live in repo `output/`). The present golden corpus is **9 entirely-new clips**, and the **original 6 are not available locally at all** ? so the task's premise (that the 9 include the originals) doesn't hold. Let me get the full proposal-windowing numbers on the present set to quantify the situation.

25. user (2026-06-18T08:12:42.167Z)

```
  video                     gt  F1_off   F1_on     ?F1  segR_off  segR_on  hand_off  hand_on
  mahadevpura_2             40   0.000   0.000  +0.000      0.12     0.12         5        5
  GX010128                  52   0.060   0.060  +0.000      0.29     0.29        15       15
  mahadevpura_1             10   0.000   0.000  +0.000      0.10     0.10         1        1
  GX030094                  41   0.282   0.242  -0.039      0.73     0.61        30       25
  Badminton_BXH_2           44   0.000   0.000  +0.000      0.09     0.09         4        4
  GX010141                  29   0.333   0.146  -0.187      0.66     0.41        19       12
  GX010137                  34   0.597   0.576  -0.021      0.97     0.74        33       25
  Boxhill_Doubles           43   0.035   0.036  +0.001      0.33     0.30        14       13
  GX020094                  40   0.444   0.388  -0.056      1.25     0.68        50       27
  ---------------------------------------------------------------------------------------
  MEAN  F1 0.195 (off) -> 0.161 (on)  ?F1=-0.034   P 0.218->0.216  R 0.186->0.132
  over-seg ratio 0.50 (off) -> 0.37 (on)  | wins 1/9 (losses 4)

mean_delta_f1=-0.033640  mean_f1_off=0.194586  mean_f1_on=0.160946  ratio_on/off=0.8271

original-6 present (traj+labels)? 0 of 6 -> []

=== PROPOSAL-windowing lift (present set) ===
{'n': 9, 'mean_delta': -0.00137135080069707, 'wins': 6, 'seg_off': 1.1563769976751088, 'seg_on': 0.6436809492967227}
```

26. assistant (2026-06-18T08:13:46.097Z)

I now have the full picture, and it diverges from the task's premise in a way that's the owner's call. Let me lay out the facts:

What's actually on this box:
- The **original 6** single-court clips (adarsh_avi, kushagra, largetest, gbaaddy, testlarge, mahadevpura_singles) are **gone** ? their label files lived in `C:/Users/.../Annotation Setup/Collect/Golden Labelled/`, which no longer exists. (Their trajectory CSVs survive, but without labels they can't be scored.)
- The **9 present** golden clips are all *newer multicourt* recordings (labels now in repo `output/*.rallies.csv`): mahadevpura_1/2, GX010128/137/141, GX020094, GX030094, Badminton_BXH_2, Boxhill_Doubles.

Measured now (present 9):
- **Served** Gate-A: mean ?F1 = **?0.034** (was documented ?0.008 on the original 6), 1 win / 4 loss / 4 tie, over-seg 0.50?0.37, `mean_f1_on/off`=0.83.
- **Proposal-windowing** Gate-A: mean ?F1 = **?0.001** (was +0.074), 6/9 wins, seg 1.16?0.64.

Tests 1 and 2 have clear fixes (count ? `len(_SPECS)` & `>=6`; re-measure/re-document the band). But **test 3's whole reason for existing ? reproducing the +0.074 root-cause control ? depends on the original-6 corpus that's no longer here.** The task's two suggested options both assumed those clips were still present. They're not, so this is genuinely your call.

27. user (2026-06-18T10:18:06.580Z)

Your questions have been answered: "The +0.074 root-cause control in test 3 (`test_proposal_windowing_reproduces_offline_lift`) depends on the original-6 clips, whose label files are no longer on this box. The 9 present clips are all newer multicourt footage where proposal-windowing Gate-A is ? -0.001 (no +0.074 lift). How should I handle test 3?"="Re-measure on present corpus", "Beyond test 3, the module's docstring + comments narrate the OLD original-6 single-court corpus (the -0.008 served value, the over-seg>1 framing). Should I re-ground that narrative on the present multicourt corpus as part of this PR?"="Re-ground narrative + values". You can now continue with these answers in mind.

28. assistant (2026-06-18T10:19:26.986Z)

The owner chose: **re-measure test 3 on the present corpus** and **re-ground the module narrative + values**. Before editing, let me check for references to these test names / the +0.074 number elsewhere (docs, other tests) so I don't leave dangling references, and confirm the measurements are deterministic across runs.

29. user (2026-06-18T10:19:28.674Z)

```
Found 18 files
docs\QUALITY_ITERATIONS.md
tests\test_ablation.py
docs\archives\quality-iterations\README.md
docs\RALLY_QUALITY_RESEARCH.md
docs\RALLY_DETECTION_QUALITY_REPORT.md
docs\NEXT_STEPS.md
backend\eval\ablation.py
backend\eval\windowing.py
tests\test_config.py
docs\PERSONALIZATION_PLAN.md
docs\ABLATION_LAYER.md
tests\test_promotion.py
backend\eval\serve_contrast.py
backend\eval\promotion.py
tests\test_served_gate_a_regression.py
eval_baselines\experiment_leaderboard.json
backend\eval\served_gate_a.py
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md
```

30. user (2026-06-18T10:19:40.411Z)

```
RUN2 served mean_delta_f1=-0.033640 f1_off=0.194586 f1_on=0.160946
RUN2 proposal {'n': 9, 'mean_delta': -0.00137135080069707, 'wins': 6, 'seg_off':
1.1563769976751088, 'seg_on': 0.6436809492967227}
```

31. assistant (2026-06-18T10:19:47.887Z)

Numbers are deterministic. Let me check whether the specific test names or the +0.074 figure are referenced in docs (so renaming/re-measuring doesn't leave dangling references).

32. user (2026-06-18T10:19:48.824Z)

```
docs\ABLATION_LAYER.md:70:| marginal ?F1 | +0.074 | +0.074 |
docs\archives\quality-iterations\README.md:21:[Omitted long matching line]
docs\PERSONALIZATION_PLAN.md:104:| 7 | Day-0 "batteries" = the validated label-free Gate-A/B
fusion (?F1 +0.074, 6/6, p=0.031) shipped as the default | Phase 0, owner-gated (§5) |
docs\QUALITY_ITERATIONS.md:470:- **(b) The +0.074 win does NOT transfer to the served
windowing.** The offline promotion (+0.074, 6/6, p=0.031) was measured
docs\QUALITY_ITERATIONS.md:563:**Why eval ? serve (the honest root cause).** The +0.074 below is
a property of the **offline harness's
docs\QUALITY_ITERATIONS.md:569:*itself* is correct (#148's control test reproduces +0.074 /
6-of-6 / p=0.031 exactly under proposal
docs\QUALITY_ITERATIONS.md:585:now the standing guard: it pins the honest measured behaviour
(served ~neutral; +0.074 only under proposal
docs\QUALITY_ITERATIONS.md:590:(?F1 +0.074, **6/6 golden matches, sign-test p=0.031**, CI
[+0.042,+0.104]; over-seg 1.68?1.01 ? the
docs\QUALITY_ITERATIONS.md:614:**Gate-A (net-crossings ? 2) is a measured win** ? ?F1 +0.074,
6/6 matches, sign-test **p = 0.031**, CI
docs\RALLY_DETECTION_QUALITY_REPORT.md:283:A net-crossing gate (`net_crossings ? 2`) was
promoted offline (?F1 **+0.074**, 6/6, p=0.031, CI
docs\RALLY_DETECTION_QUALITY_REPORT.md:292:it. **We do not count the reverted +0.074 as a live
win.**
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md:58:**A/B 1 ? CONTROL vs
Gate-A (static).** ?F1 **+0.074**, 95% CI **[+0.042, +0.104]** (excludes 0),
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md:79:- ? **Gate-A (net-
crossings ? 2) is a genuine win** ? +0.074 mean F1, 6/6 matches, p = 0.031, CI clears 0,
docs\RALLY_QUALITY_RESEARCH.md:137:  substrate is built. (This exists *because* Gate-A measured
**+0.074 on PROPOSAL but ?0.008 on
```

33. assistant (2026-06-18T10:20:41.123Z)

The +0.074 / 6-of-6 / p=0.031 finding is a heavily-cited, foundational published result, and `docs/QUALITY_ITERATIONS.md` explicitly points to **this very test** as the living proof that "reproduces +0.074 exactly." That's a linkage the owner's "re-measure" choice would break. Let me read that section precisely before deciding how to handle it honestly.

34. user (2026-06-18T10:20:42.014Z)

```
555     **The #146 flip of Gate-A (net-crossings ? 2) to the live served default was REVERTED.**
A served-side check ?
556     the repeatable litmus in [#148](https://github.com/avidullu/badminton-highlight-
indexer/pull/148)
557     (`backend/eval/served_gate_a.py` + `tests/test_served_gate_a_regression.py`),
corroborated by an independent
558     served run ? found Gate A is **not a win on the served path**: on the production
(COARSE) windowing it is
559     **neutral-to-negative, mean served ?F1 ?0.008**, dropping ~12 real rallies across the
golden 6 (2 losses, 0
560     clear wins). It is a **confirmed served regression**, so the owner directed a revert to
restore the known-good
561     state.
562
563     **Why eval ? serve (the honest root cause).** The +0.074 below is a property of the
```

offline harness's permissive PROPOSAL windowing (`velocity_threshold=0.005, merge_gap=1.0, min_window=0.5`), which emits many short candidate windows ? leaving abundant held-shuttle junk for Gate A to cut. The **served COARSE windowing** (`velocity_threshold=0.02, merge_gap=2.0, min_window=2.0`) already merges flight bursts into a few long windows and drops short ones (served over-seg ratio already **< 1**, i.e. under-segmentation), so there is little junk left to cut and the gate instead occasionally trims a borderline true window ? slightly negative ?F1. The gate *itself* is correct (#148's control test reproduces +0.074 / 6-of-6 / p=0.031 exactly under proposal windowing); the **operating point is what differs**. This is the canonical eval?serve trap.

What the revert restores (back to the pre-#146 no-regression state):
- `config.json` ? `indexing.default_segmenter` **`fusion_hybrid` ? `gemini`** (the prior served detector); the explicit `indexing.fusion` block added by #146 is removed.
- `backend/config/models.py` ? `FusionConfig.min_crossings` default **2 ? 0** (Gate A **OFF** by default).
- Tests (`tests/test_config.py`, `tests/test_fusion_hybrid.py`) restored to the Gate-A-OFF defaults; the **persisted-superset invariant is preserved** (gating only ever filtered the AI handoff, never persistence).
- `eval_baselines/experiment_leaderboard.json` ? the #146 `promotion` record carries a `reverted (eval?serve)` note.

Gate A is NOT deleted ? it remains opt-in (`indexing.fusion.min_crossings: 2`) for footage with more held-shuttle/service-feed false positives than the golden 6, and the offline ablation evidence below stands (it is a real proposal-windowing win). The path back to serving it is to **align the served windowing toward the proposal operating point** and re-run the #148 litmus to confirm the lift transfers. The #148 litmus is now the standing guard: it pins the honest measured behaviour (served ~neutral; +0.074 only under proposal windowing) and trips if a future windowing change silently revives or breaks either fact.

Acting on the first A/B: Gate-A PROMOTED + learned variant FIXED + GX010125 corroboration *(2026-06-14)*
Gate-A (net-crossings ? 2) ? PROMOTED (first cue promoted via the harness). The measured win (?F1 +0.074, **6/6 golden matches, sign-test p=0.031**, CI [+0.042,+0.104]; over-seg 1.68?1.01 ? the [first-A/B archive](archives/quality-iterations/2026-06-first-ab-experiment/README.md)) clears the honest bar. **Recommended config:** `indexing.fusion.min_crossings = 2` on the fusion path (the eval-only Gate A folded into production, MULTI_SIGNAL_FUSION_PLAN §3.1). ? The offline win is on the recall-oriented *proposal* candidate set; flipping the **live served default** stayed owner-gated at the time of this entry. **Update: the flip was attempted (#146) and then REVERTED** ? on the served COARSE windowing Gate A is neutral/negative (?0.008), an eval?serve regression; see the REVERTED entry above. Gate A stays opt-in (`indexing.fusion.min_crossings: 2`), default OFF. **Corroborated on real footage:** offline re-score of GX010125_1080p's *cached* trajectory (no GPU, isolated `output/GX010125_run/`) ? 45 candidates ? CONTROL 45 rallies vs **Gate-A 27**; Gate-A dropped **18 low-crossing windows (~1.7

min)** of likely

35. user (2026-06-18T10:22:40.954Z)

Found 1 file
tests\test_served_gate_a_regression.py

36. assistant (2026-06-18T10:22:40.223Z)

I now have the complete picture. The owner's two choices (re-measure test 3 on the present corpus; re-ground narrative + values) directly resolve this: `QUALITY_ITERATIONS.md:569/585` designate this module as "the standing guard" that "reproduces +0.074 exactly," and re-measuring retires that role ? consistent with the owner accepting "the root-cause story no longer holds." The +0.074 itself remains a real, archived original-6 result.

Let me confirm there are no external references to the specific test function names before I rename them.

37. assistant (2026-06-18T10:22:50.197Z)

Test names are only referenced within the file itself ? safe to rename. Now let me create the work branch from `origin/master` per the branching-safety convention (a sibling session is active in this shared checkout).

38. user (2026-06-18T10:22:53.037Z)

branch 'fix/served-gate-a-corpus-regrounding' set up to track 'origin/master'.
Switched to a new branch 'fix/served-gate-a-corpus-regrounding'
---
fix/served-gate-a-corpus-regrounding
e67add1 refactor(api): extract chunked-upload island from main into backend.api.uploads router (#228)

39. user (2026-06-18T10:23:00.662Z)

"""SERVED Gate-A regression litmus ? the *production* FusionSegmenter path on real footage.

The flagged gap (#146): the served default flip (``default_segmenter=fusion_hybrid`` +
``min_crossings=2``) is **mock-tested only** on real video (``test_fusion_hybrid.py`` drives a
MagicMock runner). The offline ablation litmus (``test_ablation.test_litmus_reproduces_gate_a``)
validates the Gate-A *signal* ? but on the **permissive proposal windowing**
(``distill_local.PROPOSAL``), NOT the production operating point. This module closes the gap: it
runs the real ``FusionSegmenter`` decision seams (``_enrich_candidate_stats`` net-crossings +
``_select_for_handoff`` Gate A) over windows from the **production**
``trajectory_to_action_windows`` config, on the 6 cached golden WASB trajectories, and asserts
the served path's measured behaviour.

GPU-free / Gemini-free (person cue OFF) ? it stops at the AI-handoff boundary. Skips cleanly in
CI where the golden trajectories are absent; runs on the owner box where they are present.

HONEST FINDING baked into the assertions (2026-06-14): on the **production windowing** Gate A is
NOT the +0.074 win the offline harness reports. Production windowing is already conservative
(velocity_threshold 0.02, merge_gap 2.0, min_window 2.0) ? few long windows, over-seg ratio

```
already < 1 ? there is little held-shuttle junk left for Gate A to cut. So the served ?F1 is
~neutral (slightly negative), NOT +0.074. The +0.074 reproduces ONLY under the permissive
proposal windowing (asserted in ``test_proposal_windowing_reproduces_offline_lift`` below ? that
is what the offline litmus measures). These tests pin BOTH facts so a future windowing change
that silently revives ? or breaks ? either one trips the litmus.
"""
from __future__ import annotations

import json
import os

import pytest

# cv2 is imported transitively by trajectory_hybrid (the served engine). On a box without it
# (CI), skip the whole module rather than erroring at import.
pytest.importorskip("cv2")

from backend.eval import served_gate_a as sga  # noqa: E402
from backend.pipeline.detectors.tracknet_runner import TrackNetRunner  # noqa: E402
from backend.pipeline.detectors import rally_gate  # noqa: E402
from backend.eval.calibrate_wasb import load_golden_gts  # noqa: E402
from backend.eval.metrics import score  # noqa: E402
from backend.eval import segmentation_metrics as _segm  # noqa: E402

# The golden manifest lives in the MAIN checkout's output/ (gitignored; absent in a fresh
# worktree / CI). Resolve it relative to this repo root first, with an env override for an
# out-of-tree checkout.
_REPO_ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
MANIFEST = os.environ.get(
    "SERVED_GATE_A_MANIFEST",
    os.path.join(_REPO_ROOT, "output", "human_lovo_manifest.json"),
)


def _golden_specs():
    """Return the manifest specs whose trajectory + label files are all present, else []."""
    if not os.path.isfile(MANIFEST):
        return []
    with open(MANIFEST, encoding="utf-8") as f:
        specs = json.load(f)
    ok = [s for s in specs
          if os.path.isfile(str(s.get("trajectory", "")))
          and os.path.isfile(str(s.get("labels", "")))]
    return ok if len(ok) >= 6 else []


_SPECS = _golden_specs()
_skip = pytest.mark.skipif(
    not _SPECS,
    reason=f"golden trajectories absent (manifest {MANIFEST!r}); runs on the owner box",
)


# -------------------------------------------------------------------- served-path litmus

@_skip
```

```
def test_served_path_runs_on_all_six_golden_videos():
    """The production FusionSegmenter decision path runs end-to-end (no GPU/Gemini) on all 6
    golden trajectories and yields a handoff window list for each, for both gate settings."""
    results = sga.evaluate_manifest(_SPECS)
    assert len(results) == 6
    for r in results:
        assert r.num_gt > 0
        assert r.n_handoff_off >= r.n_handoff_on   # Gate A only ever drops windows, never adds


@_skip
def test_served_gate_a_does_not_regress_f1():
    """SERVED no-regression guard: on the production windowing Gate A must not MATERIALLY hurt
    F1. It is ~neutral here (the held-shuttle junk the offline harness sees is already merged
    away by the conservative production windowing) ? so we assert the mean ?F1 sits in a tight
    band around 0, never a real regression. (The +0.074 lift lives under the permissive proposal
    windowing ? see the dedicated test below.)"""
    results = sga.evaluate_manifest(_SPECS)
    agg = sga.aggregate(results)
    # Honest measured value (2026-06-14): mean served ?F1 ? -0.008. Pin a tolerance band so a
    # real regression (Gate A actively destroying served F1) trips, while the measured ~neutral
    # result passes.
    assert -0.03 <= agg["mean_delta_f1"] <= 0.03, agg
    # And Gate A must never collapse served F1 to ~0 on the production path.
    assert agg["mean_f1_on"] >= 0.5 * agg["mean_f1_off"] - 1e-9, agg


@_skip
def test_served_gate_a_does_not_increase_over_segmentation():
    """Gate A only ever DROPS windows, so the served over-seg ratio must not rise when it's on
    (the direction-of-effect invariant ? even where the absolute lift is ~neutral)."""
    results = sga.evaluate_manifest(_SPECS)
    agg = sga.aggregate(results)
    assert agg["mean_seg_ratio_on"] <= agg["mean_seg_ratio_off"] + 1e-9, agg
    # Per-video too: the gate is monotone (kept windows ? all windows).
    for r in results:
        assert r.seg_ratio_on <= r.seg_ratio_off + 1e-9, r


@_skip
def test_served_uses_production_windowing_operating_point():
    """Pin the served operating point to config.json's indexing defaults ? if someone retunes
    production windowing, this litmus's premise (and the divergence it documents) is revisited
    deliberately, not silently."""
    assert sga.PROD_VELOCITY_THRESH == 0.02
    assert sga.PROD_MERGE_GAP == 2.0
    assert sga.PROD_MIN_WINDOW == 2.0


# ---------------------------------------------- the +0.074 lives under PROPOSAL windowing

# distill_local.PROPOSAL ? the permissive, recall-oriented candidate windowing the
```

```
OFFLINE
    # ablation/fusion_compare harness builds on (low velocity thresh, short merge/min-dur ? many
    # small windows). The +0.074 Gate-A lift is a property of THIS windowing, not the served one.
    _PROPOSAL = (0.005, 1.0, 0.5)


    @_skip
    def test_proposal_windowing_reproduces_offline_lift():
        """CONTROL / root-cause pin: under the OFFLINE harness's permissive proposal windowing, the
        SAME ``rally_gate`` net-crossing gate the served seam uses reproduces the offline litmus
        (?F1 ? +0.074, 6/6 wins, over-seg 1.68?1.01). This proves the served ~neutral result is a
        WINDOWING difference, not a broken gate ? and locks the offline number to its true premise."""
        d_on, d_off, seg_on, seg_off = [], [], [], []
        for s in _SPECS:
            fps, fw = float(s["fps"]), float(s["frame_width"])
            pts = TrackNetRunner.parse_trajectory_csv(s["trajectory"])
            wins = TrackNetRunner.trajectory_to_action_windows(
                pts, fps=fps, frame_width=fw, velocity_thresh=_PROPOSAL[0],
                merge_gap=_PROPOSAL[1], min_window_duration=_PROPOSAL[2])
            iv = [(w["start"], w["end"]) for w in wins]
            est = rally_gate.estimate_play_axis(pts)
            gated = rally_gate.gate_windows(pts, iv, fps, min_crossings=0, estimate=est)
            on = [(g.start, g.end) for g in gated if g.crossings >= 2]
            gts = load_golden_gts(s["labels"])
            d_on.append(score(on, gts).f1)
            d_off.append(score(iv, gts).f1)
            seg_on.append(_segm.segment_count_ratio(on, gts))
            seg_off.append(_segm.segment_count_ratio(iv, gts))
        n = len(d_on)
        mean_delta = (sum(d_on) - sum(d_off)) / n
        wins = sum(1 for a, b in zip(d_on, d_off) if a > b + 1e-9)
        assert mean_delta == pytest.approx(0.074, abs=0.01), mean_delta
        assert wins == 6, wins
        assert sum(seg_off) / n == pytest.approx(1.68, abs=0.05)
        assert sum(seg_on) / n == pytest.approx(1.01, abs=0.05)

```

40. assistant (2026-06-18T10:22:59.854Z)

On the new branch (HEAD e67add1, off origin/master). Since I switched branches, let me re-read the test file on this branch to ensure my edits target the current content.

41. assistant (2026-06-18T10:23:07.282Z)

The file matches master. Now I'll make the edits. First, the module docstring ? re-grounding it on the present multicourt corpus.

42. user (2026-06-18T10:23:24.626Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_served_gate_a_regression.py has been updated successfully. (file state is current in your context ? no need to Read it back)

43. assistant (2026-06-18T10:23:27.904Z)

Now test 1 ? rename and assert against the present count:

44. user (2026-06-18T10:23:33.031Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_served_gate_a_regression.py has been updated successfully. (file state is current in your context ? no need to Read it back)

45. assistant (2026-06-18T10:23:38.443Z)

Now test 2 ? rename to reflect "no *material* regression," re-measure the band, and document the new value:

46. user (2026-06-18T10:23:49.114Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_served_gate_a_regression.py has been updated successfully. (file state is current in your context ? no need to Read it back)

47. assistant (2026-06-18T10:23:55.731Z)

Now the `_PROPOSAL` section header/comment and test 3 ? re-measure on the present corpus and re-ground:

48. user (2026-06-18T10:24:17.529Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_served_gate_a_regression.py has been updated successfully. (file state is current in your context ? no need to Read it back)

49. assistant (2026-06-18T10:24:21.214Z)

Now let me run the served-gate-a tests to verify all pass on the local golden files:

50. user (2026-06-18T10:24:39.081Z)

```
============================= test session starts =============================
platform win32 -- Python 3.13.13, pytest-9.0.3, pluggy-1.6.0 --
C:\Users\avidu\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\avidu\Projects\badminton-highlight-indexer
configfile: pytest.ini
plugins: anyio-4.13.0, cov-7.1.0
collecting ... collected 5 items

tests/test_served_gate_a_regression.py::test_served_path_runs_on_all_golden_videos PASSED [ 20%]
tests/test_served_gate_a_regression.py::test_served_gate_a_does_not_materially_regress_f1 PASSED
[ 40%]
tests/test_served_gate_a_regression.py::test_served_gate_a_does_not_increase_over_segmentation
PASSED [ 60%]
tests/test_served_gate_a_regression.py::test_served_uses_production_windowing_operating_point
PASSED [ 80%]
tests/test_served_gate_a_regression.py::test_proposal_windowing_gate_a_is_f1_neutral_and_cuts_ov
er_seg PASSED [100%]

============================= 5 passed in 15.64s ==============================
```

51. assistant (2026-06-18T10:24:45.608Z)

All 5 pass. Now the pre-LGTM gate: lint, type check, and full suite with coverage. Let me run ruff and mypy first (parallel), then the full suite.

52. user (2026-06-18T10:24:46.373Z)

All checks passed!

53. user (2026-06-18T10:24:51.047Z)

```
backend\annotations\base.py:85: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs  [annotation-unchecked]
backend\sports\__init__.py:7: note: By default the bodies of untyped functions are not checked,
consider using --check-untyped-defs  [annotation-unchecked]
backend\pipeline\validators\base.py:36: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs  [annotation-unchecked]
backend\providers\__init__.py:7: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs  [annotation-unchecked]
Success: no issues found in 107 source files
```

54. assistant (2026-06-18T10:24:59.826Z)

ruff clean; mypy clean ("Success: no issues found"). Now the full suite with coverage:

55. user (2026-06-18T10:27:07.735Z)

```
backend\pipeline\detectors\wasb_infer.py             358    215    40%
backend\pipeline\detectors\wasb_runner.py            108     18    83%
backend\pipeline\personalization\__init__.py           0      0   100%
backend\pipeline\personalization\serving.py           26      0   100%
backend\pipeline\segmenters\__init__.py                8      0   100%
backend\pipeline\segmenters\base.py                   42      5    88%
backend\pipeline\segmenters\fusion_hybrid.py          77      1    99%
backend\pipeline\segmenters\gemini.py                294     71    76%
backend\pipeline\segmenters\tracknet_hybrid.py        18      0   100%
backend\pipeline\segmenters\trajectory_hybrid.py     170     15    91%
backend\pipeline\segmenters\wasb_hybrid.py            18      2    89%
backend\pipeline\segmenters\yolo_hybrid.py           225     47    79%
backend\pipeline\stitcher\compiler.py                224     44    80%
backend\pipeline\validators\__init__.py               14      0   100%
backend\pipeline\validators\base.py                   39      3    92%
backend\providers\__init__.py                         16      1    94%
backend\providers\base.py                              9      1    89%
backend\providers\gemini.py                           98     19    81%
backend\reporting\__init__.py                         26      2    92%
backend\reporting\harvest.py                         111      8    93%
backend\reporting\run_signals.py                      14      0   100%
backend\results.py                                    27      0   100%
backend\sports\__init__.py                            17      1    94%
backend\sports\badminton.py                           13      1    92%
backend\sports\base.py                                12      3    75%
backend\tools\__init__.py                              0      0   100%
backend\tools\export.py                               75      2    97%
backend\tools\rally_slicer.py                        104     47    55%
backend\utils\__init__.py                              0      0   100%
backend\utils\geometry.py                             29      0   100%
backend\utils\hashing.py                               9      0   100%
backend\utils\logging_setup.py                        11      0   100%
backend\utils\paths.py                                27      3    89%
backend\utils\run_telemetry.py                       126     12    90%
backend\utils\video.py                                25      3    88%
training\__init__.py                                   0      0   100%
training\gen0\__init__.py                              0      0   100%
training\gen0\harness.py                             108     11    90%
----------------------------------------------------------------------
TOTAL                                               8092   1519    81%
Required test coverage of 70% reached. Total coverage: 81.23%
864 passed, 7 warnings in 123.23s (0:02:03)
========================================================================
 RESULT: PASS - all tests green
========================================================================
```

56. assistant (2026-06-18T10:27:46.287Z)

Full suite green: **864 passed, coverage 81.23%** (above the 70 floor), ruff clean, mypy clean.
The served gate-a tests ran (cv2 present) and passed.

Let me clean up the scratch diagnostic and verify the working tree before committing.

57. user (2026-06-18T10:27:47.973Z)

```
--- status ---
 M tests/test_served_gate_a_regression.py
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
--- diff stat ---
 tests/test_served_gate_a_regression.py | 96 +++++++++++++++++++++--------------
 1 file changed, 58 insertions(+), 38 deletions(-)
--- HEAD (must be mine) ---
fix/served-gate-a-corpus-regrounding
e67add1 refactor(api): extract chunked-upload island from main into backend.api.uploads router
(#228)
```

58. assistant (2026-06-18T10:28:07.791Z)

Only my test file is modified (tracked); `artifacts/` and `docs/KHELSUTRA_PRODUCT_OVERVIEW.md` are pre-existing untracked files (not mine ? leaving them). HEAD is the `origin/master` tip with no sibling commits. Let me stage the explicit path and commit.

59. user (2026-06-18T10:28:14.469Z)

```
[fix/served-gate-a-corpus-regrounding e59b2fd] test(eval): re-ground served Gate-A litmus on the
present (multicourt) golden set
 1 file changed, 58 insertions(+), 38 deletions(-)
--- committed ---
e59b2fd test(eval): re-ground served Gate-A litmus on the present (multicourt) golden set
```

60. assistant (2026-06-18T10:28:47.186Z)

Committed cleanly as `e59b2fd` ? `git log origin/master..HEAD` shows only my commit (no sibling commits crept in).

Summary of what I found and did

The task's premise was slightly off, in a way worth flagging: the 9 present golden clips are **not** "the original 6 plus added multicourt." The **original 6 single-court clips are gone entirely** ? their label dir (`C:/Users/.../Annotation Setup/Collect/Golden Labelled/`) no longer exists. The present golden set is **9 brand-new multicourt clips** (labels now in repo `output/`). So the original-6 corpus the litmus was calibrated on can't be measured here at all.

Per your two answers (re-measure test 3 on the present corpus; re-ground narrative + values), I made these changes to tests/test_served_gate_a_regression.py:

Test	Before	After (measured 2026-06-18, present 9)
`..._runs_on_all_golden_videos` (renamed)	`len(results) == 6`	`== len(_SPECS)` and `>= 6`
`..._does_not_materially_regress_f1` (renamed)	band ±0.03 around ?0.008	band `?0.06..0.03` around **?0.034**; kept the `? 0.5×` guard (ratio 0.83)
`..._gate_a_is_f1_neutral_and_cuts_over_seg` (renamed)	`+0.074 / 6-of-6 / 1.68?1.01`	durable `seg_on < seg_off` invariant + snapshot **?F1??0.001 / 1.16?0.64**

The module docstring is re-grounded to the recall-starved multicourt regime, and notes the canonical **+0.074 win is the retired original-6 finding** (preserved in `docs/archives/`).

Verification (local, pre-LGTM gate): 5/5 served-litmus tests pass · full suite **864 passed, cov 81.23%** (?70) · ruff clean · mypy clean.

One thing I deliberately did *not* touch: `docs/QUALITY_ITERATIONS.md:569` and `:585` still claim this module is "the standing guard" that "reproduces +0.074 exactly." Re-measuring test 3 makes those two lines stale. That's the paper-backbone doc with your "honest results only" bar, so I left it scoped out of this test-fix PR pending your call.

Two decisions are yours:

61. user (2026-06-18T11:54:55.952Z)

Your questions have been answered: "The fix is committed on branch fix/served-gate-a-corpus-regrounding (off origin/master). How should I ship it?"="Push + open PR, you review", "docs/QUALITY_ITERATIONS.md:569/585 now claim this test 'reproduces +0.074 exactly' / is the '+0.074 standing guard' ? stale after the re-measure. How to handle?"="Fold minimal fix into this PR". You can now continue with these answers in mind.

62. assistant (2026-06-18T11:55:24.804Z)

The owner chose: **push + open PR for review** (no self-merge), and **fold the minimal doc fix into this PR**. Let me re-read the exact `QUALITY_ITERATIONS.md` lines to craft a precise, minimal honest correction.

63. user (2026-06-18T11:55:26.098Z)

563   **Why.** A multi-agent literature review (folded into
564   ANALYZERS_AND_RECONCILERS.md §13) confirmed the framework's *skeleton* is
565   standard prior art (Temporal Action Localization/Segmentation + late/score-level fusion ? cite, don't claim),
566   and flagged that our **measurement** is walking into known small-N traps that would make "promote only on
567   measured lift" promote noise or hide a real win. Four corrections, shipped as **pure, additive helpers** (no
568   existing `compare`/LOVO/`metrics.score` behaviour changed ? callers opt in):
569   - **C1 ? never a single mean.** `backend/eval/eval_stats.py`: `mean_with_ci`/`bootstrap_ci` (deterministic,
570   seeded), `paired_sign_test` (the exact distribution-free "won on enough *videos*?" test the corpus already
571   needs), `paired_delta_ci` (B?A with a CI that must clear 0). At n?6 a lone LOVO mean is not trustworthy.
572   - **C2 ? calibrate cue confidence.** `backend/eval/score_calibration.py`: Platt + isotonic (PAVA) calibrators
573   + `brier_score`/`expected_calibration_error`, so a CNN softmax, an audio onset, and an integer net-crossing
574   count are comparable before the arbiter and the over-confident cue can't swamp the rest.
575   - **C3 ? honest stacking.** `eval_stats.out_of_fold_predictions`/`grouped_folds` (leave-one-group-out by
576   match) so a *learned* producer's outputs used as arbiter features are out-of-fold (no leakage).
577   - **C4 ? over-segmentation-visible metrics.** `backend/eval/segmentation_metrics.py`: `f1_at_overlaps`
578   (F1@{10,25,50}), `average_precision`/`mean_average_precision` (mAP@tIoU), `segment_count_ratio`. tIoU alone
579   is blind to a rally split in 3 or two rallies merged ? exactly what the reconciler exists to fix.
580
581   **Status.** Helpers + 24 unit tests landed (pure stdlib, run in the numpy/GPU-free lanes); ruff + mypy clean.
582   **Not yet wired into the default harness flow** (wiring `compare`/`run_regression.py` to report CIs + F1@k is
583   the follow-on, owner-gated). The discipline they encode is the methodology for every future cue measurement ?
584   see §13/C1?C4 for the citations and the remaining (Tier-2/3) corrections.
585

586   ### Multi-signal fusion P2: the FusionSegmenter *(in progress, 2026-06-14)*
587   **Hypothesis.** Adjudicating shuttle-seeded windows with the net-crossing **Gate A** (`min_crossings?2`)

64. assistant (2026-06-18T11:55:37.208Z)

The line numbers shifted ? I'm now on a different branch (off `origin/master`), and `QUALITY_ITERATIONS.md` differs from what I read earlier on `feat/long-rally-lens`. Let me re-locate the actual stale text on this branch.

65. user (2026-06-18T11:55:39.308Z)

506-and drops short ones (served over-seg ratio already **< 1**, i.e. under-segmentation), so there is little junk
507-left to cut and the gate instead occasionally trims a borderline true window ? slightly negative ?F1. The gate
508:*itself* is correct (#148's control test reproduces +0.074 / 6-of-6 / p=0.031 exactly under proposal
509-windowing); the **operating point is what differs**. This is the canonical eval?serve trap.
510-
--
522-(it is a real proposal-windowing win). The path back to serving it is to **align the served windowing toward
523-the proposal operating point** and re-run the #148 litmus to confirm the lift transfers. The #148 litmus is
524:now the standing guard: it pins the honest measured behaviour (served ~neutral; +0.074 only under proposal
525-windowing) and trips if a future windowing change silently revives or breaks either fact.
526-

66. assistant (2026-06-18T11:55:42.847Z)

Found them at lines 508-509 and 524-525 on master. Let me read that exact region to match whitespace precisely.

67. user (2026-06-18T11:55:43.840Z)

494   **The #146 flip of Gate-A (net-crossings ? 2) to the live served default was REVERTED.** A served-side check ?
495   the repeatable litmus in [#148](https://github.com/avidullu/badminton-highlight-indexer/pull/148)
496   (`backend/eval/served_gate_a.py` + `tests/test_served_gate_a_regression.py`), corroborated by an independent
497   served run ? found Gate A is **not a win on the served path**: on the production (COARSE) windowing it is
498   **neutral-to-negative, mean served ?F1 ?0.008**, dropping ~12 real rallies across the golden 6 (2 losses, 0
499   clear wins). It is a **confirmed served regression**, so the owner directed a revert to restore the known-good
500   state.
501
502   **Why eval ? serve (the honest root cause).** The +0.074 below is a property of the **offline harness's
503   permissive PROPOSAL windowing** (`velocity_threshold=0.005, merge_gap=1.0, min_window=0.5`), which emits many
504   short candidate windows ? leaving abundant held-shuttle junk for Gate A to cut. The **served COARSE windowing**
505   (`velocity_threshold=0.02, merge_gap=2.0, min_window=2.0`) already merges flight bursts into a few long windows
506   and drops short ones (served over-seg ratio already **< 1**, i.e. under-segmentation), so there is little junk
507   left to cut and the gate instead occasionally trims a borderline true window ? slightly negative ?F1. The gate
508   *itself* is correct (#148's control test reproduces +0.074 / 6-of-6 / p=0.031 exactly

under proposal
509   windowing); the **operating point is what differs**. This is the canonical eval?serve trap.
510
511   **What the revert restores** (back to the pre-#146 no-regression state):
512   - `config.json` ? `indexing.default_segmenter` **`fusion_hybrid` ? `gemini`** (the prior served detector); the
513   explicit `indexing.fusion` block added by #146 is removed.
514   - `backend/config/models.py` ? `FusionConfig.min_crossings` default **2 ? 0** (Gate A **OFF** by default).
515   - Tests (`tests/test_config.py`, `tests/test_fusion_hybrid.py`) restored to the Gate-A-OFF defaults; the
516   **persisted-superset invariant is preserved** (gating only ever filtered the AI handoff, never persistence).
517   - `eval_baselines/experiment_leaderboard.json` ? the #146 `promotion` record carries a `reverted (eval?serve)`
518   note.
519
520   **Gate A is NOT deleted ? it remains opt-in** (`indexing.fusion.min_crossings: 2`) for footage with more
521   held-shuttle/service-feed false positives than the golden 6, and the offline ablation evidence below stands
522   (it is a real proposal-windowing win). The path back to serving it is to **align the served windowing toward
523   the proposal operating point** and re-run the #148 litmus to confirm the lift transfers. The #148 litmus is
524   now the standing guard: it pins the honest measured behaviour (served ~neutral; +0.074 only under proposal
525   windowing) and trips if a future windowing change silently revives or breaks either fact.
526

68. assistant (2026-06-18T11:55:56.277Z)

Now the two minimal honest corrections:

69. user (2026-06-18T11:55:58.809Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\QUALITY_ITERATIONS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

70. user (2026-06-18T11:56:05.921Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\QUALITY_ITERATIONS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

71. assistant (2026-06-18T11:56:12.708Z)

Let me verify the archive link resolves, re-check HEAD (sibling session active), then commit the doc fix.

72. user (2026-06-18T11:56:14.534Z)

```
docs/archives/quality-iterations/2026-06-first-ab-experiment/README.md
--- HEAD (must be my commit) ---
fix/served-gate-a-corpus-regrounding
e59b2fd test(eval): re-ground served Gate-A litmus on the present (multicourt) golden set
--- status ---
 M docs/QUALITY_ITERATIONS.md
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
```

73. assistant (2026-06-18T11:56:25.568Z)

Archive link resolves, HEAD is my commit (no sibling commits crept in), only the doc is modified. Committing the doc fix:

74. user (2026-06-18T11:56:29.284Z)

```
[fix/served-gate-a-corpus-regrounding de4240f] docs(eval): re-ground the +0.074 "standing guard" claims in QUALITY_ITERATIONS
 1 file changed, 13 insertions(+), 4 deletions(-)
--- branch commits vs master ---
de4240f docs(eval): re-ground the +0.074 "standing guard" claims in QUALITY_ITERATIONS
e59b2fd test(eval): re-ground served Gate-A litmus on the present (multicourt) golden set
```

75. assistant (2026-06-18T11:56:38.294Z)

Both commits are mine, no sibling commits. Now push and open the PR:

76. user (2026-06-18T11:56:41.282Z)

```
remote:
remote: Create a pull request for 'fix/served-gate-a-corpus-regrounding' on GitHub by visiting:
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/fix/served-gate-a-corpus-regrounding
remote:
branch 'fix/served-gate-a-corpus-regrounding' set up to track 'origin/fix/served-gate-a-corpus-regrounding'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      fix/served-gate-a-corpus-regrounding -> fix/served-gate-a-corpus-regrounding
```

77. user (2026-06-18T11:57:05.702Z)

https://github.com/avidullu/badminton-highlight-indexer/pull/236

78. assistant (2026-06-18T11:57:17.399Z)

Created: [badminton-highlight-indexer#236](https://github.com/avidullu/badminton-highlight-indexer/pull/236).

This is a meaningful checkpoint (PR opened), so let me update the live current-state memory. Let me read its top to see where the open-PRs entry goes.

79. user (2026-06-18T11:57:18.406Z)

```
1	---
2	name: current-state
3	description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not only on request). What just landed, what's next, open decisions."
4	metadata:
5	  node_type: memory
6	  type: project
7	  originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8	---
9
10	# Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12	**Checkpoint:** 2026-06-18 (? **TECH-DEBT SWEEP #2 ? CODE_MAP + 6 deferred refactors MERGED (#229-#235), master `5f555cf`, CI green; 1 HELD, 4 reported-not-doable**). Owner: "do the code map + fix the 11 tech debts with the same rigor, no regression." Approach = **characterization tests FIRST** (pin current behavior ? prove they pass on today's code ? then behavior-preserving refactor), worktree-isolated agent fleet, combined-state gate, merge-on-green.
13	- ? **#229** docs: CODE_MAP UPLOAD endpoints re-pointed to `backend/api/uploads.py` (post-#228; download_highlight stayed in main).
14	- ? **#230** motion:
```

`process_video`?`_detect_motion_segments`+`_populate_db_with_events` (new char tests mock cv2 module + seed `random` to pin the fabricated segment/events ? the previously-untested 204L path).
15 - ? **#231** rally_seg_eval: extract `_build_preset(args,side)` (11 char tests for the 0%-coverage report/main() paths ? module 0%?98%).
16 - ? **#232** database: `_execute_write` helper routes the 11 bool-returning single-statement writers (6 failure-branch char tests pin each EXACT log message ? byte-identical).
17 - ? **#233** golden_fixtures: split real-fixture subsystem into `golden_real_fixtures.py` with a **PEP-562 lazy `__getattr__` re-export** (handles both import orders; eager bottom-import would've cycled).
18 - ? **#234** main.py: job-worker drain/wake loop ? `backend/api/job_worker.py`; the monkeypatch-seam hazard solved by routing inter-fn calls through `backend.main` (`_main._arm_wake_timer`...) so `monkeypatch.setattr(m,...)` still intercepts (no leaked daemon timer). Added a minimal `mypy.ini` quarantine entry mirroring `[mypy-backend.main]` (same Optional-global debt). I spot-checked this one personally.
19 - ? **#235** eval: one shared parameterized `experiment.load_golden_videos` behind the 3 `load_videos` (column-lists + audio-policy args; char test pins the 3-way audio policy). Coverage 80.87% locally (?0.07pp = pure denominator effect from deleting 33 covered dup lines; absolute misses flat).
20 - **DUE DILIGENCE:** each PR = 1 clean commit + GitHub CI green; **combined-state gate** (all 7 merged into one worktree) = full suite **889 passed**, ruff+mypy(109) clean, **coverage 82.68% ? UP from 80.94%** (char tests net-added coverage); master push-CI = success. Worktrees/branches cleaned.
21 - ? **HELD (owner decision): `base-resolve-provider-dedup`** ? the gemini/yolo `_resolve_provider` bodies are NOT byte-identical (5 real diffs: signature, sport_profile capture-vs-discard, model_name source, an extra log line, return shape). A shared base method = a **behavior-changing redesign**, not a pure move ? agent correctly STOPPED. Needs owner approval + char tests around both segmenters' early-returns + the relocated log line if wanted.
22 - ? **NOT-DOABLE-AS-REFACTOR (reported, not forced):** trajectory_hybrid (already decomposed, hook-based base ? nothing to do); person_detector/tracknet grouping dedup (different tuple shapes ? redesign, real regression surface); `wasb_infer.run()` (GPU/WSL-only, needs owner-side torch-mocked test + real run); main.py routes-vs-CLI split (high blast radius, needs cv2 tests). [[working-agreement-merging]] [[decision-rigor-norm]]
23
24 **Checkpoint:** 2026-06-18 (? **REFACTORING SWEEP + BUG ? ALL 8 PRs (#221-#228) MERGED to master `e67add1`, master CI green**). Owner reviewed-and-approved by saying "merge when all suites + regression tests green." Before merging I did a **combined-state gate**: merged all 8 branches into one worktree (no conflicts, disjoint files) and ran the full suite (857 passed, ruff+mypy clean, cov 80.94%) so the AGGREGATE doesn't regress ? then squash-merged all 8; the push-triggered **master CI run = success**. NOTE: other sessions' PRs **#219 (docs/desktop-app-plan)** and **#220 (feat/long-rally-lens, awaiting owner review of numbers)** were intentionally LEFT OPEN (not mine, #220 owner-gated). Original sweep: a 2-phase discovery workflow (~18 agents): bug-hunt + tech-debt analysis over the grown files (mapped LOC + longest-functions: gemini.process_video=472L/71br worst, golden_fixtures=1113L, main.py=871L, database.py=738L). Produced a risk-rated worklist; the genuinely-risky/under-tested items were correctly DEFERRED (motion's random-heavy path, wasb GPU run(), main.py job-worker seam, golden_fixtures split, _execute_write dedup, rally_seg_eval reporting ? all "needs characterization tests first"). Shipped:
25 - **#221 BUG (`fix/config-loader-non-object-json`):** a valid-but-non-object `config.json` (`[]`/`42`/`"x"`/`null`) crashed startup (`.items()`/unpack), violating load_config's "non-fatal" contract ? main.py startup crash. One isinstance guard ? defaults+warn. Regression test proven to fail without fix. I implemented + verified this one personally.
26 - **7 behavior-PRESERVING refactors** (fanned out to worktree-isolated agents, each ran the full local gate then opened a PR): **#222** tracknet dedup+_active_frames · **#223** database _init_db?per-version helpers · **#224** compiler _trim_one_segment · **#225** gemini process_video?6 helpers (the 472L monster) · **#226** ablation export_pdf?ablation_pdf.py · **#227** yolo_hybrid process_video?phase helpers · **#228** main.py upload island?`backend/api/uploads.py` APIRouter (medium-risk; I spot-checked the global_config seam + lazy-import cycle-break myself).
27 - **VERIFIED INDEPENDENTLY (not just agent self-reports):** all 8 = exactly 1 clean commit off origin/master + **GitHub CI all green** (full suite + ruff + mypy + golden-fixtures cross-OS); local gates ~857 pass / cov ~80.7%. Pure-move refactors backed by existing tests (gemini/yolo have output-equivalence guards). **All OPEN for owner review ? NONE merged** (owner said "I'll take a look").

28 - ? **Follow-up (out of scope of #228):** `docs/CODE_MAP.md` still lists the upload symbols under `backend.main` ? update if #228 merges. The 11 DEFERRED tech-debt items (need characterization tests first) are a future sweep. [[working-agreement-merging]] [[decision-rigor-norm]]
29
30 **Checkpoint:** 2026-06-18 (? **P7 LONG-RALLY (>5s) QUALITY LENS + REEL KNOB SHIPPED ? PR [#220](https://github.com/avidullu/badminton-highlight-indexer/pull/220), both default-OFF, awaiting owner review of the NUMBERS**). Owner: "bake in a long-rally lens (eval) + product knob today." Delivered + measured n=15:
31 - ? **Eval lens** (`backend/eval/rally_seg_eval.py`): `--min-duration <s>` filters BOTH preds + golden GTs to rallies >s before matching (tIoU AND R2 paths consistently; a FILTER not a new metric; default 0=OFF). `--stratify`/`--long-tau`(5s) ? `{all vs long}×{single vs multicourt}` table. Court type = committed `MULTICOURT_CLIPS` set (manifest is **gitignored** ? committed set is the auditable source; manifest `court_type` overrides). `predict_all` windows once, re-scores per stratum.
32 - ? **Product knob**: `stitching.min_rally_duration` (s, 0=OFF) drops short rally pairs before the reel compile in `tools/generate_highlights.py` ? cleaner than `min_shot_count` (Unknown shot_count). +unit & integration tests.
33 - ? **MEASURED PAYOFF (n=15, local vs golden, milestone cap6+R4) ? the hypothesis split 3 ways (honest, NUMBERS INFORM):** (1) ? single-court over-fire is **short-FP-dominated** ? restrict to >5s collapses seg_ratio 1.43?1.04 (preds?GTs). (2) ?? **multicourt FAILS the "FPs are short" thesis**: over-fires **1.58× even on long rallies** (adjacent-court games = real long rallies ? long *spurious* windows) ? multicourt precision needs **court-localization (Gate-B/court mask, gap-closure Lever C)**, NOT a duration filter. (3) ?? strict tolP does NOT rise (0.326?0.278 single) ? the >5s pred set is **over-merge-contaminated** (kushagra med 9.9s, mahadevpura_1 63s blob; cap=6 truncation breaks the strict |?end|?1.5 match; tIoU forgiving holds ~0.42). (4) ? product knob still a clean reel win (a reel needs no strict boundary tolerance). **Gemini-long NOT computable in-checkout** (Boxhill gemini intervals owner-side only) ? lens is metric-agnostic, feed Gemini intervals through `filter_by_min_duration`. Pooled: single 608pred/416gt (long 218:210=1.04), multi 288/179 (long 158:100=1.58).
34 - ? **Gate:** suite 868 pass, cov **80.96%** (?70 floor, ~baseline 80.72%, not regressed), ruff+mypy(104) clean. ? **3 PRE-EXISTING fails** in `test_served_gate_a_regression.py` (cv2-gated ? **skip in CI**; hardcode old n=6 corpus, now 9+ local) ? PROVEN pre-existing (reproduce with my changes stashed; served_gate_a imports none of my funcs). Spawned chip `task_f38284e5` to recalibrate (separate PR).
35 - ?? **SHARED-CHECKOUT COLLISION HIT + RECOVERED:** mid-work a sibling switched HEAD `feat/long-rally-lens`?`docs/desktop-app-plan` (b9ff235, the #219 work). Caught via a `git stash` WIP message naming the wrong branch; my 7 files were uncommitted+intact and identical between branch tips, so `git checkout feat/long-rally-lens` carried them cleanly; re-verified HEAD=a010667 + `origin/master..HEAD` lists ONLY my commit before commit/push. [[gotcha-shared-checkout-branch-collisions]] held ? do NOT commit while on the wrong branch.
36 - ? Docs: `QUALITY_ITERATIONS.md` "P7" + `RALLY_DETECTION_GAP_CLOSURE.md` §7 row P7 (? measured). **Defaults stay OFF until owner reviews the numbers** ([[working-agreement-merging]]: touches eval+product config ? owner-gated flip, NOT a self-merge). [[eval-dual-set-regression]] [[decision-rigor-norm]] [[launch-report-convention]]
37
38 **Checkpoint:** 2026-06-18 (? **NEW TRACKED SUBPROJECT SCOPED ? `docs/DESKTOP_APP_PLAN.md`, PR [#219](https://github.com/avidullu/badminton-highlight-indexer/pull/219) OPEN (docs-only, owner-gated, all `[design]` ? NOTHING built)**). Owner-requested exploratory doc for future sessions: package the local pipeline as a **cross-platform desktop app** (Win/mac/Linux) for non-tech recreational players ? plug in the camera USB/SD card, click once, get back ONLY the highlight reels, **fully on-device** (no upload, no original-video bloat). Follows `PROJECT_DOC_TEMPLATE` (status hdr · D1?D6 decisions · §7 tracker = source-of-truth · DoD).
39 - **Centerpiece = the de-WSL native-compute port** (the real "port to Linux/Mac" subproject): WASB/TrackNet run today in **WSL2+CUDA (Windows-only)**; must run natively per-OS ? Linux/Win **CUDA**, macOS (no CUDA) **MPS/CPU**. Grounded in the `DetectorRunner` ABC (`backend/pipeline/detectors/base.py`) which already documents a `LinuxNativeRunner`/`ONNXRunner` as "the main de-risk goal"; the torch inference core (`wasb_infer.py`) just needs lifting out of WSL (identical `Frame,Visibility,X,Y` CSV ? windowing/stitch unchanged).
40 - **MAKE-OR-BREAK unknown (the P0 spike):** compute on a typical **NO-GPU enthusiast laptop**. WASB ~**3.4× real-time** on a laptop 2070S (measured, COST_SCALING); CPU far slower; Apple MPS unverified. P0 = measure CPU/MPS for a ~10-min clip ? go/no-go BEFORE any app code. If CPU-only too slow ? needs a **smaller exported owned model** (OWNED_MODEL export-clean

ONNX/CoreML) or a tiered fast/slow design.
41 - **§7 tracker:** P0 compute-feasibility spike ? P1 de-WSL native detector ? P2 app shell (Tauri vs Electron vs CLI+installer over the Python/FastAPI backend) ? P3 packaging/code-signing per OS. Only P0 is un-gated; P1?P3 owner-greenlit after P0. **Licensing locked (D4):** MIT/Apache/BSD bundle only; **ffmpeg = LGPL build** (no GPL x264 ? openh264/hardware encoders); WASB weights MIT ?.
42 - Realizes the **L0 fully-local** tier of `docs/VIDEO_LOCALITY_MODEL.md` (OQ5 packaging seed) + is the **`LocalDesktop` ComputeBackend** of `PLATFORM_ARCHITECTURE.md` §3.2. Indexed in `docs/README.md` (new "Product / platform plans" section). Branched off `origin/master` (a010667), explicit-path staged (left a sibling-WIP `backend/eval/rally_seg_eval.py` mod UNTOUCHED). ? **OWNER GATE:** do NOT build the app / start P0 without explicit approval. [[decision-rigor-norm]] [[gotcha-shared-checkout-branch-collisions]]
43
44 **Checkpoint:** 2026-06-18 (? **R1 MILESTONE LAUNCHED ? #204 MERGED** `67b6927`; flip re-verified at **n=13**, owner-approved). Owner: "verify #204 still stands at n=13, merge if findings stand" ? it stands (stronger), verified 3 ways, merged. **Local-windowing defaults now ON in production: cap=6 + R4** (`max_window_duration 6`, `window_inactivity_end 1.5`, `inactivity_rally_thresh 0.20`, `inactivity_min_density 0.30`). W2 / W1.5 hard-cap / R5 stay default-OFF.
45 - ? **n=13 re-verify (2 new Boxhill clips GX010137/GX010141):** baseline?milestone tolF1 **?+0.222, 12/0/1, p=0.0005** (stronger than n=11's 10/0); cap6 beats cap12 **12/0/1, p=0.0005**; cap6?+R4 +0.040, **9/1/3**, p=0.0215 (R4's 1 loss = GX010137, a tolerance artifact ? strict f1 & |?end| both IMPROVE there). Net guard **PASS**, merge_rate 0.651?0.161, **0 per-video regressions**. Both new clips improve (0.448?0.685, 0.333?0.431). tIoU ladder 0.236?0.474.
46 - ? **Corroborated 2 ways:** sibling **#214 `lovo_resweep.py`** matches to 3dp + LOVO cap=6 **13/13** on tIoU; my **5-lens adversarial workflow** `wf_66c34d9d-057` ? **STANDS** (every LOO fold clears; worst-fold drop GX010128 still ?+0.202 p=0.0010 ? no single clip drives it). Reproduce: `scratch/r1_evidence.py`.
47 - ? **#204 shipped clean:** rebased onto master (resolved a config/R5 conflict), 4 defaults flipped, 3 default-OFF tests + nightly docstrings updated, mandatory **`docs/R1_MILESTONE_LAUNCH_REPORT.md`** (both rulers + guard + caveats per [[launch-report-convention]]) + QUALITY_ITERATIONS "R1 LAUNCHED". Suite 855 green, cov 80.72%, ruff+mypy clean, 7/7 CI ? merged. **#216 re-baselines the nightly** (post-flip `default`==`milestone`; was self-flagging STALE ? re-snapshot n=13, both tolF1 0.359; data-only).
48 - **Caveats (Launch Report):** W1 cap = dominant lever / R4 = marginal +0.040; gate is the NET rate-based guard (strict=False BY DESIGN); GX010141 worst-behaved (future over-merge target); cap=4-vs-6 wobble = noise (LOVO re-fit to cap=4 is *worse*; tIoU firmly cap=6). [[eval-dual-set-regression]] [[decision-rigor-norm]]
49 - ? **#217 MERGED (owner did the honours)** ? nightly **all-OFF floor**: additively pinned a `baseline_off` config (SERVED + all quality knobs forced OFF) so every nightly report shows the delta the milestone buys (floor tolF1 **0.137** vs milestone **0.359**) ? a launched win keeps earning its place (the dual-eval-set discipline). default/milestone untouched. Re-baselined to 3 configs. The R1 Launch Report's "optional follow-up" is now DONE.
50 - ? **WORKTREE CLEANUP DONE (this session's R-effort):** removed my merged worktrees (nightly-floor, verify-n13, r1-flip2-leftover, r2-impl, r4-endrule, tier0-eval, w2-netcross, docs-golden/-rally-quality/-today, local-noai) + deleted 11 local + 5 remote merged branches. **KEPT (plain `git worktree remove` refused ? uncommitted WIP): w2-safeguard, hardchop** ? left untouched (not mine to clobber). Remaining worktrees are all sibling-active/recent/protected (boxhill17, pr-stitch, gap-doc, recdoc, resweep, bhi-execpolicy, agent-a771ff, tier1-windowing). NB: main checkout is now on a sibling's `feat/long-rally-lens` branch ? did not touch.
51
52 **Checkpoint:** 2026-06-18 (? **2 MORE BUG-FIX PRs SHIPPED & MERGED ? #212 + #213**; found by adversarial bug-hunt after owner said "if there are any other fixes, send + merge their PRs"). After #209, ran a 2-pass adversarial hunt (by-area ? by-class + completeness critic, ~15 agents) over `origin/master`. Pass 1 = 0 findings (clean); pass 2 = 3 confirmed real defects (all owner_gated=false, I verified each against actual code, not trusting the agents):
53 - ? **#212 (`c7baf03`) ? DB migration crash-safety:** v3/v4 used bare non-idempotent `ALTER TABLE ADD COLUMN` while v1/v2/v5 are `IF NOT EXISTS`. A DDL ALTER auto-commits independently of the `with conn:` scope, so an interrupted upgrade (after ALTER, before the end-of-block `PRAGMA user_version` bump) leaves the column present + version stale ? next open re-runs the ALTER ? `OperationalError: duplicate column name` ? **`__init__` throws forever = permanently bricked DB**. Fix: `_add_column_idempotent()` swallows ONLY 'duplicate column name'. Regression test proven to FAIL without the fix. Fully testable in CI (pure sqlite).
54 - ? **#213 (`bd33085`) ? scripts unwrap segmenter Result:** `scripts/run_phase3_eval.py` (`--segmenter` default=**motion**, the documented baseline) + `run_process_and_stitch.py` did

`len()`/truthiness on a raw `Success`/`Failure` dataclass (no `__len__`) ? `TypeError` on the default eval path; the `if not segments` guard was dead code. Other segmenters return bare lists (masked it). Fix: new generic `backend/results.unwrap_or_failure()` (DRY of backend/main.py's inline unwrap) used in both scripts + `tests/test_results.py`. Verified vs the REAL motion segmenter. ? E2E motion pass needs GPU/cv2 (owner-side); fix correctness covered by unit test + real-segmenter Failure check + main.py parity.

55 - **DUE DILIGENCE each:** my full local gate (suite 848/851 pass, ruff, mypy 104, cov ~80.7%) + GitHub CI all 8 checks green + adversarial 4-lens review workflow per PR (both **verdict=merge, 0 must-fix**, independently re-verified). Addressed the reviews' should-fixes BEFORE merge (re-raise-branch test for #212; empty-Success test + no-overclaim docstring fix for #213). Worktrees/branches cleaned.

56 - ? **NOT auto-merged (left for owner/other sessions):** chip **PR #210** (sibling fusion_compare/fusion_audio loaders, same KeyError class) is STILL its spawned session's to merge ? I deliberately didn't touch it. Optional cleanup the hunt flagged (NOT bugs): fold the 4 hand-rolled Success/Failure unwraps (main.py×2, generate_highlights.py) onto `unwrap_or_failure`; the litmus `>=6`/`==6` fragility ([[#209]] review). **NEW dev-env fact:** this container now HAS numpy/scipy/cv2/torch ? pure-logic eval AND segmenter-Failure paths run here (CLAUDE.md "lacks numpy/cv2/torch" is STALE; full GPU pipeline still can't run). [[working-agreement-merging]] [[decision-rigor-norm]]

57

58 **Checkpoint:** 2026-06-18 (? **ABLATION LITMUS MIXED-SCHEMA FIX ? MERGED #209 (`4bd9d2d`), issue #208 closed**). The pre-existing `test_ablation.py` litmus `KeyError: 'shuttle'` (flagged 2026-06-17 as OUT-OF-SCOPE `task_5878d224`) is FIXED + shipped to master. Root cause: `ablation.load_videos()` read `c["shuttle"]`/`c["person"]` unconditionally, so once R3/R4 corpus growth added 5 clips persisted via the velocity-windowing/distill path (leaner rows, no cue dicts: Badminton_BXH_2, GX010128, GX030094, mahadevpura_1, mahadevpura_2) it crashed whenever the full n=11 corpus sat in `output/` (the 6 original fusion-schema files: adarsh_avi_singles, gbaaddy, kushagra_singles, largetest_doubles, mahadevpura_singles, testlarge_short). Fix (option a, most-robust): `_is_fusion_schema()` predicate ? loader SKIPS leaner files (info-log, no crash); `_golden_present()` counts only fusion files so the skip-gate matches the loader. Litmus still reproduces Gate-A (?F1?0.074, 6/6, p?0.031 ? the 6 fusion files ARE the original n=6). +2 regression tests. **DUE DILIGENCE:** local full suite 848 pass/5 skip + ruff + mypy(104) + cov 80.66% (floor 70); **GitHub CI all 8 checks green**; **adversarial 4-lens review workflow ? verdict=merge, 0 must-fix** (independently re-verified the bug, the litmus reproduction, the 2-file scope). Owner reported issue #208 + opened PR #209 + squash-merged on green. Worktree cleaned; local+remote branch deleted. ? **Sibling latent bug still open (`task_8dd4efde`):** `fusion_compare.load_videos` + `fusion_audio.load_videos_with_audio` have the IDENTICAL unconditional `c["shuttle"]` access ? not caught by any test (synthetic tmp corpora only) but would crash the same CLIs on the mixed corpus. Separate PR (the review independently confirmed this scope call). Possible follow-up nit: litmus `>=6`-gate vs `==6`-assert would go red if a 7TH fusion clip is ever added (pre-existing; this fix improved the opposite drift). [[eval-dual-set-regression]] [[working-agreement-merging]]

59

60 **Checkpoint:** 2026-06-18 ~00:10 (? **REMAINING-3 4K REELS DONE ? ALL 6 BOXHILL-17 CLIPS NOW HAVE 4K** + cost-scaling doc + 10/10 visual QA ? all on G:). Owner asked for "just 4K highlights of the remaining 3, all local no Gemini." Delivered:

61 - ? **GX010138/139/140 4K reels** (the larger 9/10/12.8-min clips): **45/47/68 rallies**, 1.77/1.89/2.72 GB, 3840×2160, watermarked. Published to G: as `? - Highlights (4K).mp4`. Robust detached Scheduled Task `KhelSutraBoxhill4KRemaining` (keep-awake + self-restart; unregistered on completion).

62 - ? **DOGFOODED `--detect-from` end-to-end on fresh videos** (the merged #205 feature): `--video <4K original> --detect-from <1080p proxy>` ? detect on proxy, stitch 4K from original, reel named after --video (NO `_1080p` naming bug this time). Log confirmed "detecting on proxy ? stitching reel from ?". Validated beyond the unit tests.

63 - ? **Visual QA workflow** (`wf_2bd49aa1-361`, 10 vision agents): **10/10 delivered reels clean** ? correct resolution, watermark VISIBLY burned in, real rally content. (Adversarial verification of the shipped artifacts.)

64 - ? **`COST_SCALING.md` (all 6 clips)** on G:. **Cost model (RTX 2070 Super Max-Q):** GPU detection (WASB) is the driver at **~3.4× real-time** (?3.4 GPU-min/footage-min, ~0.055 s/frame); NVENC proxy ~0.2× RT; 4K stitch (CPU libx264) ~16 s/rally; end-to-end ~5× clip length; peak VRAM ~5.7 GB (fits 8 GB). ? the per-video GPU-busy sampler OVER-counts (background desktop GPU during the CPU stitch counted as busy) ? use the WASB wall-clock as the clean GPU number. 6-clip totals: 52.2 min footage ? 176 GPU-min WASB + 68 CPU-min stitch, 250 rallies, ~11.6 GB of 4K reels.

65 - ? **GX010137 INGESTED as GOLDEN #13 (n=13) + SCORED (2026-06-18) ? best clip so far,

big for the gap story.** golden 34 rallies (post-annotator-bug-fix labels). **LOCAL F1@0.5 0.82 (R 0.88!) vs GEMINI 0.89 (R 0.79):** on normal match play **local OUT-RECALLS Gemini** (finds more real rallies), but fires 14 FP vs Gemini's **0** (Gemini = perfect precision, misses 7). **? CONFIRMS opposite failure modes (local high-recall/low-precision, Gemini reverse) at n=2 ? PRECISION is local's #1 lever (Gate-A/colocation), fusion the durable win.** Local's gap to Gemini SHRINKS on normal play (hard clip ??0.16 ? normal ??0.07). Boundaries tight (local 0.39/0.44, tighter than Gemini). Folded into the gap-closure doc §3 (PR #211, `f04f3f3`). ? **OWNER found + fixed ~3 rally-annotator BUGS** (GX010137 post-fix; **GX010141 may be PRE-fix ? re-export+re-score to confirm its less-flattering numbers**). Corpus aggregate n=13: tIoU F1@0.5 0.474, R2 tolF1 0.359. Scorer `scratch/boxhill17_score_golden.py` CLIPS now has both; add the next clip + re-run.
66 - ? **GOLDEN #14/#15 INGESTED + SCORED (2026-06-18, n=15): Video 12 Boxhill_Doubles (MULTICOURT, 43 golden) + Video 4 GX020094 (singles, 40 golden), both 29.97fps.** Pre-staged WASB trajectories (KhelSutraGoldenPrep task) made it instant. **PRECISION-LEVER SMOKING GUN ? multicourt Boxhill_Doubles: local fired 87 vs 43 golden, tolP 0.20 (?80% FALSE POSITIVES = background-court games!), tolR 0.40, F1@0.5 0.43, seg_ratio 2.02×.** Video 4 singles also over-fires (local 59 vs 40, tolP 0.31). Boundaries tight on both (|?|?~1s). **? 4 ground-truth clips deep, the story is identical: gap = PRECISION (over-fire, worst on multicourt) + some recall, NEVER boundaries.** Strongest case yet for Gate-A/colocation precision lever (gap-closure doc G2). NO Gemini on these yet (offer 3-way). Ingest via `scratch/ingest_golden.py`; local in `output/golden_prep.db`. Corpus n=15: tIoU F1@0.5 0.468, R2 tolF1 0.353. **PREP DONE (KhelSutraGoldenPrep unregistered): Video 1 (GX010135, vid f10da909f0a4) + Video 9 (GX010132, vid 430d77131262) WASB trajectories + local rallies cached** (output/wasb/ + output/golden_prep.db; 4 prep proxies in output/golden_prep_proxies/; fps 29.97 for 12/4, 59.94 for 1/9) ? when owner labels them (golden CSV lands at annotation folder `[Done] Video N\<proxy>.rallies.csv` ? use `-LiteralPath`/python for the `[ ]` brackets) ingest+score is INSTANT (`ingest_golden.py`; local DB = golden_prep.db). **Video 7 (61min) SKIPPED per owner; finishing remaining badminton by EoD.** 2 SUBPROJECTS spun off as chips: long-rally eval lens (task_26c56d93) + cross-platform desktop app (task_0ba70da6).
67 - ? **LOVO THRESHOLD RE-SWEEP at n=13 ? thresholds HOLD, "retrain" buys nothing yet (MERGED #214 `55e4c89`, archived `docs/archives/quality-iterations/2026-06-18-lovo-resweep-n13/`).** Owner asked: now corpus is bigger, worth re-calibrating? Honest answer via leave-one-video-out: **tIoU picks cap=6 unanimously (13/13), ?(LOVO?fixed6) +0.000; R2 tolF1 cap=4 edges cap=6 by 0.005 but LOOCV re-fit doesn't generalize (??0.004=noise). R4 still significant (p=0.001/0.021); milestone?baseline (p<0.001).** ? corpus is in MEASURE (regression) territory not RETRAIN territory at this n; real retrain payoff = the learned fusion scorer, NOT these thresholds. **REUSABLE `scratch/lovo_resweep.py`** (CPU-only, re-run after each label batch). Taught owner LOOCV-vs-k-fold (leave-1 while n small + folds cheap ? stratified k-fold at hundreds). Watch: cap-4-vs-6 tolF1 wobble. ? owner found+fixed ~3 rally-annotator bugs (GX010141 pre-fix ? re-export to confirm its numbers).
68 - ? **GX010141 INGESTED as GOLDEN #12 (n=12) + verified.** `scratch/ingest_golden.py` (REUSABLE ? `--name/--golden-csv/--traj/--fps/--width/--tag`): copies labels?`output/GX010141.rallies.csv` + traj?`Collect/Trajectories/GX010141_june17_traj.csv`, writes `output/GX010141_golden_features.json` (gts+candidates), appends `output/human_lovo_manifest.json` (n=11?12). **Alignment OK** (29 gts max 371.5s ? 385s trajectory). `rally_seg_eval` now scores it IDENTICALLY to the standalone scorer (tIoU@0.5 F1 0.510, R2 tolF1 0.431/P0.50/R0.38/|?start|0.78/|?end|0.68) ? clean integration. **? `docs/GOLDEN_VIDEOS.md` registry doc still trails reality** (shows n=9 + GX030094/BXH2 "in progress"; manifest is n=12 incl those 2 + GX010141) ? sync as a PR (batch with the next clips, or now). When GX010137/020140(+large) land: `ingest_golden.py` each ? re-run `rally_seg_eval` (the milestone holds) ? one GOLDEN_VIDEOS.md sync PR.
69 - ? **GOLDEN GX010141 LANDED + SCORED (2026-06-18, `F:\?\Badminton - Boxhill - 17 June\GX010141.rallies.csv`, 29 rallies) ? CORRECTS my earlier optimistic agreement read.** Real accuracy (`scratch/boxhill17_score_golden.py` ? published `VS_GOLDEN.md`): **Gemini BEATS local on THIS clip** ? tIoU@0.5 F1 Gemini **0.67** (P0.94/R0.52) vs local **0.51** (P0.59/R0.45); tolF1(?2/1.5) Gemini 0.53 vs local 0.43. **Local's over-production = FALSE POSITIVES, not missed-by-Gemini rallies** (local 22 preds ? 13 tIoU-hits / 11 FP; Gemini 16 ? 15 hits / only 4 FP). ? **GX010141 is the disagreement-OUTLIER I deliberately picked first (local's likely-WORST clip)** ? do NOT generalize from n=1; the high-agreement clips (137/020140) next will likely favor local. Nuances: (a) BOTH miss ~half the 29 golden (recall ~45-52% ? detection is HARD on this continuous-drilling clip); (b) local & Gemini hit DIFFERENT golden rallies (only 3 overlap each other) ? **complementary ? fusion could lift recall**; (c) **boundaries TIGHT for both** (local |?start|0.78/|?end|0.68s) ? R2/R3/R4 windowing holds; the gap is DETECTION (recall + local precision), NOT boundaries. Honest correction logged ([[working-agreement-attribution]]).
70 - ? **OWNER labeling more (took a break): GX010137 + GX020140 next (the 2 with Gemini) +

maybe 1 large-rally-count clip; then I INGEST the batch as golden + update `docs/GOLDEN_VIDEOS.md`.** Scorer is generic ? add each clip's `(stem, proxy, golden_csv, has_gemini)` to `CLIPS` in `boxhill17_score_golden.py` + re-run. video_ids/cached rallies: local GX010137=39/GX020140=29/GX010138=45/GX010139=47/GX010140=68; Gemini (first 3) 27/16/16. Labels apply 1:1 to 1080p+4K (same timeline). [[golden-set-vendoring]] [[eval-dual-set-regression]] [[eval-boundary-tolerance]]

80. assistant (2026-06-18T11:57:50.534Z)

This current-state log already flagged these exact 3 fails (line 34) and spawned chip `task_f38284e5` to recalibrate ? my PR #236 is that work. Let me add a checkpoint entry at the top and dismiss the now-addressed chip.

81. user (2026-06-18T11:58:01.434Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

82. user (2026-06-18T11:58:11.368Z)

No pending task with id task_f38284e5. Task ids are not persisted across app restarts, so a chip from before a restart can no longer be withdrawn. Nothing was changed.

83. assistant (2026-06-18T11:58:17.523Z)

The chip predates a restart so it can't be withdrawn programmatically ? harmless; the current-state note records that #236 closes it. Let me do a final read of the complete test file to confirm it's coherent end-to-end.

84. user (2026-06-18T11:58:18.512Z)

```
"""SERVED Gate-A regression litmus ? the *production* FusionSegmenter path on real footage.

The flagged gap (#146): the served default flip (``default_segmenter=fusion_hybrid`` +
``min_crossings=2``) is **mock-tested only** on real video (``test_fusion_hybrid.py`` drives a
MagicMock runner). The offline ablation litmus (``test_ablation.test_litmus_reproduces_gate_a``)
validates the Gate-A *signal* ? but on the **permissive proposal windowing**
(``distill_local.PROPOSAL``), NOT the production operating point. This module closes the gap: it
runs the real ``FusionSegmenter`` decision seams (``_enrich_candidate_stats`` net-crossings +
``_select_for_handoff`` Gate A) over windows from the **production**
``trajectory_to_action_windows`` config, on the cached golden WASB trajectories, and asserts
the served path's measured behaviour.

GPU-free / Gemini-free (person cue OFF) ? it stops at the AI-handoff boundary. Skips cleanly in
CI where the golden trajectories are absent; runs on the owner box where they are present, over
**whatever golden clips are currently on disk** (``_golden_specs`` requires >= 6).

CORPUS NOTE ? re-grounded 2026-06-18 (the golden set is mutable and grows over time). The
original 6 *single-court* golden clips this litmus was first calibrated on (2026-06-14) have been
retired from the box ? their labels are gone ? so the present golden set is the 9 **multicourt**
clips now in ``output/`` (mahadevpura_1/2, GX0101xx, GX0x0094, Badminton_BXH_2, Boxhill_Doubles).
```

```
On that corpus the served path is **recall-starved**: WASB tracks a single shuttle while several
courts rally at once, so ``trajectory_to_action_windows`` emits FEWER windows than there are
rallies (served over-seg ratio < 1, i.e. under-segmentation). Because Gate A only ever DROPS
windows, on this under-segmented corpus it is mildly **F1-negative** (mean served ?F1 ? -0.034),
not the ~neutral -0.008 the original 6 showed. Under the permissive PROPOSAL windowing the same
gate is ~F1-neutral (?F1 ? -0.001) while still cutting over-segmentation (1.16?0.64).

The canonical **+0.074 / 6-of-6 / p=0.031** Gate-A win was a property of the *original-6* corpus
and is NOT reproduced here (those clips are no longer on the box); it is preserved in the archive
(``docs/archives/quality-iterations/2026-06-first-ab-experiment/``). These tests now pin the
**present-corpus** behaviour so a future windowing/corpus change that silently shifts it trips the
litmus. The pinned magnitudes are a date-stamped snapshot of the present golden set and will need
re-measurement as the corpus grows.
"""
from __future__ import annotations

import json
import os

import pytest

# cv2 is imported transitively by trajectory_hybrid (the served engine). On a box without it
# (CI), skip the whole module rather than erroring at import.
pytest.importorskip("cv2")

```

85. assistant (2026-06-18T11:58:24.351Z)

The file reads coherently. Let me confirm the final git state is clean and the PR is up.

86. user (2026-06-18T11:58:27.818Z)

```
--- status ---
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
--- branch vs origin ---
de4240f docs(eval): re-ground the +0.074 "standing guard" claims in QUALITY_ITERATIONS
e59b2fd test(eval): re-ground served Gate-A litmus on the present (multicourt) golden set
--- remote sync ---
de4240f
de4240f
--- PR ---
236 OPEN test(eval): re-ground served Gate-A litmus on the present (multicourt) golden set
https://github.com/avidullu/badminton-highlight-indexer/pull/236
```

87. assistant (2026-06-18T11:58:44.511Z)

Done. PR is up and everything checks out.

What shipped ? [badminton-highlight-indexer#236](https://github.com/avidullu/badminton-highlight-indexer/p

Branch `fix/served-gate-a-corpus-regrounding` off `origin/master`, 2 clean commits, local tree clean, remote in sync.

The 3 stale assertions in tests/test_served_gate_a_regression.py ? fixed and re-measured:

Test (renamed for honesty)	Before	After (measured 2026-06-18)
`..._runs_on_all_golden_videos`	`len == 6`	`== len(_SPECS)` and `>= 6`
`..._does_not_materially_regress_f1`	band ±0.03 / ?0.008	band `?0.06..0.03` / **?0.034**, kept `?0.5×` guard
`..._gate_a_is_f1_neutral_and_cuts_over_seg`	`+0.074 / 6-of-6 / 1.68?1.01`	durable `seg_on<seg_off` + snapshot **?0.001 / 1.16?0.64**

The one thing you should know that the task premise didn't anticipate: the original-6 single-court clips aren't on the box at all ? their label directory was deleted, so they can't be measured locally. The present golden set is 9 *entirely new* multicourt clips. That's why test 3 couldn't simply "pin to the original 6 by name" ? those files are gone. Per your call, I re-measured on the present corpus, re-grounded the module narrative, and folded the matching honest correction into `docs/QUALITY_ITERATIONS.md` (the two lines that claimed this test "reproduces +0.074 exactly" ? the +0.074 itself is preserved as the real archived original-6 result).

Verified locally (the real check ? CI skips this litmus, cv2-gated):
- 5/5 served-litmus tests pass
- Full suite **864 passed**, coverage **81.23%** (?70 floor)
- `ruff check .` clean · `mypy backend training` clean

I also recorded a checkpoint in the current-state memory and noted this PR closes the `task_f38284e5` chip that an earlier session spawned for exactly this recalibration.

Two caveats worth flagging for your review:
1. **The pinned magnitudes are a snapshot** of today's 9-clip set and will go stale again as the golden corpus grows ? I documented this in both the module docstring and the commit. If that recurring churn bothers you, the more durable design is to keep only the direction-of-effect invariants (like the `seg_on<seg_off` one I added) and drop the magnitude pins; say the word and I'll do that.
2. **?0.034 is a genuine (small) served regression** on this multicourt corpus, not the ~neutral the original 6 showed ? it's the recall-starved regime (Gate A can only remove already-scarce windows). The test now honestly admits it rather than masking it.

88. user (2026-06-18T11:58:46.921Z)

All the files ever used are located under the root dir "F:\GoPro Hero Black 12\KhelSutra" so feel free to index these and identify the ones whose originals may be missing although this is a feedback to me on organizing the videos so that this doesn;t repeat

89. assistant (2026-06-18T11:59:15.721Z)

Good idea ? let me index `F:\GoPro Hero Black 12\KhelSutra` and cross-reference it against the golden manifest to find which originals are missing. First, let me map the directory structure and inventory the relevant files.

90. user (2026-06-18T11:59:20.170Z)

=== TOP-LEVEL ===

```
Type Name                            SizeMB LastWriteTime
---- ----                            ------ -------------
file roora_batch_log.txt               1.60 6/15/2026 5:46:02 PM
file roora_june15_proxy_manifest.csv   0.00 6/15/2026 5:46:02 PM
DIR  2026 March 6                           3/6/2026 2:00:29 PM
DIR  Badminton - Boxhill - 17 June          6/17/2026 8:14:26 AM
DIR  Badminton 1 June 2026                  6/2/2026 8:07:26 AM
DIR  Badminton 12 June                      6/14/2026 8:20:57 AM
DIR  Badminton 12 June 2026                 6/12/2026 11:17:35 AM
```

```
DIR  Badminton 14 May 2026                 5/15/2026 6:29:42 AM
DIR  Badminton 2 May 2026                  5/4/2026 11:10:32 AM
DIR  Badminton 22 May 2026                 5/22/2026 10:49:30 AM
DIR  Badminton 4 May 2026                  5/5/2026 6:09:04 AM
DIR  Badminton 9 June 2026                 6/9/2026 8:33:14 PM
DIR  Bdminton 15 June                      6/15/2026 10:36:50 AM
DIR  GoPro 20 May 2026                     5/20/2026 8:39:58 AM
DIR  Highlights Generator Data             5/20/2026 9:01:52 AM
DIR  June 3 2026                           6/3/2026 11:45:38 AM
DIR  May 19 2026                           5/22/2026 6:05:18 AM
DIR  Misc                                  6/15/2026 1:53:26 PM
DIR  Roora June 15                         6/15/2026 1:26:14 PM
DIR  Tabletennis 10 June                   6/10/2026 6:23:16 PM
DIR  Tabletennis 11 June                   6/12/2026 11:17:25 AM
DIR  Training                              6/13/2026 7:40:05 PM
```

91. assistant (2026-06-18T11:59:26.463Z)

Many dated folders. Let me build the full inventory of video originals and label/trajectory CSVs across the tree.

92. user (2026-06-18T11:59:30.596Z)

```
=== VIDEO FILES (mp4/mov/lrv) ===
Adarsh and Avi.MP4             10,915MB  .\Training\Golden Labelled
Adarsh v Avi.MP4                5,277MB  .\Badminton 12 June\100GOPRO\Badminton
Adarsh v Vasant 2.MP4           4,220MB  .\Badminton 12 June\100GOPRO\Badminton
Adarsh v Vasant.MP4             6,957MB  .\Badminton 12 June\100GOPRO\Badminton
All 1 v 1.MP4                  10,989MB  .\Badminton 12 June\100GOPRO\Badminton
Avi Single.MP4                  2,329MB  .\June 3 2026\100GOPRO
Badminton BXH 1.MP4             6,323MB  .\GoPro 20 May 2026\100GOPRO
Badminton BXH 2.MP4            10,420MB  .\Roora June 15\Video 10 - Badminton Multicourt
Badminton BXH 2.MP4            10,420MB  .\GoPro 20 May 2026\100GOPRO
Badminton BXH 3.MP4            10,987MB  .\GoPro 20 May 2026\100GOPRO
Badminton BXH 4.MP4             8,981MB  .\GoPro 20 May 2026\100GOPRO
Bike going to Mayura.MP4          632MB  .\2026 March 6\100GOPRO\Bike
Bike return from Mayura.MP4       557MB  .\2026 March 6\100GOPRO\Bike
Boxhill Doubles.MP4            10,998MB  .\Roora June 15\Video 12 - Badminton Multicourt
Chhetri Opinion.MP4               123MB  .\2026 March 6\100GOPRO
Compressed_VeryLargeTestVideo.mp4       12MB  .\Highlights Generator Data\Tests
Cricket Half pitch - 2.MP4         28MB  .\2026 March 6\100GOPRO\Cricket
Cricket Half Pitch - 3.MP4        768MB  .\2026 March 6\100GOPRO\Cricket
Cricket Half pitch.MP4          1,298MB  .\2026 March 6\100GOPRO\Cricket
FinalOutputOfKushagraYashAviFirstVideo.mp4      683MB  .\Highlights Generator Data
FinalOutputOfTestLargeVideo_GeminiGen2Prompt.mp4      599MB  .\Highlights Generator Data\Tests
FinalOutputOfTestLargeVideo_GeminiGenPrompt.mp4      304MB  .\Highlights Generator Data\Tests
FinalOutputOfTestLargeVideo.mp4      456MB  .\Highlights Generator Data\Tests
FinalOutputOfVasanthAdarshAviVeryLargeVideo_CoachPrompt.mp4    1,628MB  .\Highlights Generator
Data\Tests
FinalOutputOfVasanthAdarshAviVeryLargeVideo_GeminiGen2Prompt.mp4    1,784MB  .\Highlights
Generator Data
FinalOutputOfVeryLargeTestVideo_CoachPrompt.mp4    1,504MB  .\Highlights Generator Data\Tests
FinalOutputOfVeryLargeTestVideo_GeminiGen2Prompt.mp4    2,223MB  .\Highlights Generator Data
FirstOutputOfTestLargeVideo.mp4       53MB  .\Highlights Generator Data\Tests
gBaaddy20May2026.MP4            6,747MB  .\Training\Golden Labelled
GX010040.MP4                      216MB  .\2026 March 6\100GOPRO\ELT Pickleball
GX010043.MP4                      287MB  .\2026 March 6\100GOPRO\ELT Pickleball
GX010044.MP4                      398MB  .\2026 March 6\100GOPRO\ELT Pickleball
GX010051.MP4                      758MB  .\2026 March 6\100GOPRO
GX010053.MP4                      187MB  .\2026 March 6\100GOPRO
GX010055.MP4                    1,795MB  .\Roora June 15\Video 7 - Badminton Multicourt
GX010056.MP4                    1,471MB  .\Misc
GX010057.MP4                    1,185MB  .\Misc
GX010083.MP4                   10,996MB  .\2026 March 6\100GOPRO\Bakar
GX010091.MP4                   10,998MB  .\Badminton 2 May 2026\100GOPRO
```

GX010092.MP4	10,998MB	.\Badminton 2 May 2026\100GOPRO
GX010093.MP4	10,998MB	.\Badminton 2 May 2026\100GOPRO
GX010094.MP4	10,997MB	.\Badminton 2 May 2026\100GOPRO
GX010096.MP4	10,999MB	.\Badminton 4 May 2026\100GOPRO
GX010097.MP4	10,998MB	.\Badminton 4 May 2026\100GOPRO
GX010098.MP4	7,735MB	.\Badminton 4 May 2026\100GOPRO
GX010099.MP4	10,989MB	.\Badminton 14 May 2026\100GOPRO
GX010104.MP4	10,989MB	.\May 19 2026\100GOPRO
GX010104.MP4	10,989MB	.\Roora June 15\Video 8 - Table tennis
GX010104.MP4	10,989MB	.\Tabletennis 10 June
GX010105.MP4	5,327MB	.\May 19 2026\100GOPRO
GX010105.MP4	5,327MB	.\Tabletennis 10 June
GX010106.MP4	7,685MB	.\Badminton 22 May 2026\gBaddy
GX010107.MP4	10,915MB	.\Badminton 22 May 2026\gBaddy
GX010108.MP4	10,915MB	.\Badminton 1 June 2026\100GOPRO
GX010109.MP4	5,237MB	.\Badminton 1 June 2026\100GOPRO
GX010110.MP4	7,159MB	.\Badminton 1 June 2026\100GOPRO
GX010115.MP4	10,987MB	.\Badminton 9 June 2026\100GOPRO
GX010116.MP4	7,798MB	.\Badminton 9 June 2026\100GOPRO
GX010117.MP4	6,784MB	.\Badminton 9 June 2026\100GOPRO
GX010118.MP4	7,307MB	.\Badminton 9 June 2026\100GOPRO
GX010119.MP4	8,411MB	.\Badminton 9 June 2026\100GOPRO
GX010120.MP4	2,082MB	.\Badminton 9 June 2026\100GOPRO
GX010121.MP4	10,987MB	.\Badminton 9 June 2026\100GOPRO
GX010122.MP4	10,989MB	.\Tabletennis 10 June
GX010123.MP4	205MB	.\Tabletennis 11 June
GX010124.MP4	10,968MB	.\Tabletennis 11 June
GX010124.MP4	10,968MB	.\Roora June 15\Video 2 - Tabletennis
GX010128.MP4	10,802MB	.\Badminton 12 June 2026
GX010129.MP4	7,673MB	.\Badminton 12 June 2026
GX010130.MP4	10,989MB	.\Badminton 12 June\100GOPRO\Pickleball
GX010131.MP4	10,893MB	.\Badminton 12 June\100GOPRO\Pickleball
GX010132.MP4	10,990MB	.\Roora June 15\Video 9 - Badminton
GX010135.MP4	10,989MB	.\Bdminton 15 June\100GOPRO
GX010135.MP4	10,989MB	.\Roora June 15\Video 1 - Badminton
GX010136.MP4	10,990MB	.\Bdminton 15 June\100GOPRO
GX010137.MP4	6,449MB	.\Badminton - Boxhill - 17 June
GX010138.MP4	7,784MB	.\Badminton - Boxhill - 17 June
GX010139.MP4	8,545MB	.\Badminton - Boxhill - 17 June
GX010140.MP4	10,990MB	.\Badminton - Boxhill - 17 June
GX010141.MP4	5,505MB	.\Badminton - Boxhill - 17 June
GX020083.MP4	10,999MB	.\2026 March 6\100GOPRO\Bakar
GX020091.MP4	7,128MB	.\Badminton 2 May 2026\100GOPRO
GX020092.MP4	10,999MB	.\Badminton 2 May 2026\100GOPRO
GX020093.MP4	1,243MB	.\Badminton 2 May 2026\100GOPRO
GX020094.MP4	10,999MB	.\Roora June 15\Video 4 - Badminton
GX020094.MP4	10,999MB	.\Badminton 2 May 2026\100GOPRO
GX020096.MP4	10,999MB	.\Badminton 4 May 2026\100GOPRO
GX020097.MP4	11,000MB	.\Badminton 4 May 2026\100GOPRO
GX020099.MP4	9,005MB	.\Badminton 14 May 2026\100GOPRO
GX020104.MP4	4,263MB	.\May 19 2026\100GOPRO
GX020104.MP4	4,263MB	.\Bdminton 15 June\100GOPRO
GX020107.MP4	6,747MB	.\Badminton 22 May 2026\gBaddy
GX020108.MP4	2,749MB	.\Badminton 1 June 2026\100GOPRO
GX020115.MP4	5,377MB	.\Badminton 9 June 2026\100GOPRO
GX020121.MP4	3,197MB	.\Badminton 9 June 2026\100GOPRO
GX020122.MP4	10,988MB	.\Tabletennis 10 June
GX020124.MP4	7,736MB	.\Tabletennis 11 June
GX020125.MP4	4,256MB	.\Badminton 12 June 2026
GX020130.MP4	9,802MB	.\Badminton 12 June\100GOPRO\Pickleball
GX020130.MP4	9,802MB	.\Roora June 15\Video 5 - Pickleball
GX020135.MP4	10,989MB	.\Bdminton 15 June\100GOPRO
GX020136.MP4	10,712MB	.\Bdminton 15 June\100GOPRO
GX020140.MP4	5,515MB	.\Badminton - Boxhill - 17 June
GX030083.MP4	3,265MB	.\2026 March 6\100GOPRO\Bakar

```
GX030092.MP4                    1,657MB  .\Badminton 2 May 2026\100GOPRO
GX030094.MP4                   10,120MB  .\Roora June 15\Video 6 - Badminton
GX030094.MP4                   10,120MB  .\Badminton 2 May 2026\100GOPRO
GX030096.MP4                    4,767MB  .\Badminton 4 May 2026\100GOPRO
GX030097.MP4                    1,440MB  .\Badminton 4 May 2026\100GOPRO
KushagraYashAviFirstVideo.MP4    7,685MB  .\Training\Golden Labelled
LargeTestVideo.MP4             10,998MB  .\Training\Golden Labelled
mahadevpura-0.MP4              10,989MB  .\Badminton 12 June 2026
mahadevpura-1.MP4               2,491MB  .\Badminton 12 June 2026
mahadevpura-2.MP4               8,800MB  .\Badminton 12 June 2026
mahadevpura-2.MP4               8,800MB  .\Roora June 15\Video 3 - Badminton Multicourt
MahadevpuraSingles.MP4            187MB  .\Training\Golden Labelled
Mom Jan 26 Singing.MP4          2,211MB  .\2026 March 6\100GOPRO
OutputOfTestLargeVideo_CoachPrompt.mp4      100MB  .\Highlights Generator Data\Tests
OutputOfVeryLargeTestVideo.mp4    1,392MB  .\Highlights Generator Data\Tests
OutputOfVeryLargeTestVideo2.mp4    2,551MB  .\Highlights Generator Data\Tests
Pickleball chestie footage.MP4    1,049MB  .\2026 March 6\100GOPRO\ELT Pickleball
Pickleball ELT 1.MP4           10,990MB  .\Roora June 15\Video 11 - Pickleball
Pickleball ELT 1.MP4           10,990MB  .\GoPro 20 May 2026\100GOPRO
Pickleball ELT 2.MP4            3,967MB  .\GoPro 20 May 2026\100GOPRO
Pickleball recording ELT.MP4      148MB  .\2026 March 6\100GOPRO\ELT Pickleball
Shivratri 2026.MP4              1,275MB  .\2026 March 6\100GOPRO
TestLargeVideo.MP4              1,440MB  .\Training\Golden Labelled
Vasant v Avi & Vijay v Avi 2.MP4   10,988MB  .\June 3 2026\100GOPRO
Vasant vs [Avi and Adarsh].MP4   10,990MB  .\Badminton 12 June\100GOPRO\Badminton
Vijay and Doubles.MP4          10,200MB  .\June 3 2026\100GOPRO
Vijay v Avi - 1.MP4            10,990MB  .\June 3 2026\100GOPRO
Vijay v Avi - 2.MP4             3,523MB  .\June 3 2026\100GOPRO
Vijay v Vasant.MP4              8,173MB  .\June 3 2026\100GOPRO
```

93. assistant (2026-06-18T12:00:04.911Z)

Key finding already visible: the **original videos are NOT missing**. All six original-golden source videos are sitting in `.\Training\Golden Labelled\` on F: (Adarsh and Avi, KushagraYashAviFirstVideo, LargeTestVideo, gBaaddy20May2026, TestLargeVideo, MahadevpuraSingles). What vanished was the **C: working-copy of the label CSVs**. Let me confirm whether the labels also live on F:, and inventory all CSVs.

94. user (2026-06-18T12:00:10.795Z)

```
=== Training\Golden Labelled (FULL) ===

Name                                       SizeKB LastWriteTime
----                                       ------ -------------
Adarsh and Avi.MP4                    11176809.30 6/1/2026 8:38:28 AM
Adarsh and Avi.rallies.csv                   4.00 6/8/2026 5:20:16 PM
gBaaddy20May2026.MP4                   6909276.00 5/22/2026 8:40:40 AM
gBaaddy20May2026.rallies.csv                 2.00 6/8/2026 9:53:40 PM
KushagraYashAviFirstVideo.MP4          7869350.30 5/22/2026 8:02:16 AM
KushagraYashAviFirstVideo.rallies.csv        1.40 6/8/2026 5:41:24 PM
KushagraYashAviFirstVideo.report.json        3.20 6/10/2026 5:38:50 PM
KushagraYashAviFirstVideo.report.md          1.60 6/10/2026 5:38:50 PM
KushagraYashAviFirstVideo.summary.md         0.50 6/10/2026 5:38:50 PM
LargeTestVideo.MP4                    11261604.30 5/4/2026 5:18:56 PM
LargeTestVideo.rallies.csv                   2.10 6/8/2026 8:28:20 PM
MahadevpuraSingles.MP4                  191584.40 1/16/2026 8:00:28 AM
MahadevpuraSingles.rallies.csv               1.30 6/9/2026 8:08:44 AM
MahadevpuraSingles.report.json               3.10 6/10/2026 4:29:16 PM
MahadevpuraSingles.report.md                 1.50 6/10/2026 4:29:16 PM
MahadevpuraSingles.summary.md                0.50 6/10/2026 4:29:16 PM
TestLargeVideo.MP4                     1475027.30 5/4/2026 8:39:14 PM
TestLargeVideo.rallies.csv                   0.30 6/8/2026 9:33:16 PM


=== ALL .rallies.csv anywhere under root ===
```

```
Adarsh and Avi.rallies.csv              .\Training\Golden Labelled
gBaaddy20May2026.rallies.csv            .\Training\Golden Labelled
GX010137.rallies.csv                    .\Badminton - Boxhill - 17 June
GX010141.rallies.csv                    .\Badminton - Boxhill - 17 June
KushagraYashAviFirstVideo.rallies.csv   .\Training\Golden Labelled
LargeTestVideo.rallies.csv              .\Training\Golden Labelled
MahadevpuraSingles.rallies.csv          .\Training\Golden Labelled
TestLargeVideo.rallies.csv              .\Training\Golden Labelled
```

95. assistant (2026-06-18T12:01:08.707Z)

Now the picture is complete and reassuring: the original-6 clips' **originals AND their label CSVs are both safe on F:** in `Training\Golden Labelled\`. Nothing was actually lost ? the manifest just points at a stale C: working-copy path. Let me confirm two things: whether the **new** clips' labels have any F: backup, and whether the C: repo `output/` labels are git-tracked (i.e., have any durable backup at all).

96. user (2026-06-18T12:01:13.772Z)

```
=== Roora June 15 subfolders (structure sample) ===
Video 1 - Badminton                     1 files
Video 10 - Badminton Multicourt         1 files
Video 11 - Pickleball                   1 files
Video 12 - Badminton Multicourt         1 files
Video 2 - Tabletennis                   1 files
Video 3 - Badminton Multicourt          1 files
Video 4 - Badminton                     1 files
Video 5 - Pickleball                    1 files
Video 6 - Badminton                     1 files
Video 7 - Badminton Multicourt          1 files
Video 8 - Table tennis                  1 files
Video 9 - Badminton                     1 files

=== any CSV under Roora June 15 ===
```

97. user (2026-06-18T12:01:15.620Z)

```
=== are new-clip labels git-tracked or ignored? ===
GX010128           .gitignore:39:output/       output/GX010128.rallies.csv
GX010137           .gitignore:39:output/       output/GX010137.rallies.csv
mahadevpura_1      .gitignore:39:output/       output/mahadevpura_1.rallies.csv
Boxhill_Doubles    .gitignore:39:output/       output/Boxhill_Doubles.rallies.csv
GX020094           .gitignore:39:output/       output/GX020094.rallies.csv
--- is it committed in git? ---
(empty above = NOT committed)
```

98. assistant (2026-06-18T12:01:59.013Z)

I now have a complete, and frankly reassuring, picture. Let me do one last cross-check: where do the golden labels live on C: (any backup outside the gitignored repo `output/`?).

99. user (2026-06-18T12:02:03.305Z)

```
=== all *.rallies.csv on C: under Projects ===
./badminton-highlight-indexer/output/Badminton_BXH_2.rallies.csv
./badminton-highlight-indexer/output/Boxhill_Doubles.rallies.csv
./badminton-highlight-indexer/output/GX010128.rallies.csv
./badminton-highlight-indexer/output/GX010137.rallies.csv
./badminton-highlight-indexer/output/GX010141.rallies.csv
./badminton-highlight-indexer/output/GX020094.rallies.csv
./badminton-highlight-indexer/output/GX030094.rallies.csv
./badminton-highlight-indexer/output/mahadevpura_1.rallies.csv
./badminton-highlight-indexer/output/mahadevpura_2.rallies.csv
```

100. assistant (2026-06-18T12:03:23.488Z)

I've indexed the whole tree and cross-referenced it against the golden manifest. The headline is good news: **no originals are missing.** And the "lost" original-6 labels aren't lost either ? they're safe on F:.

Cross-reference: all 15 golden clips vs F: archive

Clip (manifest)	Original video on F:	Original	Golden label
adarsh_avi_singles	`Training\Golden Labelled\Adarsh and Avi.MP4`	?	? **on F:** (`?\Adarsh and Avi.rallies.csv`) · ? C: path broken
kushagra_singles	`Training\Golden Labelled\KushagraYashAviFirstVideo.MP4`	?	? on F: · ? C:
largetest_doubles	`Training\Golden Labelled\LargeTestVideo.MP4`	?	? on F: · ? C:
gbaaddy	`Training\Golden Labelled\gBaaddy20May2026.MP4`	?	? on F: · ? C:
testlarge_short	`Training\Golden Labelled\TestLargeVideo.MP4`	?	? on F: · ? C:
mahadevpura_singles	`Training\Golden Labelled\MahadevpuraSingles.MP4`	?	? on F: · ? C:
mahadevpura_1	`Badminton 12 June 2026\mahadevpura-1.MP4`	?	?? **C: `output/` only**
mahadevpura_2	`Badminton 12 June 2026\mahadevpura-2.MP4` (+ Roora V3)	?	?? **C: `output/` only**
GX010128	`Badminton 12 June 2026\GX010128.MP4`	?	?? **C: `output/` only**
GX030094	`Roora June 15\Video 6\` + `Badminton 2 May\` GX030094.MP4	? (2 copies)	?? **C: `output/` only**
Badminton_BXH_2	`Roora June 15\Video 10\` + `GoPro 20 May\` Badminton BXH 2.MP4	? (2 copies)	?? **C: `output/` only**
Boxhill_Doubles	`Roora June 15\Video 12\Boxhill Doubles.MP4`	?	?? **C: `output/` only**
GX020094	`Roora June 15\Video 4\` + `Badminton 2 May\` GX020094.MP4	? (2 copies)	?? **C: `output/` only**
GX010137	`Badminton - Boxhill - 17 June\GX010137.MP4`	?	? C: `output/` **+ on F:** (Boxhill)
GX010141	`Badminton - Boxhill - 17 June\GX010141.MP4`	?	? C: `output/` **+ on F:** (Boxhill)

What actually happened (the #236 root cause)

The original-6 weren't "retired." Their labels live next to their videos on F: in `Training\Golden Labelled\` (each `*.MP4` paired with `*.rallies.csv`, last edited June 8?9). The manifest just points at a **C: working-copy path** ? `C:\Users\avidu\Projects\Annotation Setup\Collect\Golden Labelled\` ? that got cleared. The trajectories for those 6 are still on C:; only the C: label copies vanished. **Fully recoverable from F:.**

The real risk (this is the part worth acting on)

The irony: the "loss" that triggered #236 was *recoverable*. But the **7 current golden labels that are genuinely single-copy** are the new ones ? `mahadevpura_1/2`, `GX010128`, `GX030094`, `Badminton_BXH_2`, `Boxhill_Doubles`, `GX020094`. Their labels exist **only** as files in C: `output/`, which is:
- **gitignored** (`.gitignore:39: output/`) and **not committed** ? no version-control backup, and
- **not copied to F:** ? no archive backup (only `GX010137`/`GX010141` made it to F:).

So the 7 hand-made golden labels for your *current* eval corpus have exactly one copy, on C:, in a directory your own cache-cleanup tooling targets. That's the disappearance that already bit you ? except for these there'd be no F: fallback. These should be backed up today.

Organizational feedback ? so this doesn't repeat

1. **Co-locate the label with its original, always.** The original-6 did this right (`X.MP4` + `X.rallies.csv` together in `Golden Labelled\`). The new-9 broke it ? labels ended up in a gitignored repo scratch dir divorced from the video. Make the rule: every golden clip = `<name>.MP4` + `<name>.rallies.csv` side-by-side on F:.

2. **GoPro filenames are not unique ? stop referencing clips by them.** `GX010104` exists 3×, `GX0x0094`/`GX010135`/`Badminton BXH 2` each appear in 2 folders. The camera recycles `GX01xxxx` numbers across cards/sessions; today they're copies (sizes match), but the only disambiguator is the folder name ? which the manifest throws away. The codebase already has the right primitive: **`video_id` = MD5 of the file** (`backend/utils/hashing.py`). Key golden clips by `video_id` (or rename-on-ingest to `YYYY-MM-DD_session_descriptive`), not by bare GoPro name.

3. **Version-control the golden labels ? they're irreplaceable and tiny (1?4 KB).** Move them out of gitignored `output/` into a committed `golden/labels/` dir (or a small dedicated repo / git-LFS). A hand-labeled rally CSV is the single most expensive artifact you produce; it shouldn't be the least protected.

4. **One authoritative, checked-in index ? pointing at the F: archive, not C: working copies.** The manifest hardcodes absolute C: paths that drift the moment a working copy is cleaned. Have it record, per clip: canonical F: path, `video_id`/MD5, label path, fps, `court_type`, provenance. Make F: the source of truth and resolve from there.

5. **Track vendor-prep provenance.** `Roora June 15\Video N - Sport\` are renamed copies of selected originals. Keep them as a separate export, but record which original each "Video N" maps to, so a relabeled vendor clip can always be traced back to its source.

Two decisions follow from this ? and the first one affects whether PR #236 should change before you merge it:

101. user (2026-06-22T09:15:52.374Z)

hello. can you sync the repo to remote head and identify what were the next steps of this session?

102. assistant (2026-06-22T09:16:20.647Z)

Let me sync to the remote head and check the state of things (including whether PR #236 has moved).

103. user (2026-06-22T09:16:26.231Z)

```
=== current branch / status ===
serving/exposure-safety-preflight
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
?? docs/PACKAGING_PLAN.md
=== local master vs origin/master ===
a3b209a
689c738
=== origin/master recent ===
689c738 docs(pkg): mark P2c ? + the loop closed ? v1 LIGHT is feature-complete (#313)
293e5c0 feat(pkg): re-bundle backend/ui ? serve the P2c first-run wizard from indexer --app
(#312)
f8b2c3b feat(pkg): P2b-install ? batteries-included install + truly-local 'motion' first-run
default (gate O8) (#310)
0ed4cc6 fix(segmenter): fail the job on an unrunnable config instead of a silent 0-rally success
(Finding B) (#311)
a42c132 docs(pkg): track the batteries-included app plan (PACKAGING_PLAN.md) + index pointers
(#309)
=== my branch vs origin/master ===
6fdd1d4 feat(serving): exposure-safety boot posture ? fail fast on a half-configured exposed
engine (default-OFF)
```

104. assistant (2026-06-22T09:17:24.051Z)

Important: the shared checkout has moved a lot since my last turn ? it's now on `serving/exposure-safety-preflight`, and `origin/master` has jumped from `e67add1` to `689c738` (packaging PRs #309?#313 merged). Let me gather facts before touching anything, per the shared-checkout safety convention.

105. user (2026-06-22T09:17:29.585Z)

```
=== is local master ff-able to origin/master? ===
YES (local master is ancestor of origin/master ? ff-safe)

=== serving/exposure-safety-preflight pushed? (HEAD 6fdd1d4) ===

=== my #236 branch still here? ===
  fix/served-gate-a-corpus-regrounding

=== PR #236 state ===
#236 MERGED merged=2026-06-19T07:23:40Z
test(eval): re-ground served Gate-A litmus on the present (multicourt) golden set
https://github.com/Khelsutra/badminton-highlight-indexer/pull/236
```

106. assistant (2026-06-22T09:18:12.078Z)

Facts gathered ? here's the key context before I sync:

- **Local `master` is ff-safe** to `origin/master` (it's a clean ancestor).
- **PR #236 is MERGED** (2026-06-19) ? my served Gate-A re-grounding is in master. (Note: the repo moved to the `Khelsutra` org.)
- ?? The current branch `serving/exposure-safety-preflight` has an **unpushed local commit `6fdd1d4`** that is **not mine** (it predates my work ? the exposure-safety WIP that was HEAD at session start). I won't touch it.

Given the shared checkout, I'll fast-forward local `master` to the remote head **without** switching the working tree off that unpushed sibling branch:

107. user (2026-06-22T09:18:15.005Z)

```
fatal: refusing to fetch into branch 'refs/heads/master' checked out at
'C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing'
=== local master now ===
a3b209a
=== working tree still on (untouched) ===
serving/exposure-safety-preflight
6fdd1d4 feat(serving): exposure-safety boot posture ? fail fast on a half-configured exposed
engine (default-OFF)
```

108. assistant (2026-06-22T09:19:17.544Z)

I hit a real shared-checkout constraint, so I deliberately did **not** force anything:

Sync status:
- ? Fetched ? the remote head is now local: `origin/master` = **`689c738`** (packaging PRs #309?#313 landed since my last turn).
- ?? I **could not** fast-forward the local `master` branch: it's checked out in a **sibling worktree** (`.claude/worktrees/tier1-windowing`, sitting at the older `a3b209a`), so git refuses to move it ? and that's not mine to disturb.
- ?? This main checkout is parked on `serving/exposure-safety-preflight`, which carries an **unpushed, non-mine commit `6fdd1d4`**. I left it untouched.

So the remote head is synced into your local refs, but the *working tree* isn't on it ? moving it safely is your call (below), per the shared-checkout convention (report, don't force).

Next steps of this session (where we left off)

This session did two things:

1. **Fixed the 3 stale served Gate-A litmus assertions ? [PR #236](https://github.com/Khelsutra/badminton-highlight-indexer/pull/236), now MERGED** (re-grounded on the present 9-clip multicourt corpus; doc fix folded in).

2. **Indexed `F:\GoPro Hero Black 12\KhelSutra`** and found that *nothing was actually lost*. That surfaced three open threads ? the actual "next steps":

   - **(A) Decide on a full-corpus re-measure of the litmus.** The original-6 labels are **recoverable from F:** (`Training\Golden Labelled\`, video + `.rallies.csv` co-located). Since #236 merged pinned to *present-9*, restoring the original-6 and re-measuring the litmus on the **full 15-clip corpus** would be a clean follow-up PR ? if you want it.
   - **(B) Back up the 7 single-copy golden labels ? most urgent.** `mahadevpura_1`, `mahadevpura_2`, `GX010128`, `GX030094`, `Badminton_BXH_2`, `Boxhill_Doubles`, `GX020094` exist **only** in C: `output/` (gitignored, uncommitted, no F: copy). One disk wipe or cache-cleanup and they're gone for good. Copy them to F: alongside their originals.
   - **(C) Organizational fixes to prevent recurrence:** co-locate each label with its original on F:; key clips by `video_id`/MD5 instead of recycled GoPro names (`GX010104` exists 3×); version-control the tiny golden label CSVs out of gitignored `output/`; maintain one authoritative index that points at the F: archive; record Roora vendor-prep provenance.

How would you like me to position this checkout?

109. user (2026-06-22T09:20:03.539Z)

Your questions have been answered: "This checkout is on serving/exposure-safety-preflight (unpushed sibling commit 6fdd1d4); local master is held by the tier1-windowing worktree. How should I position the working tree relative to the freshly-fetched origin/master (689c738)?"="Cut a fresh branch off origin/master", "Want me to act on the open threads from the F: indexing audit?"="Back up the 7 single-copy labels to F:". You can now continue with these answers in mind.

110. assistant (2026-06-22T09:20:51.607Z)

Both chosen. Before I cut the fresh branch, I need to check a snag: this checkout has untracked `docs/PACKAGING_PLAN.md`, `docs/KHELSUTRA_PRODUCT_OVERVIEW.md`, and `artifacts/`, but `origin/master` now *tracks* `PACKAGING_PLAN.md` (#309). Switching could be blocked or could clobber WIP ? let me compare before doing anything.

111. user (2026-06-22T09:20:55.909Z)

```
=== which of these paths does origin/master track? ===
TRACKED in origin/master: docs/PACKAGING_PLAN.md
not in origin/master: docs/KHELSUTRA_PRODUCT_OVERVIEW.md
(artifacts/ tracked entries above, if any)

=== diff untracked working copy vs origin/master version ===
DIFFERS from origin/master: docs/PACKAGING_PLAN.md
```

112. assistant (2026-06-22T09:22:16.095Z)

The blocker is precise: only `docs/PACKAGING_PLAN.md` conflicts ? it's untracked here, **differs from master's tracked version**, so it's WIP I must preserve (the other two paths aren't in master, no conflict). The user accepted switching this checkout, so I'll preserve that WIP in gitignored `scratch/` first, then switch. Let me re-verify state right before acting.

113. user (2026-06-22T09:22:19.913Z)

```
=== re-verify: branch + clean tree (no tracked mods) ===
serving/exposure-safety-preflight
(empty above = no tracked modifications ? safe to switch)
=== origin/master head ===
689c738
=== untracked now ===
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
?? docs/PACKAGING_PLAN.md
```

114. assistant (2026-06-22T09:22:30.216Z)

Tree is clean. Preserving the WIP `PACKAGING_PLAN.md` into gitignored `scratch/`, then cutting the fresh branch off `origin/master`:

115. user (2026-06-22T09:22:32.872Z)

```
WIP preserved -> scratch/PACKAGING_PLAN.serving-wip.md
branch 'work/golden-corpus-hygiene' set up to track 'origin/master'.
Switched to a new branch 'work/golden-corpus-hygiene'
=== now on ===
work/golden-corpus-hygiene
689c738 docs(pkg): mark P2c ? + the loop closed ? v1 LIGHT is feature-complete (#313)
=== tree state ===
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
```

116. assistant (2026-06-22T09:24:50.106Z)

On the fresh branch `work/golden-corpus-hygiene` at remote head (689c738); the serving WIP commit `6fdd1d4` is safe on its branch, and your WIP `PACKAGING_PLAN.md` is preserved at `scratch/PACKAGING_PLAN.serving-wip.md`.

Now the label backup. One judgment call: 3 of the 7 originals (`GX020094`, `GX030094`, `Badminton BXH 2`) exist in **two** F: folders each, so I can't be certain which physical copy a label was derived from ? and it doesn't matter, since the timeline is identical across copies. Rather than risk pairing a label with the wrong copy, I'll consolidate all 7 into the **existing golden-labels archive** `Training\Golden Labelled\` (which already holds the original-6 labels and videos ? the de-facto golden archive). Let me verify sources/target first, without overwriting anything.

117. user (2026-06-22T09:24:56.446Z)

```
=== target exists? ===
False

=== source files (C:) ===
mahadevpura_1.rallies.csv          465 bytes  src OK
mahadevpura_2.rallies.csv         1719 bytes  src OK
GX010128.rallies.csv              2286 bytes  src OK
GX030094.rallies.csv              1800 bytes  src OK
Badminton_BXH_2.rallies.csv       1970 bytes  src OK
Boxhill_Doubles.rallies.csv       1915 bytes  src OK
GX020094.rallies.csv              1715 bytes  src OK

=== already present at target? (will NOT overwrite) ===
Join-Path:
Line |
   8 |  foreach ($n in $names) { $p = Join-Path $dst $n; if
(Test-Path -Liter ?
     |
~~~~~~~~~~~~~~~~~
     | Cannot find drive.
A drive with the name 'F' does not exist.
Test-Path:
Line |
   8 |  ? ames) { $p = Join-Path $dst $n; if (Test-Path
-LiteralPath $p) { "ALR ?
     |
~~~~~~~~~~~~~~~~~~~~~~~~~
     | Value cannot be
null. (Parameter 'The provided Path argument was null or an empty collection.')
absent (will copy): mahadevpura_1.rallies.csv
Join-Path:
Line |
   8 |  foreach ($n in $names) { $p = Join-Path $dst $n; if
```

```
(Test-Path -Liter ?
     |
~~~~~~~~~~~~~~~~~
     | Cannot find drive.
A drive with the name 'F' does not exist.
Test-Path:
Line |
   8 |  ? ames) { $p = Join-Path $dst $n; if (Test-Path
-LiteralPath $p) { "ALR ?
     |
~~~~~~~~~~~~~~~~~~~~~~~~~
     | Value cannot be
null. (Parameter 'The provided Path argument was null or an empty collection.')
absent (will copy): mahadevpura_2.rallies.csv
Join-Path:
Line |
   8 |  foreach ($n in $names) { $p = Join-Path $dst $n; if
(Test-Path -Liter ?
     |
~~~~~~~~~~~~~~~~~
     | Cannot find drive.
A drive with the name 'F' does not exist.
Test-Path:
Line |
   8 |  ? ames) { $p = Join-Path $dst $n; if (Test-Path
-LiteralPath $p) { "ALR ?
     |
~~~~~~~~~~~~~~~~~~~~~~~~~
     | Value cannot be
null. (Parameter 'The provided Path argument was null or an empty collection.')
absent (will copy): GX010128.rallies.csv
Join-Path:
Line |
   8 |  foreach ($n in $names) { $p = Join-Path $dst $n; if
(Test-Path -Liter ?
     |
~~~~~~~~~~~~~~~~~
     | Cannot find drive.
A drive with the name 'F' does not exist.
Test-Path:
Line |
   8 |  ? ames) { $p = Join-Path $dst $n; if (Test-Path
-LiteralPath $p) { "ALR ?
     |
~~~~~~~~~~~~~~~~~~~~~~~~~
     | Value cannot be
null. (Parameter 'The provided Path argument was null or an empty collection.')
absent (will copy): GX030094.rallies.csv
Join-Path:
Line |
   8 |  foreach ($n in $names) { $p = Join-Path $dst $n; if
(Test-Path -Liter ?
     |
~~~~~~~~~~~~~~~~~
     | Cannot find drive.
A drive with the name 'F' does not exist.
Test-Path:
Line |
   8 |  ? ames) { $p = Join-Path $dst $n; if (Test-Path
-LiteralPath $p) { "ALR ?
     |
~~~~~~~~~~~~~~~~~~~~~~~~~
     | Value cannot be
null. (Parameter 'The provided Path argument was null or an empty collection.')
absent (will copy): Badminton_BXH_2.rallies.csv
```

```
Join-Path:
Line |
   8 |  foreach ($n in $names) { $p = Join-Path $dst $n; if
(Test-Path -Liter ?
     |
~~~~~~~~~~~~~~~~~
     | Cannot find drive.
A drive with the name 'F' does not exist.
Test-Path:
Line |
   8 |  ? ames) { $p = Join-Path $dst $n; if (Test-Path
-LiteralPath $p) { "ALR ?
     |
~~~~~~~~~~~~~~~~~~~~~~~~~
     | Value cannot be
null. (Parameter 'The provided Path argument was null or an empty collection.')
absent (will copy): Boxhill_Doubles.rallies.csv
Join-Path:
Line |
   8 |  foreach ($n in $names) { $p = Join-Path $dst $n; if
(Test-Path -Liter ?
     |
~~~~~~~~~~~~~~~~~
     | Cannot find drive.
A drive with the name 'F' does not exist.
Test-Path:
Line |
   8 |  ? ames) { $p = Join-Path $dst $n; if (Test-Path
-LiteralPath $p) { "ALR ?
     |
~~~~~~~~~~~~~~~~~~~~~~~~~
     | Value cannot be
null. (Parameter 'The provided Path argument was null or an empty collection.')
absent (will copy): GX020094.rallies.csv
```

118. assistant (2026-06-22T09:25:16.029Z)

F: is **not currently mounted** ("A drive with the name 'F' does not exist") ? the external GoPro drive has been disconnected since the earlier indexing turn. The C: source files are all confirmed present (sizes intact), but the backup target is gone. The drive may have remounted under a different letter (your notes mention some footage on `G:`). Let me check what's actually connected.

119. user (2026-06-22T09:25:20.818Z)

```
=== filesystem drives ===

Name UsedGB FreeGB DisplayRoot
---- ------ ------ -----------
C    622.30 123.40
E      0.00   0.00
Temp 622.30 123.40


=== does KhelSutra exist on any drive letter? ===
```

120. assistant (2026-06-22T09:26:07.342Z)

The backup is blocked by hardware, not by anything in the repo: **the external GoPro archive drive is disconnected** ? it's not on F: or any other letter (only C: and an empty E: card slot are mounted). So I can't write the labels to the archive right now.

Here's where everything stands:

? Done this turn
- **Synced to remote head:** fetched `origin/master` = **`689c738`** (packaging PRs #309?#313).
- **Cut a fresh branch off remote head:** now on **`work/golden-corpus-hygiene`** at `689c738`.
- **Preserved your WIP:** the untracked `docs/PACKAGING_PLAN.md` (which *differed* from the merged #309 version) is safe at **`scratch/PACKAGING_PLAN.serving-wip.md`** ? I moved it aside so the branch switch wouldn't clobber it. The serving branch's unpushed commit `6fdd1d4` is untouched on its own ref.

? Blocked: the F: label backup
- The 7 source labels are all **present and intact on C:** (`output/`, 465?2286 bytes each) ? verified.
- The **target archive is offline** ? `GoPro Hero Black 12\KhelSutra` isn't on any mounted drive. I have the exact copy plan staged (consolidate into `Training\Golden Labelled\`, no overwrites) and will run it the instant the drive is back.

This is, ironically, the exact failure mode the audit warned about: the only copy of these 7 hand-labeled CSVs lives in a gitignored folder on one disk, and their backup target is an intermittently-connected external drive.

To unblock, pick one:
1. **Reconnect the GoPro/external drive** (tell me the letter if it's not F:) ? I run the backup immediately, as planned.
2. **Let me commit the 7 labels to git now** as a stronger interim backup ? it survives both the drive being away *and* a C: disk failure, and it's exactly org-fix (C) (version-control the tiny label CSVs out of gitignored `output/`). This is a repo/data-placement change, so I'm asking before doing it. I can still also copy to the drive once it's reconnected.

? Remaining open threads (unchanged)
- **(A)** Optional follow-up PR: restore the original-6 labels from the archive and re-measure the served Gate-A litmus on the full 15-clip corpus (today it's pinned to the present 9 via the merged #236).
- **(C)** The organizational fixes (co-locate labels with originals, key by `video_id` not recycled GoPro names, version-control the label CSVs, one authoritative index pointing at the archive, vendor-prep provenance) ? capturable as a tracked doc when you want.

How would you like to handle the backup?

121. user (2026-06-22T09:28:27.157Z)

f: is now mounted and my goal is to be able to run all the training and inference from the cloud completely with all the data in GCS buckets. Would be good to document this so that other sessions also know where to pick up the data from so that I can remove my local machine as a potential blocker

122. assistant (2026-06-22T09:29:25.587Z)

F: is back ? let me run the staged label backup now, and in parallel investigate the cloud/GCS setup so I can design the "GCS as source of truth" documentation properly.

123. user (2026-06-22T09:29:32.277Z)

TARGET MISSING (F: not mounted?): F:\GoPro Hero Black 12\KhelSutra\Training\Golden Labelled

124. assistant (2026-06-22T09:29:38.284Z)

Still not found. Let me check what's actually mounted now and where the KhelSutra tree is.

125. user (2026-06-22T09:29:43.135Z)

=== drives ===

Name FreeGB DisplayRoot
---- ------ -----------

```
C    122.90
E      0.00
F    748.80
Temp 122.90


=== F:\ root (if any) ===

Name
----
[Khelsutra] GoPro Hero Black 12
Data
KhelSutra_work
Install Security for Mac.zip
Install Security for Windows.exe


=== locate KhelSutra on any drive ===
```

126. assistant (2026-06-22T09:29:52.805Z)

```
F: is mounted (748 GB) but **the tree was reorganized again** ? the top folder is now
`[Khelsutra] GoPro Hero Black 12` (renamed), plus new `Data` and `KhelSutra_work` folders. This
is the same path-fragility problem recurring. Let me locate the golden archive in the new
layout.
```

127. user (2026-06-22T09:29:59.090Z)

```
=== F:\[Khelsutra] GoPro Hero Black 12 ===

D KhelSutra


=== F:\Data ===

D Backup
D Emails
D Initial GoPRo with Sports Recordings
D Motorcyling
D Wife Problem
f Search.pdf


=== F:\KhelSutra_work ===

D 19june_proxies
D kushagra_june12_proxies


=== search F: for 'Golden Labelled' folder (any depth) ===
F:\[Khelsutra] GoPro Hero Black 12\KhelSutra\Training\Golden Labelled
=== search F: for any *.rallies.csv (any depth) ===
F:\KhelSutra_work\kushagra_june12_proxies\mahadevpura-1.rallies.csv
F:\[Khelsutra] GoPro Hero Black 12\KhelSutra\Training\Golden Labelled\Adarsh and Avi.rallies.csv
F:\[Khelsutra] GoPro Hero Black 12\KhelSutra\Training\Golden
Labelled\gBaaddy20May2026.rallies.csv
F:\[Khelsutra] GoPro Hero Black 12\KhelSutra\Training\Golden
Labelled\KushagraYashAviFirstVideo.rallies.csv
F:\[Khelsutra] GoPro Hero Black 12\KhelSutra\Training\Golden Labelled\LargeTestVideo.rallies.csv
F:\[Khelsutra] GoPro Hero Black 12\KhelSutra\Training\Golden
Labelled\MahadevpuraSingles.rallies.csv
F:\[Khelsutra] GoPro Hero Black 12\KhelSutra\Training\Golden Labelled\TestLargeVideo.rallies.csv
F:\[Khelsutra] GoPro Hero Black 12\KhelSutra\Badminton - Boxhill - 17 June\GX010137.rallies.csv
F:\[Khelsutra] GoPro Hero Black 12\KhelSutra\Badminton - Boxhill - 17 June\GX010141.rallies.csv
```

128. assistant (2026-06-22T09:30:29.337Z)

Found it ? the archive moved to `F:\[Khelsutra] GoPro Hero Black 12\KhelSutra\` (top folder was renamed). That's *another* instance of the exact path-fragility your GCS goal would eliminate. Let me run the backup to the correct path, and in parallel inventory the cloud setup and existing GCS references in the repo.

129. user (2026-06-22T09:30:35.565Z)

```
target OK: F:\[Khelsutra] GoPro Hero Black 12\KhelSutra\Training\Golden Labelled

COPIED + verified  : mahadevpura_1.rallies.csv
COPIED + verified  : mahadevpura_2.rallies.csv
COPIED + verified  : GX010128.rallies.csv
COPIED + verified  : GX030094.rallies.csv
COPIED + verified  : Badminton_BXH_2.rallies.csv
COPIED + verified  : Boxhill_Doubles.rallies.csv
COPIED + verified  : GX020094.rallies.csv
```

130. user (2026-06-22T09:30:46.764Z)

```
=== gcloud present? ===
Google Cloud SDK 571.0.0
=== active account / project ===
avi.dullu@gmail.com
gen-lang-client-0306026826
=== buckets (gcloud storage ls) ===
gs://khelsutra-rally-corpus/
```

131. user (2026-06-22T09:30:47.470Z)

```
Found 36 files
tests\test_api_endpoints.py
pyproject.toml
docs\PACKAGING_PLAN.md
backend\ui\assets\index-GSXqL6iY.js
backend\main.py
docs\COMPUTE_DECOUPLED_SERVING\README.md
docs\CLOUD_GPU_DRYRUN_GUIDE.md
backend\config\models.py
deploy\gcp\README.md
deploy\gcp\provision.sh
tests\test_compute_backend.py
deploy\cloudrun\README.md
backend\compute\base.py
backend\compute\cloud_run.py
deploy\cloudrun\Dockerfile
docs\PER_VIDEO_WORKSPACE.md
backend\utils\paths.py
tests\test_input_source.py
backend\storage\__init__.py
backend\storage\ref.py
docs\archives\decisions\DECISIONS.md
docs\VIDEO_LOCALITY_MODEL.md
docs\SETUP_NEW_MACHINE.md
docs\HOSTING_PLAN.md
docs\COMPUTE_DECOUPLED_SERVING\architecture.svg
tests\test_training_gen0.py
docs\archives\past_projects\DECOUPLED_COMPUTE\HANDOFF.md
docs\archives\past_projects\DECOUPLED_COMPUTE\README.md
docs\archives\past_projects\DECOUPLED_COMPUTE\DEPLOY_REPORT_GCP_2026-06-20.md
docs\archives\past_projects\DECOUPLED_COMPUTE\04-BUILD-AND-TEST-PLAN.md
tests\test_workspace.py
deploy\gcp\teardown.sh
```

```
tests\test_storage.py
backend\storage\base.py
backend\storage\gcs.py
tests\test_provider_gemini.py
```

132. assistant (2026-06-22T09:31:23.369Z)

Backup complete ? all **7 labels copied and hash-verified** into the F: golden archive. And the cloud picture is now clear:

- **gcloud authed** (avi.dullu@gmail.com, project `gen-lang-client-0306026826`); **one bucket: `gs://khelsutra-rally-corpus/`**.
- The repo **already has cloud scaffolding**: `backend/storage/{gcs,ref,base}.py` (a StorageBackend abstraction), `backend/compute/{cloud_run,base}.py`, `deploy/gcp/`, `deploy/cloudrun/`, and docs like `HOSTING_PLAN.md`, `PER_VIDEO_WORKSPACE.md`, `VIDEO_LOCALITY_MODEL.md`, `SETUP_NEW_MACHINE.md`.

Before writing anything, I need to understand what already exists (so I extend conventions rather than reinvent). Let me inspect the bucket and the storage abstraction.

133. user (2026-06-22T09:31:31.245Z)

```
=== top-level prefixes in bucket ===
gs://khelsutra-rally-corpus/highlights/
gs://khelsutra-rally-corpus/labels/
gs://khelsutra-rally-corpus/models/
gs://khelsutra-rally-corpus/outputs/
gs://khelsutra-rally-corpus/promo/
gs://khelsutra-rally-corpus/videos/
gs://khelsutra-rally-corpus/weights/

=== recursive object count + total size ===
2214223255   gs://khelsutra-rally-corpus
```

134. user (2026-06-22T09:31:31.275Z)

```
1       # Hosting Plan ? khelsutra.guru
2
3       **Status:** PROPOSAL ? awaiting owner ratification.
4       **Date:** 2026-06-10 · **Constraints:** budget $10?25/mo · GCP credits available ·
DigitalOcean
5       acceptable · alpha backend = owner's GPU box behind a tunnel · 10?25 alpha testers ·
registrar = GoDaddy.
6
7       > **? Update (2026-06-20, ADR-013): DigitalOcean is DROPPED for COMPUTE** ? GCP is the
sole compute
8       > stack (training + serving). The **"All-DigitalOcean"** compute option below is
retired; DO Spaces
9       > remains only a *possible* S3-compatible storage target. Edge/site hosting choices here
are unaffected.
10      **Companion:** `docs/UI_PROPOSAL.md`.
11
12      ## 1. Recommendation in one line
13
14      **Alpha: free edge + your box** (Cloudflare DNS/Pages/Access/Tunnel ? FastAPI on the GPU
box) at
15      **$0/mo**, saving GCP credits; **Beta: GCP control plane + your box as pull-worker**
(Cloud Run +
16      GCS + Firebase Auth, credits-covered), keeping the edge where it's free.
17
18      Rationale for the Cloudflare pieces despite the GCP preference: GCP has no free
equivalent of the
19      Tunnel + Access combo (IAP requires a load balancer, ~$18/mo standing cost), and the
alpha needs
```

exactly three things GCP charges for but Cloudflare gives free: a secure tunnel to a home box with
a custom domain, email-allowlist auth (free ?50 users ? fits 10?25 testers), and static hosting.
Credits are better spent at beta on compute/storage, which is where GCP is excellent.

2. Alpha topology ($0/mo)

```
tester browser
   ?  https
   ?
Cloudflare edge (free): DNS + TLS + Access (email-OTP allowlist)
   ??? app.khelsutra.guru  ? Cloudflare Pages (React SPA; auto-deploy from GitHub)
   ??? api.khelsutra.guru  ? Cloudflare Tunnel (outbound-only `cloudflared` service on the box)
                                  ?
                                  ?
                       FastAPI :8000 on the GPU box (Windows)
                       ? Gemini segmenter path (no GPU in request path)
                       ? async jobs (#16), SQLite WAL, output/ on disk
```

- **No inbound ports opened** on the home network; `cloudflared` dials out. Home IP never exposed.
- **Auth:** Cloudflare Access in front of BOTH hostnames; backend trusts the
  `Cf-Access-Authenticated-User-Email` header (verify the Access JWT for rigor) for per-user scoping.
- **Upload limit:** free-tier proxy caps request bodies at **100 MB** ? SPA uploads in ?90 MB
  chunks (UI_PROPOSAL B2). Match videos (1?4 GB) flow fine as chunks.
- **Box offline = service offline.** Acceptable for alpha; SPA shows a friendly "processing engine
  offline" banner (a `/api/health` ping). Beta removes this with the queue.
- **Costs:** Pages, Tunnel, Access, DNS = $0. Only the Gemini per-run cost (~$0.05/match) and
  domain renewal.

3. Beta topology (GCP credits; ~$15?20/mo post-credits)

```
browser ??? Cloudflare edge (free, unchanged) ??? app.khelsutra.guru (Pages)
                                              ??? api.khelsutra.guru ??? Cloud Run (FastAPI control plane, scale-to-zero)
                                                                       ? jobs table (Cloud SQL Postgres)
                                                                       ? signed resumable URLs
                                                                       ?
                                                                  GCS buckets (uploads / reels, lifecycle ? cold)
                                                                       ? pull jobs, outbound-only
                                                                  GPU box = first `byo_worker` (PLATFORM P3)
```

- Uploads go **browser ? GCS directly** (resumable signed URLs) ? no proxy body limit, no home
  bandwidth for ingest; the box pulls only what it processes (WASB/owned-model jobs), Cloud Run
  handles Gemini-path jobs entirely in-cloud.
- Auth graduates to **Firebase Auth** (free; Google + email sign-in; India-friendly phone OTP
  later); Access allowlist retires.

- Post-credits estimate: Cloud Run ~$0?5 (free tier: 2 M req + 180 K vCPU-s/mo, scale-to-zero) · GCS 100 GB ? $2.60 + egress $0.12/GB (reel downloads ? watch this line) · Cloud SQL smallest ? $10 ? **~$15?20/mo**, inside budget. Egress lever if it grows: serve reels from Cloudflare R2 (zero egress) while keeping everything else GCP.

4. Alternatives considered

Option	Monthly (alpha ? beta)	Pros	Cons
Recommended hybrid (CF edge + box ? +GCP)	**$0 ? ~$0 w/ credits** (~$15?20 after)	Cheapest at every phase; credits used where they matter; maps to PLATFORM P2/P3	Two vendors (CF + GCP)
All-DigitalOcean	~$11 ? ~$17 ($6 droplet + Spaces $5: 250 GiB + **1 TiB egress included**)	One simple vendor; Spaces' included egress is the best video-download deal; per-second billing	No credits; no scale-to-zero; still needs a tunnel-equivalent to reach the box (run `frp`/WireGuard on the droplet)
All-GCP from day 1	~$18+ ? credits	Single vendor, credits immediately	LB+IAP standing cost just to gate an alpha; no free tunnel to the home box; credits burn on plumbing
All-Cloudflare (Pages+Workers+R2)	~$0 ? ~$5	Zero egress on R2; cheapest forever	FastAPI doesn't run on Workers (API stays on box/Cloud Run anyway); ignores GCP credits

5. Domain setup ? GoDaddy ? Cloudflare DNS (one-time, ~15 min, you)

1. Create a free Cloudflare account ? **Add a domain** ? `khelsutra.guru` ? Free plan. Cloudflare imports existing records and shows **two nameservers** (e.g. `xxx.ns.cloudflare.com`).
2. GoDaddy ? **My Products ? khelsutra.guru ? Manage DNS ? Nameservers ? Change ? "I'll use my own nameservers"** ? paste the two Cloudflare nameservers ? Save. Propagation: minutes to 24 h. (Registration/renewal stays at GoDaddy; only DNS moves. Optional later: transfer the registration to Cloudflare Registrar for at-cost .guru renewals.)
3. Records (I manage after takeover): `app` ? Pages project; `api` ? `<tunnel-id>.cfargotunnel.com` (created automatically by `cloudflared`); apex redirect ? `app`.

6. What I need to take over management

Access	Scope	Used for
Cloudflare API token (you create: dash ? My Profile ? API Tokens)	Zone:DNS:Edit, Account:Pages:Edit, Account:Access:Apps:Edit, Account:Tunnel:Edit ? scoped to this zone	DNS records, Pages deploys (wrangler), Access policies, tunnel config
GitHub repo `khelsutra` (private)	Pages connects once (dashboard, you; build root = `web/`) ? every push auto-deploys	I write the SPA in your Projects folder; commits pushed = deployed
On the box (one-time, I hand you exact commands)	`cloudflared` installed as a Windows service + `config.yml`	The tunnel itself
GCP (beta only)	A `khelsutra` project + service account (Cloud Run admin, Storage admin, Cloud SQL admin)	Control-plane deploys

Day-to-day flow after setup: I build ? commit ? push ? Pages deploys the SPA automatically; API/tunnel/DNS changes via the scoped token. Secrets (Gemini key, Access policy) never live in the repo; `.env` on the box + Cloudflare/GCP dashboards.

7. Go-live security checklist (alpha)

Access allowlist ON for both hostnames ? CORS narrowed to `https://app.khelsutra.guru` ?

```
rate
limit + per-user upload quota ? `security.allowed_video_root` set ? upload size cap + extension
allowlist ? **rotate the Gemini API key** (NEXT_STEPS housekeeping ? it was shared in chat) ?
confirm no router port-forwards ? weekly backup: SQLite DB + `output/` ? GCS bucket (rclone
scheduled task; credits).

## 8. Decisions needed from owner

1. Ratify the **hybrid recommendation** (CF free edge now, GCP credits at beta)? Or force
   single-vendor (DO / GCP) despite the cost notes above?
2. OK to move **DNS to Cloudflare** (registration stays at GoDaddy)?
3. Beta reel storage: **GCS (credits)** vs **R2 (zero egress)** vs **DO Spaces (1 TiB included)** ?
   can defer until beta Drop 2.
4. Approve the access grants in §6 so I can run the setup end-to-end.

*Ratified choices get recorded in `docs/archives/decisions/DECISIONS.md`.*

```

135. user (2026-06-22T09:31:31.294Z)

```
"""Storage value objects + URI parsing for the pluggable storage layer.

A ``StorageRef`` is a backend-agnostic pointer to one object. URIs:

    <bare path>                 -> local   (today's behaviour; e.g. C:/v/x.mp4, /data/x.mp4)
    local://<path>              -> local   (local:///abs or local://rel/p)
    s3://<bucket>/<key>         -> s3       (AWS + R2 + DO Spaces + MinIO via endpoint_url)
    gcs://<bucket>/<key>        -> gcs      (gs:// accepted, normalized to gcs://)
    gdrive://<path>             -> gdrive   (path under your locally-MOUNTED Google Drive "My Drive")

No cloud SDKs are imported here ? this module is pure stdlib so ``import backend.storage``
stays cheap.
"""
from __future__ import annotations

import re
from dataclasses import dataclass
from datetime import datetime
from typing import Optional

_SCHEME_RE = re.compile(r"^([A-Za-z][A-Za-z0-9+.\-]*)://")
_KNOWN = ("local", "s3", "gcs", "gdrive")


@dataclass(frozen=True)
class StorageStat:
    """Lightweight object metadata (uniform across backends)."""
    size: int
    modified: Optional[datetime] = None
    etag: Optional[str] = None
    content_type: Optional[str] = None


@dataclass(frozen=True)
class StorageRef:
    """A backend + a location. ``bucket`` is the S3/GCS bucket; for ``local`` and
```

```
``gdrive`` it is
    empty and ``key`` carries the path (a filesystem path, or a path under the mounted Drive)."""
    backend: str
    bucket: str
    key: str
    uri: str

    @property
    def name(self) -> str:
        return self.key.rstrip("/").rsplit("/", 1)[-1]


def parse_uri(uri: str) -> StorageRef:
    """Parse a storage URI (or a bare filesystem path) into a ``StorageRef``.

    A string with no ``scheme://`` prefix (including Windows ``C:\\...`` paths, which
    have no ``//``) is treated as a local filesystem path ? preserving today's behaviour.
    """
    s = str(uri)
    m = _SCHEME_RE.match(s)
    if not m:
        return StorageRef("local", "", s, f"local://{s}")
    scheme = m.group(1).lower()
    rest = s[m.end():]
    if scheme == "gs":
        scheme = "gcs"
    if scheme in ("local", "gdrive"):
        # Path-keyed, no bucket: local FS, or a path under the mounted Google Drive "My Drive".
        return StorageRef(scheme, "", rest, s)
    if scheme in _KNOWN:
        bucket, _, key = rest.partition("/")
        return StorageRef(scheme, bucket, key, s)
    raise ValueError(f"unsupported storage URI scheme: {scheme}:// (in {s!r})")


def resolve_input_uri(raw: str, *, input_backend: str = "local", input_root: str = "") -> str:
    """Resolve a user-supplied input reference into a storage URI (COMPUTE_DECOUPLED_SERVING M2).

    Precedence (first match wins):
      1. An explicit ``scheme://`` on ``raw`` (``gs://``, ``s3://``, ``local://``, ``gdrive://``) ? as-is.
      2. An ABSOLUTE local path (``/data/x.mp4``, ``C:\\?``, ``C:/?``) ? as-is (always local).
      3. A bare/relative name + a non-empty ``input_root`` ? joined under it
         (``"gs://bucket/videos"`` + ``"x.mp4"`` ? ``"gs://bucket/videos/x.mp4"``).
      4. A bare/relative name + ``input_backend != "local"`` ? prefixed (``"gcs"`` + ``"bucket/x"`` ?
         ``"gcs://bucket/x"``).
      5. Otherwise ? ``raw`` unchanged (a relative/bare LOCAL path ? today's behaviour).

    So the default (``input_backend="local"``, ``input_root=""``) is an identity passthrough for every
    path used today, preserving byte-for-byte behaviour.
    """
    s = str(raw)
    if _SCHEME_RE.match(s):           # 1. explicit scheme wins
        return s
    # 2. an anchored LOCAL path is always local, never joined. Detect this OS-INDEPENDENTLY ?
    #    os.path.isabs is platform- AND Python-version-dependent (on Windows + py3.13 a
```

```
POSIX-style
        #    "/data/x" is NOT absolute). Anchored == a leading "/" or "\\", or a Windows drive prefix "X:".
        if s.startswith(("/", "\\")) or (len(s) >= 2 and s[1] == ":" and s[0].isalpha()):
            return s
        if input_root:                    # 3. join a relative name under the configured root
            return input_root.rstrip("/") + "/" + s.lstrip("/")
        if input_backend and input_backend != "local":   # 4. prefix a bucket/key with the declared backend
            return f"{input_backend}://{s}"
        return s                          # 5. bare/relative local path ? unchanged

```

136. assistant (2026-06-22T09:32:00.778Z)

The bucket already has a layout (`labels/ videos/ models/ weights/ outputs/ highlights/ promo/`, ~2.06 GB total) and the engine already understands `gs://` inputs via `backend/storage/ref.py`. So this is largely about **completing the data migration and documenting the canonical map** ? not building from scratch. Let me see what's actually in each prefix, whether a layout doc already exists, and what `SETUP_NEW_MACHINE.md` tells sessions today.

137. user (2026-06-22T09:32:17.517Z)

```
=== labels/ ===
gs://khelsutra-rally-corpus/labels/Badminton BXH 2_proxy.rallies.csv
gs://khelsutra-rally-corpus/labels/Boxhill Doubles_proxy.rallies.csv
gs://khelsutra-rally-corpus/labels/GX010128.rallies.csv
gs://khelsutra-rally-corpus/labels/GX020094_proxy.rallies.csv
gs://khelsutra-rally-corpus/labels/GX030094_proxy.rallies.csv
gs://khelsutra-rally-corpus/labels/mahadevpura-1.rallies.csv
gs://khelsutra-rally-corpus/labels/mahadevpura-2_proxy.rallies.csv

=== videos/ ===
gs://khelsutra-rally-corpus/videos/GX030094_proxy.mp4
gs://khelsutra-rally-corpus/videos/mahadevpura-1.MP4
gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4

=== models/ ===
gs://khelsutra-rally-corpus/models/wasb_badminton_best.pth.tar

=== weights/ ===
gs://khelsutra-rally-corpus/weights/0.1.0-l4/

=== outputs/ ===
gs://khelsutra-rally-corpus/outputs/05e0e577029daafcf425ca70c0c4f0a3/
gs://khelsutra-rally-corpus/outputs/parity/

=== highlights/ ===
gs://khelsutra-rally-corpus/highlights/curated_top10_longest.mp4
gs://khelsutra-rally-corpus/highlights/p3_app.log
gs://khelsutra-rally-corpus/highlights/p3_packaging_highlights.mp4
gs://khelsutra-rally-corpus/highlights/p3_run.log

=== promo/ ===
gs://khelsutra-rally-corpus/promo/full_cuj_real.mp4
gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-screencast.mp4
gs://khelsutra-rally-corpus/promo/khelsutra-i18n-screencast.mp4
gs://khelsutra-rally-corpus/promo/promo_cuj.gif
gs://khelsutra-rally-corpus/promo/promo_cuj.mp4
```

138. user (2026-06-22T09:32:17.674Z)

Found 11 files

docs\PACKAGING_PLAN.md
docs\CLOUD_GPU_DRYRUN_GUIDE.md
deploy\gcp\README.md
deploy\gcp\provision.sh
docs\archives\decisions\DECISIONS.md
docs\SETUP_NEW_MACHINE.md
docs\archives\past_projects\DECOUPLED_COMPUTE\HANDOFF.md
docs\archives\past_projects\DECOUPLED_COMPUTE\README.md
docs\archives\past_projects\DECOUPLED_COMPUTE\DEPLOY_REPORT_GCP_2026-06-20.md
tests\test_workspace.py
deploy\gcp\teardown.sh

139. assistant (2026-06-22T09:33:21.158Z)

The bucket is **partially populated from past cloud dry-runs** ? and the gaps are exactly what blocks "run everything from cloud":

- `labels/` ? 7 new-clip labels (inconsistently named, some with `_proxy` suffix); **missing the original-6 + GX010137/141**
- `videos/` ? only 3 files; **the corpus originals/proxies are almost entirely local-only**
- **`trajectories/` ? doesn't exist**; every WASB trajectory the litmus/eval needs is local C:-only
- the manifest references absolute **C: paths**, so it isn't cloud-portable

No doc currently maps the bucket layout. Let me read the two most relevant existing docs and the consumption config so my documentation is accurate, not aspirational.

140. user (2026-06-22T09:33:21.705Z)

```
# Setting up a new machine ? zero ? training & inference

**The canonical "spin up a box and make the models ready" runbook.** Copy-pasteable; followable by
any human.

> **? COMPUTE = GCP (ADR-013, 2026-06-20). DigitalOcean is DROPPED for compute** ? use
> **[`deploy/gcp/README.md`](../deploy/gcp/README.md)** for the real GPU/training path (GCP is the sole
> stack: DO GPU isn't enabled on the account + isn't cleanly credit-funded; Paperspace is
> subscription-gated). The provider-neutral steps here (bootstrap / scratch / gpu_setup, the suite,
> secret-free inference) still apply on **any** Linux box; the DigitalOcean **provisioning** commands
> below are retained only as a generic example.

> **Last verified:** 2026-06-20 on a DigitalOcean Droplet (Ubuntu 24.04, 2 vCPU / 3.8 GB, **no GPU**).
> Bootstrap succeeded (torch 2.12.1+cpu); full mocked suite **922 passed / 11 skipped**; a **secret-free**
> CPU inference (no GPU, no API key) detected **2 rallies** and persisted them to SQLite.
>
> ? Assumes the DECOUPLED_COMPUTE PRs are merged: the CPU-mock detector seam, `deploy/digitalocean/bootstrap.sh`,
> the storage layer, and the worker dispatcher (#251). The **bucket** steps (§7) are config-driven and
> **verified against real DO Spaces + GCS** ? see the appendix.

## What you'll have at the end

A machine that can: run the test suite (green), run **secret-free CPU inference** (rallies, no GPU/keys),
run **real Gemini inference** (with a key), and run **Gen-0 model training** ? plus pointers to the GPU /
```

native-detector path and bucket-driven train/infer.

---

1. Credentials to create once (least-privilege)

#	Credential	Where	Needed for
1	**SSH keypair** (public key added to the provider)	`ssh-keygen -t ed25519 -f ~/.ssh/rally` ? paste `.pub` into DO ? Settings ? Security	logging into the box
2	**DigitalOcean API token** (read+write)	DO ? API ? Tokens	`doctl` provisioning (optional ? you can click-create)
3	**GitHub repo read access**	a read-only **deploy key** *or* a fine-grained **PAT** (`Contents: Read`)	`git clone` on the box (or use the `scp` path in §3)
4	**`GEMINI_API_KEY`**	Google AI Studio / Vertex	**real** Gemini inference only (§6b). The secret-free path (§6a) needs none. **Rotate before sharing.**
5	**Bucket creds** ? DO Spaces key/secret, or GCS SA JSON, or Drive SA JSON + folder	provider console	bucket-driven train/infer (§7, *pending PR-B*)
6	**GPU box** (Paperspace free M4000 / GPU Droplet)	provider console	real WASB/TrackNet detectors + faster runs (§8)

Secret hygiene: never put a secret in the repo, a PR, or `config.json`. Set them as env vars on the box
(or a git-ignored `.env`). Scope each as narrowly as the UI allows and rotate/revoke when done.

---

2. Provision the machine

> **? COMPUTE = GCP only (2026-06-20, ADR-013 supersedes the ADR-011 split):**
> **BOTH training and serving run on GCP** ? provision via **`deploy/gcp/README.md`** (GPU VM today;
> Cloud Run+GPU scale-to-zero for serving, ADR-012). **DigitalOcean is dropped for compute** (DO GPU
> isn't enabled on the account + DO charges newer customers' cards for GPU, so it isn't cleanly
> credit-funded; Paperspace is subscription-gated). The GCS bucket `gs://khelsutra-rally-corpus` + SA +
> `$100` budget are the storage/cost anchors. The native WASB runner (`detector_impl="native"`) runs the
> same on any Linux GPU box. **The DigitalOcean provisioning steps below are a generic example only.**

- **CPU box** (suite + secret-free stub inference + Gen-0 training plumbing): cheapest **Ubuntu 24.04**
  Droplet, **2 vCPU / 4 GB**, region nearest you, **attach the SSH key** (#1). ~$0.06/hr ? destroy when done.
- **GPU box** (real WASB/TrackNet detectors ? real trajectory extraction + training): a **DO GPU Droplet**.
  ? You **cannot add a GPU to a CPU Droplet** ? it's a separate machine type, in **GPU regions**
  (`nyc2`/`tor1`/`atl1`, **not** `blr1`), with **dual local NVMe**. **RTX 4000 Ada (20 GB, ~$0.76/hr)** is
  plenty for WASB + Gen-0; the `gpu-*-base` image preinstalls the NVIDIA driver. **Co-locate the Space in the
  GPU box's region** for cheap corpus-pull.

CLI provision (after `doctl auth init` with token #2 ? **verify size/image slugs first** via `doctl compute size list`):
```bash
# CPU box:
doctl compute droplet create rally-box --image ubuntu-24-04-x64 --size s-2vcpu-4gb \
  --region blr1 --ssh-keys <your-key-fp> --wait
# GPU box (RTX 4000 Ada):
```

```
doctl compute droplet create rally-gpu --image gpu-h100x1-base --size gpu-4000adax1-20gb \
  --region tor1 --ssh-keys <your-key-fp> --wait
doctl compute droplet list   # note the IP -> ssh -i ~/.ssh/rally root@<IP>
```

---

3. Get the code onto the box

Option A ? git clone (private repo, so authenticate with #3):
```bash
ssh -i ~/.ssh/rally root@<IP>
git clone git@github.com:avidullu/badminton-highlight-indexer.git ~/rally   # via deploy key
#   or: git clone https://<PAT>@github.com/avidullu/badminton-highlight-indexer.git ~/rally
cd ~/rally
```

Option B ? push a local checkout (no GitHub access needed; what the maintainer used):
```bash
# on your machine, from the repo root:
git archive --format=tar.gz -o /tmp/rally.tgz HEAD
scp -i ~/.ssh/rally /tmp/rally.tgz root@<IP>:/root/
ssh -i ~/.ssh/rally root@<IP> 'mkdir -p ~/rally && tar -xzf ~/rally.tgz -C ~/rally'
```

---

4. Bootstrap (one command)

```bash
cd ~/rally
./deploy/digitalocean/bootstrap.sh          # CPU box
#   or
./deploy/digitalocean/bootstrap.sh --gpu     # GPU box (installs CUDA torch)
```

It mirrors CI: installs `ffmpeg` + opencv runtime, creates `.venv`, installs CPU-or-CUDA `torch`, then
`pip install -e .[dev]`, and prints a device check (`cuda_available True/False`).

<details><summary>Manual equivalent (if you can't run the script)</summary>

```bash
sudo apt-get update
sudo apt-get install -y python3 python3-venv python3-pip git ffmpeg libgl1
sudo apt-get install -y libglib2.0-0t64 || sudo apt-get install -y libglib2.0-0
python3 -m venv .venv && . .venv/bin/activate && python -m pip install -U pip
pip install torch torchvision --index-url https://download.pytorch.org/whl/cpu   # or .../whl/cu124 on GPU
pip install -e .[dev]
```
</details>

---

5. Verify the machine (the smoke gate)

```bash
. .venv/bin/activate
python -m pytest -q          # expect all green (922 passed / 11 skipped on the current tree)
```

```
A green suite proves the whole pipeline runs on this box with **no GPU, no WSL, no bucket, no API key**.

---

## 6. Run inference

### 6a. Secret-free CPU inference (no GPU, no API key) ? the "mock a GPU with a CPU" path

Drop a permissive smoke `config.json` in the repo root (the CLI reads `./config.json` from the cwd):
```bash
cd ~/rally && cat > config.json <<'JSON'
{
  "sport": "badminton",
  "indexing": { "default_segmenter": "wasb_hybrid", "detector_impl": "stub", "skip_ai_handoff": true },
  "validation": { "min_fps": 0, "min_blur_threshold": 0, "max_scene_cuts_per_minute": 1000,
                  "relevance": { "court_pixels_min_ratio": 0.0 } }
}
JSON

# a throwaway 12s 1280x720 test clip
ffmpeg -y -f lavfi -i testsrc=duration=12:size=1280x720:rate=30 -pix_fmt yuv420p /root/sample.mp4

# run ? env -u drops any Gemini key to prove none is needed
env -u GEMINI_API_KEY RALLY_DETECTOR_IMPL=stub \
  indexer --video-path /root/sample.mp4 --segmenter wasb_hybrid --output-dir ~/rally/out
```
Expect: `detector_impl=stub ? CPU mock runner`, `Phase 3 SKIPPED (fully-local)`, and
`Indexed 2 play segments` in `~/rally/out/sports_indexer.db`. *(`detector_impl=stub` swaps the GPU+WSL
detector for a deterministic CPU one; `skip_ai_handoff=true` makes candidate windows the rallies, so no
Gemini handoff happens.)*

6b. Real inference (Gemini segmenter)

```bash
export GEMINI_API_KEY="<your-rotated-key>"          # secret stays in the env, never in the repo
# use a real badminton clip (must pass validation: ?1280x720, ?30fps, a green court, etc.)
indexer --video-path /root/match.mp4 --segmenter gemini --output-dir ~/rally/out
#   or the hybrid: --segmenter wasb_hybrid (WASB needs a GPU box, §8) with skip_ai_handoff=false
```

---

7. Object storage ? bucket-driven train/infer (S3-compatible / GCS / Drive)

The worker (`python -m backend.worker`) reads/writes by **URI**, so you choose a backend with **config +
credentials ? no code change**. A bare path / `local://` is a passthrough (today's behaviour); `s3://` and
`gcs://` go to a bucket. Install only the extra you use (cloud SDKs are lazy-imported).

> **Verified end-to-end 2026-06-20:** the worker bucket round-trip (pull video ? infer ? push outputs) passed
> against **real DigitalOcean Spaces** (sgp1), a real **GCS** client (emulator), and
```

MinIO.

7a. S3-compatible ? AWS S3 / DigitalOcean Spaces / Cloudflare R2 / MinIO

One backend for all four ? they differ only by `endpoint_url` + region. `pip install -e .[storage-s3]`.

Credentials ? `~/.aws/credentials` (boto3's default chain; **never** in the repo):
```ini
[default]
aws_access_key_id = <key>
aws_secret_access_key = <secret>
```
Config ? `config.json` `storage.s3`:
```jsonc
"storage": { "backend": "local", "s3": {
  "endpoint_url": "https://sgp1.digitaloceanspaces.com",  // DO Spaces (VERIFIED); region = the Space's, e.g. sgp1
  "region": "sgp1"
  // AWS S3 : omit endpoint_url; set region (e.g. ap-south-1)
  // R2     : endpoint_url https://<account>.r2.cloudflarestorage.com ; region "auto"
  // MinIO  : endpoint_url http://host:9000 ; region us-east-1 ; "addressing_style": "path"
}}
```
Run (secret-free flavor; use `--segmenter gemini` for real detection):
```bash
env -u GEMINI_API_KEY RALLY_DETECTOR_IMPL=stub python -m backend.worker infer \
  --video-uri s3://<bucket>/videos/clip.mp4 --out-uri s3://<bucket> \
  --segmenter wasb_hybrid --set indexing.skip_ai_handoff=true
```

7b. Google Cloud Storage

`pip install -e .[storage-gcs]`. **Credentials** ? a service-account JSON (SA with *Storage Object Admin*):
```bash
export GOOGLE_APPLICATION_CREDENTIALS=/path/to/sa.json
```
Config ? `storage.gcs.project` (no endpoint ? default `googleapis.com`):
```jsonc
"storage": { "backend": "local", "gcs": { "project": "<gcp-project-id>" } }
```
Same `worker infer` command with `gcs://<bucket>/?` URIs. *(On-box test with no GCP creds:
`pip install gcp-storage-emulator && gcp-storage-emulator start --port 9023 --in-memory` +
`export STORAGE_EMULATOR_HOST=http://127.0.0.1:9023`.)*

7c. Google Drive *(corpus source ? assumes a MOUNTED Drive)*

Baked-in assumption: you mount Google Drive for Desktop. That's a fair requirement for using
Drive files locally ? it needs no service account, no API quota, no OAuth, and Drive streams + caches
bytes on access. Once mounted, reference files by a path under your *My Drive* with the `gdrive://`
scheme ? a portable, OS-agnostic alias over the mount (so the same URI works on any mounted box):
```bash
python -m backend.worker infer \
  --video-uri "gdrive://Sharing/Annotation Videos/Golden Label Videos Request - June 15/[Done] Video 4 - Badminton/GX020094_proxy.mp4" ?
```
The backend resolves the mount root automatically ? Windows `G:\My Drive`, macOS

`~/Library/CloudStorage/GoogleDrive-<acct>/My Drive` ? or set it explicitly (it's a path, not a secret):
```bash
export GDRIVE_MOUNT_ROOT="G:\My Drive"          # or: storage.gdrive.mount_root in config
```
The shared golden-corpus folder `drive/folders/1sHGL7iacnmM80ChT-p2cKy638gK504T_` mounts at
`G:\My Drive\Sharing\Annotation Videos\Golden Label Videos Request - June 15\`. *(You can also just pass
the raw `G:\My Drive\?` path ? the default `local` backend handles it identically; `gdrive://` only adds
mount-root portability.)* If Drive isn't mounted the backend fails fast with a clear "mount your Drive"
error. ? **A headless GPU VM has no mount** ? there, stage Drive?bucket and read `gcs://`, or `gdown
<file-id> -O clip.mp4` to a local path.

Bucket layout (for the bucket backends): `videos/ labels/ manifest.json weights/ outputs/<video_id>/`
(see `docs/DECOUPLED_COMPUTE/04`).

---

8. Run training (Gen-0 rally head)

CPU-runnable, minutes ? but it needs a **labeled corpus** (videos + `.rallies.csv` labels + a manifest),
which a fresh box doesn't have. Bring it via §7's bucket (`worker train`, pending) or `scp` it up, then:
```bash
python -m training.gen0.harness --manifest <manifest.json> --floor <baseline> --version 0.1.0
```
The harness runs train ? eval ? ship-gate and emits a versioned weight bundle. *(The Gen-2 VLM is a
separate, GPU-heavy, separately-budgeted effort ? out of scope here.)*

---

9. GPU box ? port procedure (NVMe + native WASB)

The GPU box is **cattle, not a pet**: durable state lives in the bucket + git, so a fresh box is running in
minutes. Port = provision (§2) ? code on (§3) ? NVMe ? `bootstrap.sh --gpu` ? `gpu_setup.sh` ? pull corpus.

```bash
# on the box, after §3 put the code at ~/rally:
cd ~/rally
sudo bash deploy/digitalocean/mount_scratch.sh && export TMPDIR=/scratch   # detect+mount the spare NVMe
./deploy/digitalocean/bootstrap.sh --gpu                                    # CUDA torch; expects cuda_available True
. .venv/bin/activate && pip install -e .[storage-s3]
bash deploy/digitalocean/gpu_setup.sh                                       # native WASB conda env (no WSL) ? M3.5 enabler
# secrets (the only two non-reproducible items): Gemini key + bucket creds
mkdir -p /root/.keys && printf '%s' '<rotated-key>' > /root/.keys/GEMINI_API_KEY && chmod 600 /root/.keys/GEMINI_API_KEY
install -m600 /dev/stdin ~/.aws/credentials <<'INI'
[default]
aws_access_key_id = <key>
aws_secret_access_key = <secret>
INI
```

```
```

- **NVMe scratch** (`mount_scratch.sh`): DO GPU Droplets ship a second raw NVMe; the script formats+mounts it
  at `/scratch` and the pipeline's sync-first decode runs there (`export TMPDIR=/scratch` or `--scratch /scratch`).
  Ephemeral, not in fstab. On a CPU box it just makes `/scratch` on the boot disk.
- **Native WASB** (`gpu_setup.sh`): replicates `docs/TRACKNET_WSL_SETUP.md`'s conda env **natively on Linux**
  (no WSL) ? the M3.5 enabler. ? Needs `wasb_badminton_best.pth.tar` present and the **WASB-C3** weight
  provenance cleared before sharing any derived model.
- **The detector itself:** real `--segmenter wasb_hybrid` extraction needs the Linux-native `DetectorRunner`
  (tracked M3.5, `docs/DECOUPLED_COMPUTE/04`). Until it lands, validate the train/infer plumbing with
  `detector_impl=stub`.
- **Teardown** = `doctl compute droplet delete rally-gpu`. Re-spin tomorrow identically.

---

10. Teardown

```bash
rm -f ~/rally/config.json ~/sample.mp4        # remove the throwaway smoke artifacts
doctl compute droplet delete rally-box        # or destroy in the console
```
Revoke any temporary token/key (#2?#5) and delete the SSH key from the provider when the box is gone.

Appendix ? verification log

Step	Command	Result (2026-06-20, DO Ubuntu 24.04, 2 vCPU, no GPU)
Bootstrap	`bootstrap.sh`	? torch 2.12.1+cpu, deps + editable build
Smoke	`pytest -q`	? 922 passed, 11 skipped (49s)
Secret-free infer (local)	§6a	? 2 rallies, persisted to SQLite, `GEMINI_API_KEY` unset
S3 worker round-trip ? real DO Spaces (sgp1)	§7a	? pulled `s3://rally-e2e/videos/?` ? 2 rallies ? pushed `outputs/<md5>/intervals.csv`
GCS worker round-trip (emulator)	§7b	? pulled `gcs://?` ? 2 rallies ? pushed `gcs://?/outputs/<md5>/?`
S3 backend vs MinIO (put/get/list/stat + presigned GET)	§7a	?
```

### 141. user (2026-06-22T09:33:22.315Z)

```
GCP provisioning runbook ? DECOUPLED_COMPUTE GPU + storage (asia-south1)

> **? GCP is the SOLE compute stack (ADR-013, 2026-06-20).** DigitalOcean is **dropped for compute**
> (DO GPU Droplets aren't enabled on the account + aren't cleanly credit-funded ? DO charges newer
> customers' cards for GPU; Paperspace is subscription-gated). **All training + serving runs on GCP.**
> The `deploy/digitalocean/` scripts are kept only as **provider-agnostic** Linux-GPU setup that this
> runbook reuses. ? Capacity note: asia-south1 (Mumbai) GPUs are frequently stocked-out ? sweep zones /
> fall back to asia-southeast1 (Singapore) etc.; weights still land in the asia-south1 bucket.

Co-located **GPU VM + GCS bucket in ONE region** (asia-south1 / Mumbai) for the first
```
```

real (non-mock) WASB training + inference. This is the **DO?GCP pivot** (ADR-010): DigitalOcean GPU Droplets are North-America-only (no Singapore/India GPU), so we moved GPU + storage to GCP, which has GPUs in asia-south1 (Mumbai) and asia-southeast1 (Singapore). Owner picked **Mumbai** (closest to Bangalore ? lowest latency).

> **Cost discipline (read first):** a running GPU VM is the only meaningful cost. Use the cheapest
> adequate GPU (**T4**), **STOP or DELETE the VM the moment a run finishes** (a stopped VM bills only
> for its disk), and keep the **$100 budget alert** live. `teardown.sh` switches everything off.

---

0. Live resources (created 2026-06-20 ? the "switch-off" inventory)

Resource	Identifier	Cost when idle	How to switch off
Project	`gen-lang-client-0306026826` (billing **Khelsutra Billing** `01420D-419EE0-53D83C`)	?	n/a (shared Gemini project)
GCS bucket	`gs://khelsutra-rally-corpus` (asia-south1, UBLA)	~$0.02/GB/mo storage	`gcloud storage rm -r` (? data loss)
Service account	`rally-worker@gen-lang-client-0306026826.iam.gserviceaccount.com` (`roles/storage.objectAdmin`)	$0	`gcloud iam service-accounts delete`
SA key	`~/.gcp/rally-worker-sa.json` (local only, **never** committed)	$0	delete the file + the key
Budget alert	`rally-100usd-guard` ? $100, alerts at 50/90/100%	$0	`gcloud billing budgets delete`
GPU VM	**NOT created** ? blocked on `GPUS_ALL_REGIONS` quota (see §2)	**~$0.35/hr (T4) running, ~$0.01/hr stopped**	`teardown.sh` (stop or delete)
APIs enabled	serviceusage, compute, cloudbilling, billingbudgets, iam, cloudresourcemanager	$0	leave on (free)

Run `bash deploy/gcp/teardown.sh` (see §6) to stop/delete the billable bits between sessions.

---

1. One-time project bootstrap (owner, console ? already done 2026-06-20)

A bare Gemini/AI-Studio project has the **Service Usage API disabled**, which blocks *all* gcloud provisioning (gcloud's own `services enable` needs that API ? chicken-and-egg). These were enabled in the console; recorded here so a fresh project can be bootstrapped the same way:

1. **Service Usage API** ? https://console.developers.google.com/apis/api/serviceusage.googleapis.com/overview?project=gen-lang-client-0306026826
2. **Link a billing account** (a payment method) ? https://console.cloud.google.com/billing/linkedaccount?project=gen-lang-client-0306026826
3. After (1), the rest enable via gcloud (done): `gcloud services enable compute.googleapis.com cloudbilling.googleapis.com billingbudgets.googleapis.com iam.googleapis.com cloudresourcemanager.googleapis.com`

`provision.sh` (§5) is idempotent and re-runs all of step (3) + the bucket/SA/budget.

---

2. ? GPU quota ? the one remaining owner action

New GCP projects start at **`GPUS_ALL_REGIONS` = 0**, the *global* cap that gates GPU creation even
when a regional per-type quota is non-zero. Confirmed 2026-06-20:

```
asia-south1  NVIDIA_T4_GPUS = 1   NVIDIA_L4_GPUS = 1   (regional: OK)
GLOBAL       GPUS_ALL_REGIONS = 0  ? BLOCKS the VM
```

Owner: IAM & Admin ? **Quotas** ? filter **"GPUs (all regions)"** ? Edit ? request **1** ? submit.
(For a single T4 this is usually approved in minutes; sometimes a short review.) Re-check:

```bash
gcloud compute project-info describe --project gen-lang-client-0306026826 \
  --flatten="quotas[]" --format="table(quotas.metric,quotas.limit)" | grep GPUS_ALL_REGIONS
```

When that reads `1`, run §3.

> **? Spot does NOT skip the quota.** TESTED 2026-06-20: a **Spot** T4 create also fails with
> `Quota 'GPUS_ALL_REGIONS' exceeded. Limit: 0.0 globally` ? `GPUS_ALL_REGIONS` gates **both**
> on-demand AND preemptible GPUs, so the increase above is required either way. (A pending request
> was filed via the Cloud Quotas API: `POST ?/quotaPreferences` `GPUS-ALL-REGIONS-per-project`
> ? 1.) **Note (2026-06-20, ADR-013 supersedes the ADR-011 split): GCP is the SOLE compute stack ?
> BOTH training and serving run on GCP; DigitalOcean is dropped for compute.** Spot is still cheaper
> once the quota is granted, and the resumable cache makes preemption recoverable.

---

3. Create the GPU VM (after quota ? 1)

> **? Recommended: `bash deploy/gcp/create_gpu_vm.sh`** ? the one-command way. It sweeps zones for
> L4 capacity (Mumbai ? Singapore), uses the driver-ready image, attaches the `rally-worker` SA, and
> **always bakes in the Ops Agent** (`--metadata-from-file startup-script=deploy/gcp/ops-agent-startup.sh`)
> so Observability/GPU graphs populate automatically (no more "Ops Agent missing"). It prints the
> zone + the SSH/DELETE commands. `RALLY_GPU=t4` for a T4; `RALLY_VM_NAME=?` to rename. The manual
> command below is kept for reference / customization.

T4 is plenty for WASB + Gen-0. A local SSD gives a fast `/scratch`. *(The native runner defaults to
streaming ? it decodes frames in-process, skipping the slow cv2 PNG extraction; bit-identical to disk,
verified on a real GPU 2026-06-20. Set `indexing.wasb.stream_video=false` only if a container/codec cv2
can't seek; `WASB_SCRATCH` still backs the disk path.)*

```bash
gcloud compute instances create rally-gpu \
  --project=gen-lang-client-0306026826 \
  --zone=asia-south1-a \
  --machine-type=n1-standard-8 \
```

```
    --accelerator=type=nvidia-tesla-t4,count=1 \
    --maintenance-policy=TERMINATE --provisioning-model=STANDARD \
    --image-family=ubuntu-2404-lts-amd64 --image-project=ubuntu-os-cloud \
    --boot-disk-size=120GB --boot-disk-type=pd-balanced \
    --local-ssd=interface=NVME \
    --service-account=rally-worker@gen-lang-client-0306026826.iam.gserviceaccount.com \
    --scopes=cloud-platform
```

Spot variant (cheaper; still needs the §2 quota ? Spot is NOT a quota bypass): same command with `--provisioning-model=SPOT` (replacing `STANDARD`). Add `--instance-termination-action=STOP` to keep the disk on preemption so a re-run resumes. Cheaper, can be preempted.

> **Validated image (2026-06-20):** `--image-family=common-cu129-ubuntu-2204-nvidia-580
> --image-project=deeplearning-platform-release` ships the NVIDIA driver (no install-on-boot wait;
> `nvidia-smi` works immediately) ? used for the first real L4 run. `gpu_setup.sh` installs miniconda
> on it. **T4 was capacity-stocked-out in `asia-south1-a`; an L4 (`g2-standard-4`, no `--accelerator`
> flag ? L4 is built into g2) in `asia-south1-c` landed** ? try L4/other zones if T4 is unavailable.

3b. Observability ? Ops Agent (GPU graphs + logs in the console)
The VM **Observability ? GPU** graphs + Logs need the **Ops Agent** (the console's OS-policy install is blocked on DLVM images). Install it directly + grant the worker SA the write roles (it otherwise only has storage), then GPU/NVML metrics + syslog flow to Cloud Monitoring/Logging (tab populates in ~2-3 min):
```bash
P=gen-lang-client-0306026826; SA=rally-worker@$P.iam.gserviceaccount.com
# 1. ENABLE the APIs (bare Gemini project has them OFF ? the agent installs but its exports get
#    PermissionDenied/SERVICE_DISABLED until these are on; THIS was the empty-graphs cause 2026-06-20):
gcloud services enable logging.googleapis.com monitoring.googleapis.com --project $P
# 2. let rally-worker publish metrics + logs:
gcloud projects add-iam-policy-binding $P --member="serviceAccount:$SA" --role=roles/monitoring.metricWriter --condition=None
gcloud projects add-iam-policy-binding $P --member="serviceAccount:$SA" --role=roles/logging.logWriter   --condition=None
# 3. on the box (created with --scopes=cloud-platform):
curl -sSO https://dl.google.com/cloudagents/add-google-cloud-ops-agent-repo.sh && sudo bash add-google-cloud-ops-agent-repo.sh --also-install
# (if the agent was installed BEFORE the APIs were enabled: `sudo systemctl restart google-cloud-ops-agent`)
```
> Best: bake `curl ?add-google-cloud-ops-agent-repo.sh|bash --also-install` into the VM
> `--metadata=startup-script` so observability is automatic on every box.

> **Long remote steps must be detached.** A plain `ssh ? "<long cmd>"` dies if the SSH drops (seen
> mid-build, 2026-06-20) ? run them `nohup`'d on the box (`nohup bash setup.sh >log 2>&1 &`) and poll
> the log, so a dropped connection can't kill the install.

> If `asia-south1-a` reports no T4 capacity, try `-b` / `-c`. The Ubuntu image needs the NVIDIA
> driver: either add `--metadata=install-nvidia-driver=True` style startup, or after SSH run GCP's
> driver installer, **or** use a Deep-Learning-VM image (`--image-family=common-cu124-ubuntu-2204`

> `--image-project=deeplearning-platform-release`) which ships the driver. Confirm `nvidia-smi` works
> before §4.

SSH: `gcloud compute ssh rally-gpu --zone=asia-south1-a`.

---

4. Port the repo + WASB env onto the box

The `deploy/digitalocean/` scripts are **provider-agnostic** (they detect the local NVMe / install torch); reuse them on GCP:

```bash
# from your machine ? ship the repo, BAKING the source commit so the trained bundle is
# provenance-stamped. A git-archive tarball carries no .git, so training/gen0/harness.py would
# otherwise record git_sha:"unknown" (seen on the 2026-06-20 L4 run). It reads, in order:
# live `git rev-parse` ? $RALLY_GIT_SHA ? a GIT_SHA file at the repo root ? "unknown".
git rev-parse HEAD > /tmp/GIT_SHA
git archive --format=tar.gz --add-file=/tmp/GIT_SHA -o /tmp/rally.tgz HEAD   # GIT_SHA ? archive root (git ?2.38)
gcloud compute scp /tmp/rally.tgz rally-gpu:/root/ --zone=asia-south1-a
gcloud compute ssh rally-gpu --zone=asia-south1-a -- 'mkdir -p ~/rally && tar -xzf ~/rally.tgz -C ~/rally'
# (Alternative, if not baking the file: `export RALLY_GIT_SHA=$(git rev-parse HEAD)` on the box
#  in the same shell that runs `python -m backend.worker train ?`.)

# on the box:
cd ~/rally
sudo bash deploy/digitalocean/mount_scratch.sh && export TMPDIR=/scratch   # detects the local NVMe
./deploy/digitalocean/bootstrap.sh --gpu                                   # venv + torch cu124 (confirm cuda True)
. .venv/bin/activate && pip install -e .[storage-gcs]                      # GCS backend
bash deploy/digitalocean/gpu_setup.sh                                      # native `wasb` conda env + WASB-SBDT

# secrets (NOT in repo): the GCS SA JSON + a rotated Gemini key
gcloud compute scp ~/.gcp/rally-worker-sa.json rally-gpu:/root/.gcp/sa.json --zone=asia-south1-a
# on the box: export GOOGLE_APPLICATION_CREDENTIALS=/root/.gcp/sa.json
mkdir -p /root/.keys && printf '%s' '<ROTATED_GEMINI_KEY>' > /root/.keys/GEMINI_API_KEY && chmod 600 /root/.keys/GEMINI_API_KEY
```

WASB weights (`wasb_badminton_best.pth.tar`) must be present at `~/models/WASB-SBDT/pretrained_weights/` (gpu_setup.sh does NOT fetch them ? stage from Drive/bucket). Point the native runner at them: `export WASB_WEIGHTS=~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar` and `export WASB_PYTHON=$(conda run -n wasb which python)` (or the env's python path). See `docs/TRACKNET_WSL_SETUP.md` for the WASB env. **WASB-C3:** weight provenance is an open legal item before sharing any trajectories/model externally ? internal extraction + Gen-0 is fine.

---

5. Stage the corpus + run the first REAL train/infer

Config for GCS (no secrets in config ? auth via `GOOGLE_APPLICATION_CREDENTIALS`):

```json
{ "sport": "badminton",
  "indexing": { "default_segmenter": "wasb_hybrid", "detector_impl": "native" },
  "storage": { "backend": "local", "gcs": { "project": "gen-lang-client-0306026826" } } }
```

```bash
# stage the 7 Done videos + 7 labels into the bucket (from the co-located box ? fast):
#   videos -> gs://khelsutra-rally-corpus/videos/   labels -> gs://khelsutra-rally-corpus/labels/
# (owner handles the actual copy; see "Upload options" below)

# FIRST REAL training (real WASB trajectories, native runner):
RALLY_DETECTOR_IMPL=native python -m backend.worker train \
  --corpus-uri gcs://khelsutra-rally-corpus --out-uri gcs://khelsutra-rally-corpus \
  --version 0.1.0 --trajectory-source detector --segmenter wasb_hybrid --config config.json

# FIRST REAL inference on the held-out Video 1 (GX010135_proxy.mp4, no labels):
python -m backend.worker infer \
  --video-uri gcs://khelsutra-rally-corpus/videos/GX010135_proxy.mp4 \
  --out-uri  gcs://khelsutra-rally-corpus --segmenter wasb_hybrid --config config.json
```

Upload options (how to get a video in + run inference)
The worker resolves `--video-uri` via the storage layer ? pick whichever fits:
- **GCS bucket (recommended, co-located):** `gcs://khelsutra-rally-corpus/videos/clip.mp4`. Upload with
  `gcloud storage cp clip.mp4 gs://khelsutra-rally-corpus/videos/`.
- **S3 / DO Spaces / R2:** `s3://bucket/clip.mp4` (install `.[storage-s3]`, creds in `~/.aws/credentials`).
- **Local upload:** scp the file to the box and pass a bare path / `local:///root/clip.mp4` (identity passthrough, no copy).
- **Google Drive (assumes Drive is MOUNTED):** mount Google Drive for Desktop ? a fair requirement to
  use Drive files locally ? and reference them with `gdrive://<path-under-My-Drive>`. The backend resolves
  it against your local mount (auto-detected: Windows `G:\My Drive`, macOS
  `~/Library/CloudStorage/GoogleDrive-<acct>/My Drive`; override with `GDRIVE_MOUNT_ROOT` or
  `storage.gdrive.mount_root`), then it's plain local I/O ? **no service account, no API, no extra install.**
  E.g. the golden-corpus folder `?/drive/folders/1sHGL7iacnmM80ChT-p2cKy638gK504T_` mounts at
  `G:\My Drive\Sharing\Annotation Videos\Golden Label Videos Request - June 15\`, so
  `gdrive://Sharing/Annotation Videos/Golden Label Videos Request - June 15/[Done] Video 4 - Badminton/GX020094_proxy.mp4`
  (or just the raw `G:\My Drive\?` path ? the `local` backend handles that too). ? A **headless GPU VM has
  no mount** ? there, stage Drive?bucket and read `gcs://`, or `gdown <file-id> -O clip.mp4` to a local path.

---

6. Switch off between sessions (cost control)

> **? DELETE, don't just STOP.** A *stopped* VM halts compute billing but **still bills for its
> persistent disk** (pd-ssd ? $0.17/GB/mo ? a 200 GB boot disk ? **$34/mo**). Only **delete** reaches
> $0. So `teardown.sh` **defaults to `delete`** (owner cost rule, 2026-06-20). The boot disk has
> `autoDelete=true`, so delete reclaims it too ? verified 2026-06-20.

```bash
bash deploy/gcp/teardown.sh          # DELETE the VM (default; $0 ? removes VM + boot disk)
bash deploy/gcp/teardown.sh delete   # same (explicit)
bash deploy/gcp/teardown.sh stop     # STOP only ? keeps the disk ? STILL BILLS (~$/mo!); fast resume
bash deploy/gcp/teardown.sh all      # delete VM + SA + budget (keeps the bucket/data)
# cross-region VM: set the zone (teardown defaults to asia-south1-a)
RALLY_GCP_ZONE=asia-southeast1-a bash deploy/gcp/teardown.sh delete
```

After deleting, audit for orphans (a stray disk bills even with no VM):
`gcloud compute instances list` (empty) and `gcloud compute disks list` (no unattached disk).

The **bucket is never auto-deleted** (it holds the corpus). Delete it explicitly only when done:
`gcloud storage rm -r gs://khelsutra-rally-corpus`.

142. assistant (2026-06-22T09:34:45.085Z)

This is clearer than expected: the cloud architecture is **already built** (the archived DECOUPLED_COMPUTE project) ? `backend.worker train/infer` reads/writes by `gcs://` URI, `deploy/gcp/` provisions it, and a canonical bucket layout (`videos/ labels/ manifest.json weights/ outputs/<video_id>/`) was specified. What's missing is (1) the bucket has **drifted** from that layout and isn't fully populated, and (2) there's **no single authoritative doc** mapping where each piece of data lives ? it's scattered across three runbooks.

So the right deliverable is an authoritative **data map + gap inventory + migration plan**. Let me get a precise local-corpus inventory for the gap table, then write it.

143. user (2026-06-22T09:34:56.859Z)

```
Exit code 1
=== local trajectories (C:) ===
count=20  totalMB=17.5
adarsh_avi_full_traj.csv                    2.36 MB
adarsh_avi_traj.csv                         1.06 MB
adarsh_traj.csv                             1.05 MB
Badminton_BXH_2_june15_traj.csv             1.48 MB
Boxhill_Doubles_june18_traj.csv             0.80 MB
gbaaddy_traj.csv                            1.62 MB
GX010128_june12_traj.csv                    1.47 MB
GX010137_june17_traj.csv                    0.76 MB
GX010141_june17_traj.csv                    0.74 MB
GX020094_june18_traj.csv                    0.58 MB
GX030094_june15_traj.csv                    0.67 MB
kushagra_traj.csv                           1.11 MB
largetestvideo_traj.csv                     0.66 MB
mahadevpura_traj.csv                        0.40 MB
mahadevpura1_june12_traj.csv                0.44 MB
mahadevpura2_june12_traj.csv                1.39 MB
pickleball_traj.csv                         0.14 MB
sample_long_traj.csv                        0.57 MB
tabletennis_traj.csv                        0.09 MB
testlarge_traj.csv                          0.09 MB

=== manifest ===
C:\Users\avidu\Projects\badminton-highlight-indexer\output\human_lovo_manifest.json  4503 bytes

=== bucket: is there a root manifest.json or trajectories/ ? ===
ERROR: (gcloud.storage.ls) One or more URLs matched no objects.
ERROR: (gcloud.storage.ls) One or more URLs matched no objects.
```

144. assistant (2026-06-22T09:35:32.178Z)

Everything's confirmed: the corpus metadata that blocks cloud eval is **tiny and entirely local** ? 20 trajectory CSVs (17.5 MB total) + a 4.5 KB manifest with C: paths, and **none** of it is in the bucket (no `trajectories/`, no root `manifest.json`). Only the originals are large. Let me verify the `worker train/infer` entry points are real (one runbook says "pending", another shows them as used) so I document accurately, then write the map.

145. user (2026-06-22T09:35:34.759Z)

```
=== backend.worker exists? ===
backend/worker/:
__init__.py
__main__.py
__pycache__
--- train/infer + corpus-uri in worker ---
backend/worker/__main__.py:157:def cmd_infer(args: argparse.Namespace) -> int:
backend/worker/__main__.py:332:def cmd_train(args: argparse.Namespace) -> int:
backend/worker/__main__.py:358:    corpus = args.corpus_uri.rstrip("/")
backend/worker/__main__.py:457:    pi = sub.add_parser("infer", help="run inference on one video
URI and push outputs")
backend/worker/__main__.py:472:    pt = sub.add_parser("train", help="Gen-0 train-from-bucket
(labels [+videos] -> trajectories -> harness)")
backend/worker/__main__.py:473:    pt.add_argument("--corpus-uri", required=True, help="prefix
containing labels/ (+ videos/ for "
Binary file backend/worker/__pycache__/__main__.cpython-313.pyc matches
=== how the corpus manifest is loaded (path format) ===
backend/worker/__main__.py:333:    """Gen-0 train-from-bucket: pull labels (+ videos) ->
trajectories -> manifest -> shell out
backend/worker/__main__.py:417:    manifest_path = os.path.join(scratch, "manifest.json")
backend/worker/__main__.py:418:    with open(manifest_path, "w", encoding="utf-8") as f:
backend/worker/__main__.py:421:    print(f"[worker] built manifest: {len(specs)} videos
({src_label} trajectories)")
backend/worker/__main__.py:425:           "--manifest", manifest_path, "--out", out_dir, "--
version", args.version]
backend/worker/__main__.py:436:    # PUSH the versioned artifact bundle (weights.json +
manifest.json) to the bucket.
Binary file backend/worker/__pycache__/__main__.cpython-313.pyc matches
```

146. assistant (2026-06-22T09:36:14.060Z)

Now I have an accurate mental model: `worker train --corpus-uri gcs://khelsutra-rally-corpus` reads `labels/` (+`videos/`), **extracts trajectories and builds the manifest on-box at train time** (it does *not* read a pre-staged manifest), and pushes a `weights/<version>/` bundle; `worker infer --video-uri gcs://.../videos/x.mp4` writes to `outputs/<video_id>/`. So the input contract is `labels/` + `videos/`, and trajectories are a regenerable cache.

Let me write the authoritative map.

147. assistant (2026-06-22T09:37:11.774Z)

Data in GCS ? the corpus source-of-truth map

Status: LIVE / in-progress (migration not complete). **Owner-gated** decisions in §7.
Last inventoried: 2026-06-22 against `gs://khelsutra-rally-corpus`.
Goal (owner, 2026-06-22): run **all** training + inference from the cloud with **all data in GCS**,
so the local Windows box (and the intermittently-mounted external drive) is **no longer a blocker**.

> **? Other sessions: this is where the data lives.** Pull from `gs://khelsutra-rally-corpus`, not
> from anyone's `C:` / `F:` / Google Drive. If something you need isn't in the bucket yet, it's in the

> **gap table (§5)** ? flag it / migrate it, don't hard-code a local path.

This doc is the **data map**. The *machine* runbooks are
[`SETUP_NEW_MACHINE.md`](SETUP_NEW_MACHINE.md)
(spin up a box) and [`deploy/gcp/README.md`](../deploy/gcp/README.md) (provision GPU + train/infer).
The cloud engine itself was built by the (archived) DECOUPLED_COMPUTE project
(`docs/archives/past_projects/DECOUPLED_COMPUTE/`); this doc does **not** redesign it ? it records
where the bytes are and *what's still missing*.

---

1. The bucket

Bucket	**`gs://khelsutra-rally-corpus`** (single corpus bucket)
Project	`gen-lang-client-0306026826` (billing *Khelsutra Billing*)
Region	`asia-south1` (Mumbai), UBLA, co-located with the GPU VMs
Auth	SA `rally-worker@gen-lang-client-0306026826.iam.gserviceaccount.com` (`roles/storage.objectAdmin`); key at `~/.gcp/rally-worker-sa.json` (local only, **never committed**). Interactive: `gcloud auth login`.
Size today	~2.06 GB (mostly past dry-run reels/outputs ? **not** the full corpus)

The engine is **backend-agnostic by URI** (`backend/storage/ref.py`): `gcs://` (or `gs://`), `s3://`, `gdrive://`, or a bare local path. Pick the backend with **config + creds, no code change**.

---

2. Canonical layout (the target contract)

The cloud worker's input/output contract (`backend/worker/__main__.py`; lineage: `DECOUPLED_COMPUTE/04`):

Prefix	Holds	Who reads/writes it
`videos/`	source videos / proxies (`<clip>.mp4`)	**input** to `worker infer` & `worker train`
`labels/`	golden rally labels (`<clip>.rallies.csv`)	**input** ground truth to `worker train` + all eval
`trajectories/`	cached WASB trajectory CSVs	**(PROPOSED ? see §7.2)** GPU-free eval/litmus input
`weights/<version>/`	trained Gen-0 bundles (`weights.json` + a run `manifest.json`)	**output** of `worker train`
`models/`	upstream pretrained weights (`wasb_badminton_best.pth.tar`)	input to the WASB detector
`outputs/<video_id>/`	per-video inference outputs (intervals, etc.)	**output** of `worker infer`
`highlights/`, `promo/`	rendered reels + promo artifacts	output / demo

> **Note on the manifest.** `worker train` does **not** read a pre-staged corpus `manifest.json` ? it
> *builds* one on-box at train time from `labels/` (+`videos/`), extracting trajectories via the native
> detector (`--trajectory-source detector`). A `manifest.json` only appears as an **output** inside each
> `weights/<version>/` bundle. The **eval/litmus** path is what wants a stable, path-resolvable manifest
> (§7.3).

---

3. What's actually in the bucket today (2026-06-22)

Prefix	Present	Drift / notes
`labels/`	7 files: `GX010128`, `mahadevpura-1`, + `Badminton BXH 2_proxy`, `Boxhill Doubles_proxy`, `GX020094_proxy`, `GX030094_proxy`, `mahadevpura-2_proxy`	? **inconsistent naming** (`_proxy` on some, not others); ? **missing** the original-6 labels + `GX010137`, `GX010141`
`videos/`	3: `GX030094_proxy.mp4`, `mahadevpura-1.MP4`, `mahadevpura_smoke_180s.mp4`	? corpus videos/proxies almost entirely absent
`trajectories/`	?	? **does not exist** (every trajectory is local-only)
`models/`	`wasb_badminton_best.pth.tar`	? (? WASB-C3 weight-provenance still an open legal item before any external share)
`weights/`	`0.1.0-l4/`	? first real L4 train bundle
`outputs/`	one `<video_id>/`, `parity/`	? dry-run outputs
`highlights/`, `promo/`	reels + promo	?

Drift summary: the bucket has extra prefixes (`highlights/`, `promo/`, `models/` alongside `weights/`) and **inconsistent `labels/` naming**, and is **missing the bulk of the corpus inputs**
(`videos/`, all `trajectories/`, most `labels/`). It reflects past *dry runs*, not a complete corpus.

---

4. How the cloud consumes it

Provisioning + the verified commands live in [`deploy/gcp/README.md`](../deploy/gcp/README.md) §3?§5.
The data-facing entry points (`backend/worker/__main__.py`):

```bash
# config (no secrets in config; auth via GOOGLE_APPLICATION_CREDENTIALS):
#   { "storage": { "backend": "local", "gcs": { "project": "gen-lang-client-0306026826" } } }

# INFERENCE on one cloud video -> outputs/<video_id>/ in the bucket
python -m backend.worker infer \
  --video-uri gcs://khelsutra-rally-corpus/videos/<clip>.mp4 \
  --out-uri   gcs://khelsutra-rally-corpus --segmenter wasb_hybrid --config config.json

# TRAIN from the bucket (reads labels/ [+ videos/], extracts trajectories, builds manifest
# runs training.gen0.harness, pushes weights/<version>/):
RALLY_DETECTOR_IMPL=native python -m backend.worker train \
  --corpus-uri gcs://khelsutra-rally-corpus --out-uri gcs://khelsutra-rally-corpus \
  --version <v> --trajectory-source detector --segmenter wasb_hybrid --config config.json
```

A bare/relative path resolves under a configured `input_root`/`input_backend`
(`backend/storage/ref.py::resolve_input_uri`), so workflows can be written bucket-relative.

---

5. The gap ? what's still local-only (why the box is still a blocker)

All of this is **tiny except the originals** ? the metadata that blocks cloud eval is < 25 MB total:

Data	Where it is now	Size	In bucket?	Needed for
20 WASB trajectory CSVs	`C:\?\Annotation Setup\Collect\Trajectories\`	**17.5 MB**	? none	

| eval/litmus (cached), train (`--trajectory-source cached`) |
golden manifest	`output/human_lovo_manifest.json` (C:-absolute paths)	4.5 KB	?	eval corpus definition
original-6 labels	`F:\[Khelsutra]?\Training\Golden Labelled\*.rallies.csv`	~12 KB	?	train + eval
7 new-clip labels	repo `output/*.rallies.csv` (gitignored) **+** F: archive (backed up 2026-06-22)	~12 KB	? partial/inconsistent	train + eval
source videos / proxies	F: archive + Google Drive golden folder	**100s of GB**	? 3 only	infer + train (real WASB extraction)

> The corpus *metadata* (trajectories + labels + manifest, ~25 MB) is what actually blocks GPU-free
> cloud eval and the served Gate-A litmus ? and it's trivially small to migrate. The **videos** are the
> only large migration, and they already exist on **Google Drive** (the golden-corpus folder), so they
> can go **Drive ? bucket from a cloud box**, keeping the local machine out of the path entirely.

---

6. Migration plan (close the gap)

Phase 1 ? corpus metadata (?25 MB, do first, removes the eval blocker). Stage all trajectories,
all labels, and a **bucket-relative eval manifest** into `gs://khelsutra-rally-corpus`. Pending the §7
naming decision. Sketch:

```bash

# trajectories (decision-free; absent today)

gcloud storage cp "C:\?\Collect\Trajectories\*.csv" gs://khelsutra-rally-corpus/trajectories/

# labels (after the §7.1 naming decision) -> labels/&lt;canonical-name&gt;.rallies.csv

# eval manifest with gs:// (or bucket-relative) paths -> a stable manifest object (§7.3)

```

Phase 2 ? videos (large; Drive ? bucket, from a cloud box). Prefer **proxies** over 4K originals
(smaller, sufficient for detection). Run the copy from a co-located GPU VM (Drive mounted there, or
`gdown` by file-id), not from the local box. Track per-clip in §8.

Phase 3 ? flip eval/training to read GCS by default. Point the eval manifest + litmus at the
bucket (env override or `input_root`) so a fresh box/CI needs no local corpus. Re-verify the suite +
the served Gate-A litmus read from `gcs://` (and re-measure if the corpus set changes ? see PR #236).

---

7. Decisions needed (owner)

1. **`labels/` naming convention.** Reconcile the `_proxy` inconsistency. Options: (a)
`<manifest-name>.rallies.csv` matching the eval `name` (e.g. `GX020094.rallies.csv`); (b)
`<video-basename>.rallies.csv` pairing with `videos/<video>` (the original-6 convention); (c)
key by **`video_id`/MD5** (most robust ? the org-hygiene recommendation, immune to recycled
GoPro names). **Recommend (c) or (a).**
2. **Cache `trajectories/` in the bucket?** Training can regenerate them (`--trajectory-source
detector`), but the **eval/litmus** consumes cached CSVs. Caching them (17.5 MB) makes cloud/CI
eval **GPU-free**. **Recommend: yes, cache them.**
3. **Eval manifest in GCS.** The local manifest has C: absolute paths. Produce a bucket-relative
/ `gs://` manifest (e.g. `gs://?/manifests/golden.json`) so cloud eval is path-portable.

Recommend: yes.
4. **Originals migration scope** ? proxies-only vs full 4K, and how much of the 15-clip golden set vs the wider archive.

---

8. Progress tracker

Item	Status
Bucket + SA + provisioning (deploy/gcp)	? built
Cloud worker `infer`/`train` by URI	? built (`backend/worker`)
This data-map doc	? 2026-06-22
Phase 1: trajectories ? bucket	? pending §7
Phase 1: labels reconciled ? bucket	? pending §7.1
Phase 1: bucket-relative eval manifest	? pending §7.3
Phase 2: corpus videos/proxies ? bucket (Drive?bucket)	?
Phase 3: eval/litmus default to GCS	?

Definition of done: a fresh GPU box (or CI) can run inference, the eval suite, and Gen-0 training
reading **only** from `gs://khelsutra-rally-corpus` ? no `C:` / `F:` / local-Drive path required.

Related

- [`SETUP_NEW_MACHINE.md`](SETUP_NEW_MACHINE.md) ? spin up a box (storage backends in §7).
- [`deploy/gcp/README.md`](../deploy/gcp/README.md) ? GCP GPU provisioning + train/infer (§5).
- [`VIDEO_LOCALITY_MODEL.md`](VIDEO_LOCALITY_MODEL.md) ·
[`PER_VIDEO_WORKSPACE.md`](PER_VIDEO_WORKSPACE.md) ? locality tiers + `outputs/<video_id>/`.
- `docs/archives/past_projects/DECOUPLED_COMPUTE/` ? the cloud-engine design lineage.

148. user (2026-06-22T09:37:11.805Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DATA_IN_GCS.md (file state is current in your context ? no need to Read it back)

149. assistant (2026-06-22T09:37:17.956Z)

Now let me index the doc in `docs/README.md` so sessions can find it. Let me locate the right section.

150. user (2026-06-22T09:37:19.696Z)

15-
16:## Product / platform plans
17:[Omitted long matching line]
18:[Omitted long matching line]
19-[Omitted long context line]
20:[Omitted long matching line]
21-
22-## Practical / ongoing
23:[Omitted long matching line]
24-- TRACKNET_WSL_SETUP.md

151. user (2026-06-22T09:37:23.522Z)

16   ## Product / platform plans
17   - [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) ? **? Active tracked project (2026-06-21, `DRAFT` owner-gated):** the productionization successor to DECOUPLED_COMPUTE (its deferred M3/M4). **One pure core (DETECT?WINDOW?STITCH) that never branches on deployment profile**, with a `ComputeBackend` (where it runs) + `StorageBackend`

(what bytes) selected by **one startup config**. Ships **truly-local** (local/USB + local GPU, in-process stitch, zero cloud) + **cloud-serving** (upload<2GB / gdrive / bucket ? a **Cloud Run GPU job runs detect AND stitch**, metered into a per-job cost ledger) + **cpu-mock** (CI). Includes [TESTING_STRATEGY.md](COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md) (profile-PARITY proof for $0), [architecture.svg](COMPUTE_DECOUPLED_SERVING/architecture.svg), a [runlogs/](COMPUTE_DECOUPLED_SERVING/runlogs/) posterity dir, and **ADR-014** ([lineage ADR-009?010?011?012?013?014](archives/decisions/DECISIONS.md)). §7 M0?M10 tracker is the source of truth.

18 - [DECOUPLED_COMPUTE/](archives/past_projects/DECOUPLED_COMPUTE/README.md) ? ? **DELIVERED + ARCHIVED (2026-06-21).** Lifted the rally pipeline onto a **rented cloud-native GPU** doing bucket-driven `train` + `infer`: storage/compute seams (PRs #246?#259) + the **M3.5 native WASB runner**; **M0?M2 + M3.5 proven on a GCP L4** (M2 ? `weights/0.1.0-l4/`). Delivered on **GCP** ? DigitalOcean dropped for compute ([ADR-013](archives/decisions/DECISIONS.md)). **M3/M4 (serving endpoint + per-video billing + scale-to-zero) deferred ? the "Cloud Run + GPU serving" subproject** ([ADR-012](archives/decisions/DECISIONS.md)). Snapshot under `docs/archives/past_projects/DECOUPLED_COMPUTE/` (incl. `DEPLOY_REPORT_GCP_2026-06-20.md`).

19 - PACKAGING_PLAN.md ? **? Active tracked project (`IN PROGRESS`, 11 PRs merged 2026-06-21/22):** ship the engine as a **downloadable batteries-included Python app** ? `pip install` ? `indexer --app` boots a local webserver and opens the **bundled KhelSutra SPA** single-origin, **torch-free** (LIGHT tier), cross-platform, no native component (decision D10/O1). Built ON the decoupled-compute seam. **Supersedes the `DESKTOP_APP_PLAN.md` narrative.** §7 tracker = source of truth (P0a?e ?, P1a/b ?, P2-launcher/P2a/P2d/P2e ?). **? Next:** P3 GPU validation + screencast ? P2b-install ? P2c wizard (cross-repo) ? P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.

20 - DESKTOP_APP_PLAN.md ? **? Superseded narrative (by `PACKAGING_PLAN.md`; D10 = self-hosted webserver, NOT a desktop app).** Original scoping doc (2026-06-18, `[design]`): package the local pipeline as a **cross-platform desktop app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical *no-GPU* enthusiast laptop (WASB is ~3.4× real-time on a laptop 2070S)? §7 tracker = **P0 compute-feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers**. Realizes the **L0 fully-local** tier of VIDEO_LOCALITY_MODEL.md; is the **`LocalDesktop` ComputeBackend** of PLATFORM_ARCHITECTURE.md §3.2; the smaller/export-clean owned model (OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.

21

22 ## Practical / ongoing

23 - SETUP_NEW_MACHINE.md ? **? The canonical "spin up a box ? ready for training & inference" runbook.** Copy-pasteable, human-followable, **verified end-to-end on DigitalOcean (2026-06-20)** ? but **compute is now GCP-only (ADR-013)**, see [`deploy/gcp/README.md`](../deploy/gcp/README.md): credentials ? provision ? bootstrap ? suite (922 passed) ? secret-free CPU inference (no GPU/keys) ? real Gemini inference ? Gen-0 training, with pointers to the GPU/native-detector + bucket paths. The `deploy/digitalocean/*` scripts are provider-agnostic and reused on GCP.

24 - TRACKNET_WSL_SETUP.md

25 - Evaluation harness and calibration drivers (in `backend/eval/`)

26

27 ## Data layer (collection · annotation · golden sets ? **no ML**)

28 - [`data_pipeline/`](data_pipeline/README.md) ? **the isolated data pillar**: how footage is recorded (`VIDEO_RECORDING_GUIDELINES`), golden sets are created + which videos (`GOLDEN_SET_IMPLEMENTATION` · `GOLDEN_SET_PHASE1_VIDEOS` · `GOLDEN_VIDEOS`), and data is shared/ingested (`GOLDEN_DATA_SHARING`). Deliberately **no segmentation/stitching/model** detail ? that's the ML pillar (top level + `COMPUTE_DECOUPLED_SERVING/`). The **annotation-vendor (Roora)** engagement is archived under `archives/roora-vendor/`.

29

152. assistant (2026-06-22T09:37:28.728Z)

I'll index the new doc under "Practical / ongoing", next to the machine runbook it complements.

153. user (2026-06-22T09:37:34.055Z)

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154. assistant (2026-06-22T09:38:03.739Z)

Now let me commit the doc on the work branch (re-verifying HEAD first, per the shared-checkout convention) and record it in memory so future sessions actually find it.

155. user (2026-06-22T09:38:13.174Z)

```
=== re-verify HEAD (mine) ===
work/golden-corpus-hygiene
689c738 docs(pkg): mark P2c ? + the loop closed ? v1 LIGHT is feature-complete (#313)
=== stage explicit paths ===
warning: in the working copy of 'docs/DATA_IN_GCS.md', LF will be replaced by CRLF the next time
Git touches it
A  docs/DATA_IN_GCS.md
M  docs/README.md
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
=== commit ===
[work/golden-corpus-hygiene 8884c9f] docs(data): add DATA_IN_GCS.md ? the corpus source-of-truth
map (GCS)
 2 files changed, 176 insertions(+)
 create mode 100644 docs/DATA_IN_GCS.md
=== branch vs master ===
8884c9f docs(data): add DATA_IN_GCS.md ? the corpus source-of-truth map (GCS)
```

156. assistant (2026-06-22T09:38:30.237Z)

Committed as `8884c9f` (only my commit on the branch). Now let me record this in memory so future sessions actually find the map. Let me create the memory file and locate the MEMORY.md insertion point.

157. assistant (2026-06-22T09:38:44.872Z)

```
---
name: gcs-data-source-of-truth
description: GCS bucket gs://khelsutra-rally-corpus is the corpus source of truth; map at
docs/DATA_IN_GCS.md. Pull data from the bucket, not local C:/F:/Drive.
metadata:
  type: project
---

Owner goal (2026-06-22): run **all** training + inference from the cloud with **all data in
`gs://khelsutra-rally-corpus`**, so the local Windows box + the intermittently-mounted external
drive stop being blockers. **Other sessions: pull data from the bucket, not from anyone's
`C:`/`F:`/Google Drive.**

**The authoritative map = `docs/DATA_IN_GCS.md`** (indexed in `docs/README.md` ?
Practical/ongoing;
landed on branch `work/golden-corpus-hygiene`, commit 8884c9f). It records the bucket layout,
current
contents + drift, how `backend.worker infer|train` consume `gs://`, the still-local-only gap,
and a
phased migration plan + owner decisions.

Key facts:
- The **cloud engine already exists** ? don't redesign it. `python -m backend.worker infer
  --video-uri gcs://?/videos/x.mp4` and `worker train --corpus-uri gcs://khelsutra-rally-corpus`
  (reads `labels/` [+`videos/`], extracts trajectories + builds the manifest on-box, pushes
  `weights/<version>/`). URI backends in `backend/storage/ref.py`. Provisioning = `deploy/gcp/`.
  Built by the archived DECOUPLED_COMPUTE project.
- **Canonical layout:** `videos/ labels/ trajectories/(proposed) weights/<v>/ models/
outputs/<video_id>/`.
```

- **The gap (why the box is still a blocker):** the corpus *metadata* is ~25 MB and entirely local ?
20 trajectory CSVs (17.5 MB at `C:\?\Annotation Setup\Collect\Trajectories\`), the 4.5 KB
`output/human_lovo_manifest.json` (C:-absolute paths), and most labels. Only the **videos** are
large, and they're on Google Drive ? migrate **Drive?bucket from a cloud box**, not via the local box.
- **Open owner decisions (DATA_IN_GCS §7):** `labels/` naming (recycled GoPro names ? key by
`video_id`/MD5 is the robust org fix); cache `trajectories/` in the bucket for GPU-free eval; a
bucket-relative eval manifest.

Pairs with [[compute-stack-gcp-only]] (GCP is the sole compute stack, ADR-013). The path-fragility this
fixes was witnessed live (2026-06-22): the F: archive root was renamed `GoPro Hero Black 12` ?
`[Khelsutra] GoPro Hero Black 12` between turns ? absolute local paths drift; the bucket doesn't.

158. user (2026-06-22T09:38:44.903Z)

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159. user (2026-06-22T09:38:46.031Z)

26:- [Compute = GCP only; DO dropped](compute-stack-gcp-only.md) ? **standing decision (ADR-013, 2026-06-20):** GCP is the SOLE compute stack (training + serving); **DigitalOcean dropped for compute** (GPU not enabled/credit-funded; Paperspace subscription-gated). Provision via `deploy/gcp/` only; don't relitigate DO/Paperspace without a new ADR; $200 DO credits ? non-compute only.

160. user (2026-06-22T09:38:55.375Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

161. user (2026-06-22T09:39:02.617Z)

24   - [GCP Ops Agent: always enable](gcp-ops-agent-enable.md) ? owner directive (missed TWICE): every rally GPU VM needs the Ops Agent for Observability/GPU graphs ? **3 pieces** (logging+monitoring APIs ON + SA metric/log-writer roles + agent install via startup-script). BAKE into `deploy/gcp` provisioning so it's automatic.
25   - [GCP VM: always DELETE, never just stop](gcp-vm-always-delete.md) ? owner cost rule (2026-06-20): a STOPPED VM still bills its disk (~$34/mo for 200GB pd-ssd); **DELETE** ? $0 (boot disk autoDelete=true). `teardown.sh` now defaults to `delete`; audit `gcloud compute disks list` for orphans. DO droplet `claude-do-rally-test` (~$24/mo) is a shared testbed (owner-gated).
26   - [Compute = GCP only; DO dropped](compute-stack-gcp-only.md) ? **standing decision (ADR-013, 2026-06-20):** GCP is the SOLE compute stack (training + serving); **DigitalOcean dropped for compute** (GPU not enabled/credit-funded; Paperspace subscription-gated). Provision via `deploy/gcp/` only; don't relitigate DO/Paperspace without a new ADR; $200 DO credits ? non-compute only.
27   - [Cache-hygiene policy](cache-hygiene-policy.md) ? retention table (purge frames/staged vs keep CSVs/DB/reels) + the shipped tooling: self-cleaning runners (keep_frames default false), tools/cleanup_caches.py (dry-run-first), disk guard, SessionStart nudge hook. The `for e in *` WSL footgun.

162. user (2026-06-22T09:39:12.761Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\MEMORY.md has been updated successfully. (file state is current in your context ? no need to Read it back)

163. assistant (2026-06-22T09:39:24.543Z)

Now a current-state checkpoint. Let me re-read the top (it's a shared file that may have changed) to anchor the edit correctly.

164. user (2026-06-22T09:39:25.702Z)

```
---
name: current-state
description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not only on request). What just landed, what's next, open decisions."
metadata:
  node_type: memory
  type: project
  originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
---
```

Current state ? updated continuously (see [[working-agreement-checkpoints]])

> **Scope of this file = LIVE only:** the current handoff + *today's* checkpoints + open PRs/decisions. Keep it read-first-short (?1 page). Older checkpoints are archived in [[current-state-archive]] ? when today's checkpoints age out (next clean-slate day), move them there, don't let them pile up here.

Checkpoint: 2026-06-22 (**PACKAGING P2b + P2c ? BOTH MERGED; v1 LIGHT engine-complete + wizard'd**). Same session as the P3 checkpoint below; owner: "complete P2b-install" ? "start P2c wizard SPA now" ? "merge #310 now". **? P2b-install ? engine [#310](https://github.com/Khelsutra/badminton-highlight-indexer/pull/310) MERGED** (master `f8b2c3b`): `write_light_default_config` seeds a truly-local + **`motion`** (gate **O8** = resolved) torch-free default `config.json` on the first `indexer --app` (minimal, profile-driven, NEVER overwrites; placed before `load_app_config`, scoped to `--app`); `scripts/install.py` (stdlib, cross-platform ? `uv tool install` else pip + a per-OS launcher shortcut that **pins `RALLY_APP_HOME`** ? load-bearing since `load_dotenv` runs at import + writes an uninstall manifest) + `scripts/uninstall.py` (manifest-driven; KEEPS user data unless `--purge`) + README. **12-agent adversarial review (`wf_2be29ecf`) folded:** MAJOR PATH-robustness (`uv tool update-shell` + resolve+embed the abs `indexer` path + warn) + uninstall **containment guard** (`_within(path, app_home)` refuses rmtree outside app-home ? a corrupted/foreign manifest can't nuke an arbitrary dir) + `--dry-run` fidelity + tests for the destructive path. Suite **1352?**, ruff+mypy clean, cov **85.17%**. ? **CI was billing-blocked for hours** ? the engine repo bills under the **`Khelsutra` ORG** (separate from the personal account); owner fixed the ORG billing ? CI green ? merged. **? P2c first-run wizard ? khelsutra [#88](https://github.com/avidullu/khelsutra/pull/88) MERGED** (master `7ae9cee`): `web/src/components/SetupWizard.tsx` ? 3 steps (AI-key *guided-setup* since no key-write endpoint yet / compute from `GET /api/capabilities` / videos) rendered before `IngestPanel`, shows once per browser (localStorage), 6 langs (en/kn/hi/es/da/id, brand stays Latin "KhelSutra" per #77), `getCapabilities()`+`EngineCapabilities`. **223 web tests** (+ the i18n parity/brand/no-bare guards) + **browser-verified e2e against a live local engine** (all 3 steps render real caps: ffmpeg+GPU detected; offline degrades gracefully). **13-agent review (`wf_191a9b86`) folded:** MAJOR a11y (the "Get a free Gemini key" link was below the 44px tap floor ? `min-h-11`) + loading-window finish-button (gated to `offline` only) + `role=dialog`?`region` + kn brand spacing + the no-bare-strings guard now covers SetupWizard. ? a sibling reorg-merged **#87** (es/da/id catalogs) mid-build ? rebased (auto-merged clean) + re-synced `site/public/locales` via `npm run copy:locales` (a `locales-drift.test` enforces the mirror). Both built in **isolated worktrees off `origin/master`** (since removed); sibling worktrees untouched. **Also this session (owner asks): deployed the real Gemini key (from `G:\My Drive\gemini.api_key.txt`) to the droplet `/etc/khelsutra/engine.env` + validated live (55 models, $0) + ran the FULL suite on the droplet (1313 passed); injecting the key auto-started `khelsutra-engine` (enabled unit) on loopback ? owner: "keep it running" (NOT publicly exposed ? `api./app.khelsutra.guru` have no public DNS route yet).** ? **LOOP CLOSED ? engine [#312](https://github.com/Khelsutra/badminton-highlight-indexer/pull/312) MERGED** (master `293e5c0`): re-bundled `backend/ui/` from khelsutra@`7ae9cee` via `scripts/bundle_ui.sh` so **`indexer --app` now SERVES the wizard** (`/api/version` spa_commit=7ae9cee; `/` serves `index-GSXqL6iY.js`; live-boot proven). Pure vendored-asset refresh; 53 UI/skew contract tests + full suite 1355? + cov 85.25%. **? v1 LIGHT is now feature-complete ? only OWNER steps remain: a clean-Windows/macOS-box acceptance run + ffmpeg-acquisition UX** (winget/brew/apt is still a manual dep). North-star (native-WASB-in-default-download P3b/P4a + training P4b + annotation P4c) stays future + counsel(O3)/owner-GPU-gated. ?? **THIS SESSION TALLY: 3 PRs merged (#310

P2b-install, khelsutra #88 P2c wizard, #312 bundle) + P3 GPU validation ($0) + droplet key/suite. 0 open PRs, 0 VMs, all worktrees/branches cleaned.** North-star (native-WASB-in-default-download P3b/P4a + training P4b + annotation P4c) stays future + counsel(O3)/owner-GPU-gated. [[packaging-app-plan]] [[next-session-packaging-app]] [[gotcha-shared-checkout-branch-collisions]] [[working-agreement-merging]]

165. assistant (2026-06-22T09:39:52.079Z)

The shared `current-state.md` has moved on (sibling packaging work is now top). Let me insert this session's checkpoint at the top.

166. user (2026-06-22T09:40:03.866Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

167. assistant (2026-06-22T09:40:52.164Z)

Done with the documentation and the protective backup. Here's where things stand.

? Completed this turn

- **F: label backup** ? the 7 single-copy new-clip labels copied + **hash-verified** into the golden archive (at the renamed path `F:\[Khelsutra] GoPro Hero Black 12\KhelSutra\Training\Golden Labelled\`).
- **Authoritative data map** ? docs/DATA_IN_GCS.md, indexed in docs/README.md, committed on `work/golden-corpus-hygiene` (`8884c9f`).
- **Memory** ? new [[gcs-data-source-of-truth]] entry + MEMORY.md pointer + a current-state checkpoint, so future sessions pull from the bucket, not local disks.

The key finding

Your goal is **closer than it looks**: the cloud engine is *already built* (the archived DECOUPLED_COMPUTE project ? `backend.worker infer|train` read/write `gs://`, `deploy/gcp/` provisions it). What's missing is just **populated data + one map**. And the data that actually blocks GPU-free cloud eval is **~25 MB and entirely local** (20 trajectory CSVs = 17.5 MB + a 4.5 KB manifest + labels). Only the videos are large ? and they're on **Google Drive already**, so they migrate **Drive?bucket from a cloud box**, never touching your local machine.

I deliberately did **not** push/PR or upload anything yet ? the bucket layout has real choices you care about (this whole thread has been about organization), so two decisions first:

168. user (2026-06-22T09:57:52.146Z)

Your questions have been answered: "How should golden labels be named in gs://khelsutra-rally-corpus/labels/? (Today it's inconsistent: some files have a _proxy suffix, some don't.)"="Can we not use 1. but also append the "creation time" of that file in a human readable form as a prefix to the filename?", "How should I proceed with the doc and the Phase-1 (~25 MB) metadata migration?"="Push doc PR + run Phase-1 now". You can now continue with these answers in mind.

169. assistant (2026-06-22T09:59:55.619Z)

Got it ? naming = **manifest name with a human-readable creation-date prefix** (e.g. `2026-06-08_GX020094.rallies.csv`), and proceed with push + Phase-1 now. Let me write a migration planner that resolves every clip's trajectory + label (redirecting the original-6 labels to F:), stamps each label with its file date, and builds a `gs://`-pathed manifest ? dry-run first so I can verify before uploading.

170. assistant (2026-06-22T10:00:11.243Z)

"""Plan (and optionally emit) the Phase-1 golden-corpus migration to GCS.

Naming (owner decision 2026-06-22): labels = `<YYYY-MM-DD>_<name>.rallies.csv` where the date is the

```
SOURCE label file's last-modified date (its authoring date). Trajectories = `<name>.csv`
(manifest-name
keyed; the manifest links them). Builds a gs://-pathed eval manifest for cloud consumption.

Dry-run by default: prints the src->dst plan + writes the new manifest + an upload plan to
scratch/.
Nothing is uploaded here ? the PowerShell step runs the verified `gcloud storage cp`s.
"""
import datetime
import json
import os

REPO = r"C:\Users\avidu\Projects\badminton-highlight-indexer"
MANIFEST = os.path.join(REPO, "output", "human_lovo_manifest.json")
```

F: golden-labels archive (original-6 labels live here; C: working copies were removed).

```
F_GOLDEN = r"F:\[Khelsutra] GoPro Hero Black 12\KhelSutra\Training\Golden Labelled"
BUCKET = "gs://khelsutra-rally-corpus"

OUT_MANIFEST = os.path.join(REPO, "scratch", "golden.gcs.json")
OUT_PLAN = os.path.join(REPO, "scratch", "upload_plan.tsv")


def _resolve_label(spec):
    p = str(spec["labels"])
    if os.path.isfile(p):
        return p
    alt = os.path.join(F_GOLDEN, os.path.basename(p))   # original-6 fallback -> F: archive
    return alt if os.path.isfile(alt) else None


def main():
    with open(MANIFEST, encoding="utf-8") as f:
        specs = json.load(f)

    plan = []          # (src, dst) upload pairs
    new_manifest = []
    problems = []

    print(f"{'name':<20} {'date':<10} {'traj':<6} {'label':<6}  label_src")
    for s in specs:
        name = s["name"]
        traj = str(s["trajectory"])
        label = _resolve_label(s)
        traj_ok = os.path.isfile(traj)
        if not traj_ok:
            problems.append(f"MISSING trajectory: {name} -> {traj}")
        if label is None:
            problems.append(f"MISSING label: {name} -> {s['labels']}")
        date = ""
        if label:
            date = datetime.datetime.fromtimestamp(os.path.getmtime(label)).strftime("%Y-%m-%d")
        print(f"{name:<20} {date:<10} {'ok' if traj_ok else 'MISS':<6} "
              f"{'ok' if label else 'MISS':<6}  {label or s['labels']}")

        gcs_traj = f"{BUCKET}/trajectories/{name}.csv"
        gcs_label = f"{BUCKET}/labels/{date}_{name}.rallies.csv" if label else None
        if traj_ok:
            plan.append((traj, gcs_traj))
        if label:
            plan.append((label, gcs_label))
        new_manifest.append({
            "name": name,
            "trajectory": gcs_traj,
            "fps": s["fps"],
```

```
            "frame_width": s["frame_width"],
            "labels": gcs_label,
        })

    with open(OUT_MANIFEST, "w", encoding="utf-8") as f:
        json.dump(new_manifest, f, indent=2)
    with open(OUT_PLAN, "w", encoding="utf-8") as f:
        for src, dst in plan:
            f.write(f"{src}\t{dst}\n")

    print(f"\n{len(plan)} objects to upload ({sum(1 for _,d in plan if '/trajectories/' in d)}
traj, "
          f"{sum(1 for _,d in plan if '/labels/' in d)} labels)")
    print(f"manifest -> {OUT_MANIFEST}  (then gs to {BUCKET}/manifests/golden.json)")
    print(f"plan     -> {OUT_PLAN}")
    if problems:
        print("\n!! PROBLEMS:")
        for p in problems:
            print("  " + p)
    else:
        print("\nall trajectories + labels resolved OK")


if __name__ == "__main__":
    main()
```

171. user (2026-06-22T10:00:11.276Z)

```
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-
indexer\scratch\migrate_golden_to_gcs.py (file state is current in your context ? no need to
Read it back)
```

172. user (2026-06-22T10:00:15.903Z)

```
name                 date       traj   label   label_src
adarsh_avi_singles   2026-06-22 ok     ok      C:/Users/avidu/Projects/Annotation
Setup/Collect/Golden Labelled/Adarsh and Avi.rallies.csv
kushagra_singles     2026-06-22 ok     ok      C:/Users/avidu/Projects/Annotation
Setup/Collect/Golden Labelled/KushagraYashAviFirstVideo.rallies.csv
largetest_doubles    2026-06-22 ok     ok      C:/Users/avidu/Projects/Annotation
Setup/Collect/Golden Labelled/LargeTestVideo.rallies.csv
gbaaddy              2026-06-22 ok     ok      C:/Users/avidu/Projects/Annotation
Setup/Collect/Golden Labelled/gBaaddy20May2026.rallies.csv
testlarge_short      2026-06-22 ok     ok      C:/Users/avidu/Projects/Annotation
Setup/Collect/Golden Labelled/TestLargeVideo.rallies.csv
mahadevpura_singles  2026-06-22 ok     ok      C:/Users/avidu/Projects/Annotation
Setup/Collect/Golden Labelled/MahadevpuraSingles.rallies.csv
mahadevpura_2        2026-06-16 ok     ok      C:/Users/avidu/Projects/badminton-highlight-
indexer/output/mahadevpura_2.rallies.csv
GX010128             2026-06-16 ok     ok      C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX010128.rallies.csv
mahadevpura_1        2026-06-17 ok     ok      C:/Users/avidu/Projects/badminton-highlight-
indexer/output/mahadevpura_1.rallies.csv
GX030094             2026-06-17 ok     ok      C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX030094.rallies.csv
Badminton_BXH_2      2026-06-17 ok     ok      C:/Users/avidu/Projects/badminton-highlight-
indexer/output/Badminton_BXH_2.rallies.csv
GX010141             2026-06-18 ok     ok      C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX010141.rallies.csv
GX010137             2026-06-18 ok     ok      C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX010137.rallies.csv
Boxhill_Doubles      2026-06-18 ok     ok      C:/Users/avidu/Projects/badminton-highlight-
indexer/output/Boxhill_Doubles.rallies.csv
GX020094             2026-06-18 ok     ok      C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX020094.rallies.csv
```

30 objects to upload (15 traj, 15 labels)
manifest -> C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\golden.gcs.json (then
gs to gs://khelsutra-rally-corpus/manifests/golden.json)
plan   -> C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\upload_plan.tsv

all trajectories + labels resolved OK

173. assistant (2026-06-22T10:00:43.687Z)

The original-6 labels resolved on C: (the `Golden Labelled` folder is back), but their date
shows **2026-06-22** ? that's a copy-reset mtime, not their real June-8/9 authoring date (the F:
archive copies have the true dates). To make the date prefix meaningful, I'll use the **earliest
mtime across all known copies** (C: + F:). Let me fix that and re-run.

174. user (2026-06-22T10:00:48.420Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\migrate_golden_to_gcs.py
has been updated successfully. (file state is current in your context ? no need to Read it back)

175. user (2026-06-22T10:00:55.727Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\migrate_golden_to_gcs.py
has been updated successfully. (file state is current in your context ? no need to Read it back)

176. user (2026-06-22T10:00:59.838Z)

```
name                 date       traj   label   label_src
adarsh_avi_singles   2026-06-08 ok     ok      C:/Users/avidu/Projects/Annotation
Setup/Collect/Golden Labelled/Adarsh and Avi.rallies.csv
kushagra_singles     2026-06-08 ok     ok      C:/Users/avidu/Projects/Annotation
Setup/Collect/Golden Labelled/KushagraYashAviFirstVideo.rallies.csv
largetest_doubles    2026-06-08 ok     ok      C:/Users/avidu/Projects/Annotation
Setup/Collect/Golden Labelled/LargeTestVideo.rallies.csv
gbaaddy              2026-06-08 ok     ok      C:/Users/avidu/Projects/Annotation
Setup/Collect/Golden Labelled/gBaaddy20May2026.rallies.csv
testlarge_short      2026-06-08 ok     ok      C:/Users/avidu/Projects/Annotation
Setup/Collect/Golden Labelled/TestLargeVideo.rallies.csv
mahadevpura_singles  2026-06-09 ok     ok      C:/Users/avidu/Projects/Annotation
Setup/Collect/Golden Labelled/MahadevpuraSingles.rallies.csv
mahadevpura_2        2026-06-16 ok     ok      C:/Users/avidu/Projects/badminton-highlight-
indexer/output/mahadevpura_2.rallies.csv
GX010128             2026-06-16 ok     ok      C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX010128.rallies.csv
mahadevpura_1        2026-06-17 ok     ok      C:/Users/avidu/Projects/badminton-highlight-
indexer/output/mahadevpura_1.rallies.csv
GX030094             2026-06-17 ok     ok      C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX030094.rallies.csv
Badminton_BXH_2      2026-06-17 ok     ok      C:/Users/avidu/Projects/badminton-highlight-
indexer/output/Badminton_BXH_2.rallies.csv
GX010141             2026-06-18 ok     ok      C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX010141.rallies.csv
GX010137             2026-06-18 ok     ok      C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX010137.rallies.csv
Boxhill_Doubles      2026-06-18 ok     ok      C:/Users/avidu/Projects/badminton-highlight-
indexer/output/Boxhill_Doubles.rallies.csv
GX020094             2026-06-18 ok     ok      C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX020094.rallies.csv
```

30 objects to upload (15 traj, 15 labels)
manifest -> C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\golden.gcs.json (then
gs to gs://khelsutra-rally-corpus/manifests/golden.json)
plan   -> C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\upload_plan.tsv

all trajectories + labels resolved OK

177. assistant (2026-06-22T10:01:11.632Z)

Dates are now meaningful (original-6 = Jun 8?9, new-9 = Jun 16?18), all 15 trajectories + 15 labels resolved. Let me verify the generated manifest, then execute the uploads.

178. user (2026-06-22T10:01:12.458Z)


```
[
  {
    "name": "adarsh_avi_singles",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/adarsh_avi_singles.csv",
    "fps": 59.94,
    "frame_width": 1920,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-08_adarsh_avi_singles.rallies.csv"
  },
  {
    "name": "kushagra_singles",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/kushagra_singles.csv",
    "fps": 59.94,
    "frame_width": 1920,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-08_kushagra_singles.rallies.csv"
  },
  {
    "name": "largetest_doubles",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/largetest_doubles.csv",
    "fps": 29.97,
    "frame_width": 1920,
```


179. assistant (2026-06-22T10:01:25.389Z)

Manifest is correct. Executing the 30 uploads + the manifest now.

180. user (2026-06-22T10:04:18.839Z)

```
uploaded: ok=30 fail=0 (expected 30)
manifest uploaded -> gs://khelsutra-rally-corpus/manifests/golden.json
```

181. assistant (2026-06-22T10:04:47.925Z)

All 30 + manifest uploaded. Let me verify the bucket state (and check the pre-existing inconsistent label objects, which now coexist with the new canonical ones).

182. user (2026-06-22T10:04:57.609Z)

```
=== trajectories/ (expect 15) ===
15
=== manifests/ ===
gs://khelsutra-rally-corpus/manifests/golden.json
=== labels/ (NEW canonical + OLD pre-existing) ===
gs://khelsutra-rally-corpus/labels/2026-06-08_adarsh_avi_singles.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-08_gbaaddy.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-08_kushagra_singles.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-08_largetest_doubles.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-08_testlarge_short.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-09_mahadevpura_singles.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-16_GX010128.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-16_mahadevpura_2.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-17_Badminton_BXH_2.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-17_GX030094.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-17_mahadevpura_1.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-18_Boxhill_Doubles.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-18_GX010137.rallies.csv
```

gs://khelsutra-rally-corpus/labels/2026-06-18_GX010141.rallies.csv
gs://khelsutra-rally-corpus/labels/2026-06-18_GX020094.rallies.csv
gs://khelsutra-rally-corpus/labels/Badminton BXH 2_proxy.rallies.csv
gs://khelsutra-rally-corpus/labels/Boxhill Doubles_proxy.rallies.csv
gs://khelsutra-rally-corpus/labels/GX010128.rallies.csv
gs://khelsutra-rally-corpus/labels/GX020094_proxy.rallies.csv
gs://khelsutra-rally-corpus/labels/GX030094_proxy.rallies.csv
gs://khelsutra-rally-corpus/labels/mahadevpura-1.rallies.csv
gs://khelsutra-rally-corpus/labels/mahadevpura-2_proxy.rallies.csv

183. assistant (2026-06-22T10:05:33.585Z)

Phase-1 landed cleanly: **15 trajectories + 15 canonical timestamped labels + `manifests/golden.json`** are in the bucket. One thing to flag: the **7 old inconsistent label objects** (`*_proxy.rallies.csv`, bare `GX010128`/`mahadevpura-1`) now coexist with the new canonical ones ? they're superseded and would confuse `worker train` (which globs `labels/`). I'll flag those for cleanup rather than blind-delete (the `_proxy` provenance differs, and bucket deletes are destructive).

Let me update the doc to lock the decisions and mark Phase-1 done.

184. user (2026-06-22T10:05:48.226Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DATA_IN_GCS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

185. user (2026-06-22T10:05:57.391Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DATA_IN_GCS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

186. user (2026-06-22T10:06:10.894Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DATA_IN_GCS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

187. assistant (2026-06-22T10:06:30.429Z)

Let me refine the script reference (it lives in gitignored `scratch/`, so I'll describe the method rather than imply a committed tool), then commit and push.

188. user (2026-06-22T10:06:33.425Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DATA_IN_GCS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

189. user (2026-06-22T10:06:44.230Z)

```
=== re-verify HEAD ===
work/golden-corpus-hygiene
8884c9f docs(data): add DATA_IN_GCS.md ? the corpus source-of-truth map (GCS)
warning: in the working copy of 'docs/DATA_IN_GCS.md', LF will be replaced by CRLF the next time
Git touches it
[work/golden-corpus-hygiene 6ae3503] docs(data): record Phase-1 GCS migration (trajectories +
labels + manifest)
 1 file changed, 32 insertions(+), 10 deletions(-)
=== branch vs master ===
6ae3503 docs(data): record Phase-1 GCS migration (trajectories + labels + manifest)
8884c9f docs(data): add DATA_IN_GCS.md ? the corpus source-of-truth map (GCS)
```

190. assistant (2026-06-22T10:07:00.473Z)

Two commits, both mine. Pushing and opening the PR.

191. user (2026-06-22T10:07:03.768Z)

remote: https://github.com/Khelsutra/badminton-highlight-indexer/pull/new/work/golden-corpus-hygiene
remote:
branch 'work/golden-corpus-hygiene' set up to track 'origin/work/golden-corpus-hygiene'.
To https://github.com/Khelsutra/badminton-highlight-indexer.git
 * [new branch]   work/golden-corpus-hygiene -> work/golden-corpus-hygiene

192. user (2026-06-22T10:07:26.866Z)

https://github.com/Khelsutra/badminton-highlight-indexer/pull/316

193. assistant (2026-06-22T10:07:35.192Z)

PR opened: [badminton-highlight-indexer#316](https://github.com/Khelsutra/badminton-highlight-indexer/pull/316). Let me update the current-state memory to reflect Phase-1 done + the PR (re-reading first since it's a shared file).

194. user (2026-06-22T10:07:36.123Z)

14   **Checkpoint:** 2026-06-22 (**khelsutra brand-repo HYGIENE + DOC-FRESHNESS SWEEP ? 2 PRs merged, clean slate**). Session = "sync to remote head + low-hanging fruit + quick doc sweep." khelsutra was at master `e2570e4` (#93). ? **[#94] hygiene** ? `.gitignore` now covers `*.db` (runtime `site/access.db`) + `.claude/` (machine-specific; both were untracked ? committable) + corrected a stale NEXT_STEPS PR count; **caught my own `.gitignore` inline-comment bug via `git check-ignore` BEFORE commit** (`.gitignore` has no trailing-comment syntax ? comments must be own-line). ? **[#95] doc-freshness sweep** ? a **15-agent AUDIT** (one verifier/doc, every falsifiable claim checked at file:line incl. cross-repo vs engine `origin/master`) found **44 stale findings** (the pattern: docs froze at their authoring-PR while the SPA / engine API / localization / alpha-deploy shipped past them) ? a **12-agent FIXER fan-out** applied **63 surgical edits across 13 docs**; I **independently re-verified every load-bearing engine claim** on engine `origin/master` before merging (`/api/{health:308,capabilities:340,videos:374}` + compile `download_url:768` + `native_wasb_runner.py` on master + `auth.py::CF_ACCESS_HEADER` + `_allowed_hosts_from_env` REPLACE-semantics ? all real; brittle engine line-anchors ? symbol refs like `main.py::compile_highlights`). Honesty held (P6 stays `?` engine-done/SPA-pending, native-WASB "not production-verified", footage shots owner-gated, historical changelogs preserved; the 2 `fresh` docs untouched). Also pruned **6 stale merged local branches**. **All CI green (site+web+layout); 0 open PRs; only `master` locally; khelsutra master now `7dc7de5`.** *(Brand/site track ? distinct from the serving/packaging/UX/golden threads below.)* [[lesson-verify-engine-surface-before-scoping]] [[working-agreement-merging]] [[doc-status-register]]
15
16   **Checkpoint:** 2026-06-22 (GOLDEN-FIXTURES **P0 ? #314 MERGED**, master `5c2eec8`; a SEPARATE thread from packaging/UX). Owner: "all owned golden videos are owned+minor-free; pick (a); make P0 now." On scouting, P0 was bigger/riskier than the audit's "4 files": the video names are FUNCTIONAL (eval_baselines keys ? training floor + the gitignored `output/human_lovo_manifest.json` + data files; `MULTICOURT_CLIPS`; hot in QUALITY_ITERATIONS/the quality track) ? surfaced it ? owner picked **SAFE SCOPE**. Shipped: scrubbed personal `C:\Users\avidu` paths from 3 code docstrings + **`tools/leak_scan.py`** (structural PII guard: 32-hex MD5 + personal paths over backend/training/scripts/tools/eval_baselines; tests/ excluded; optional gitignored `.leakscan_private` for local name scanning) + CI `leak-scan` job + 10 tests. Clean scan; suite **1365?/7s**, cov **85.27%**, all CI green. **DEFERRED (safe-scope):** the functional video-name rename + git-history rewrite (need manifest/data/training-floor coordination + quality-track pause; do before open-sourcing). ?? **NEXT = P3 real Tier-0 fixtures** ? owner APPROVED (all owned cleared + **(a)** commit de-attributed per-frame CSV); needs the golden **trajectory CSVs + label CSVs** (`Annotation Setup/Collect/Trajectories/` + `Golden Labelled/` / OneDrive `golden-data/`) ? I locate them or owner points me, then `--freeze-real` per video. Golden-fixtures track = #186/#189/#191/#192/#193/#194 + #314, all merged+audited. [[golden-set-vendoring]]
17
18   **Checkpoint:** 2026-06-22 (? **GCS = corpus source-of-truth ? `docs/DATA_IN_GCS.md` shipped; F: label backup done; 3 owner decisions open**). Owner goal: run ALL train+infer from the cloud with ALL data in `gs://khelsutra-rally-corpus` ? remove the local box as a blocker;

document where data lives so other sessions pick up from the bucket. [[gcs-data-source-of-truth]]
19   - ? **Authoritative map `docs/DATA_IN_GCS.md`** (indexed in docs/README.md; branch `work/golden-corpus-hygiene`, commit `8884c9f`): bucket layout · current contents + **drift** · how `backend.worker infer|train` consume `gs://` · the gap · phased migration plan + DoD. The **cloud engine already exists** (archived DECOUPLED_COMPUTE: `worker infer --video-uri` / `worker train --corpus-uri`, URI backends in `backend/storage/ref.py`, `deploy/gcp/`) ? the doc records WHERE bytes are, it does NOT redesign.
20   - ? **Gap measured:** corpus *metadata* is **~25 MB & entirely local** ? 20 trajectory CSVs (17.5 MB at `C:\?\Collect\Trajectories\`), `output/human_lovo_manifest.json` (C:-absolute paths), most labels. Bucket has only 3 videos, **no `trajectories/`**, inconsistent `_proxy` label naming. Only the VIDEOS are large ? migrate **Drive?bucket from a cloud box**, not via the local box.

195. user (2026-06-22T10:07:48.257Z)

18   **Checkpoint:** 2026-06-22 (? **GCS = corpus source-of-truth ? `docs/DATA_IN_GCS.md` shipped; F: label backup done; 3 owner decisions open**). Owner goal: run ALL train+infer from the cloud with ALL data in `gs://khelsutra-rally-corpus` ? remove the local box as a blocker; document where data lives so other sessions pick up from the bucket. [[gcs-data-source-of-truth]]
19   - ? **Authoritative map `docs/DATA_IN_GCS.md`** (indexed in docs/README.md; branch `work/golden-corpus-hygiene`, commit `8884c9f`): bucket layout · current contents + **drift** · how `backend.worker infer|train` consume `gs://` · the gap · phased migration plan + DoD. The **cloud engine already exists** (archived DECOUPLED_COMPUTE: `worker infer --video-uri` / `worker train --corpus-uri`, URI backends in `backend/storage/ref.py`, `deploy/gcp/`) ? the doc records WHERE bytes are, it does NOT redesign.
20   - ? **Gap measured:** corpus *metadata* is **~25 MB & entirely local** ? 20 trajectory CSVs (17.5 MB at `C:\?\Collect\Trajectories\`), `output/human_lovo_manifest.json` (C:-absolute paths), most labels. Bucket has only 3 videos, **no `trajectories/`**, inconsistent `_proxy` label naming. Only the VIDEOS are large ? migrate **Drive?bucket from a cloud box**, not via the local box.
21   - ? **F: backup done:** the 7 single-copy new-clip labels hash-verified into `F:\[Khelsutra] GoPro Hero Black 12\KhelSutra\Training\Golden Labelled\`. ? F: root was RENAMED `GoPro Hero Black 12`?`[Khelsutra] GoPro Hero Black 12` between turns ? live proof of the path-fragility GCS fixes.
22   - ? **OPEN owner decisions (DATA_IN_GCS §7) before Phase-1 upload:** (1) `labels/` naming (recycled GoPro names ? key by `video_id`/MD5?); (2) cache `trajectories/` in the bucket for GPU-free eval?; (3) a bucket-relative eval manifest? On answers I upload trajectories+labels+manifest (~25 MB, seconds).
23   - ? Branch `work/golden-corpus-hygiene` off origin/master (`689c738`), 1 commit, **NOT pushed/PR'd yet** (awaiting owner). Serving WIP `6fdd1d4` + `scratch/PACKAGING_PLAN.serving-wip.md` preserved untouched. Prior #236 served-Gate-A re-grounding MERGED 2026-06-19; original-6 labels turned out **recoverable from F:** (not lost) ? optional follow-up: re-measure litmus on the full 15-clip corpus. [[compute-stack-gcp-only]] [[gotcha-shared-checkout-branch-collisions]]

196. user (2026-06-22T10:08:01.934Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

197. user (2026-06-22T10:08:21.244Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

198. assistant (2026-06-22T10:08:46.263Z)

All done. Here's the complete picture.

? Shipped this turn

1. Protective backup ? the 7 single-copy new-clip labels are hash-verified into the F: golden archive (at the renamed `F:\[Khelsutra] GoPro Hero Black 12\?\Golden Labelled\`).

2. The authoritative data map ? docs/DATA_IN_GCS.md (indexed in docs/README.md): bucket layout, current contents + drift, how `backend.worker infer|train` consume `gs://`, the gap, and the migration plan. **Other sessions now have one place that says where data lives.**

3. Phase-1 migration executed ? the corpus metadata that blocked GPU-free cloud eval is now in `gs://khelsutra-rally-corpus`:
Prefix	Uploaded
`trajectories/<name>.csv`	**15** (was 0)
`labels/<YYYY-MM-DD>_<name>.rallies.csv`	**15** canonical (your naming: manifest name + authoring-date prefix, using the *earliest* mtime across C:/F: copies so restored-copy resets don't lie)
`manifests/golden.json`	gs://-pathed eval manifest, 15 clips

4. PR + memory ? [badminton-highlight-indexer#316](https://github.com/Khelsutra/badminton-highlight-indexer/pull/316) is open for your review (no self-merge). Memory updated ([[gcs-data-source-of-truth]] + MEMORY.md + current-state) so future sessions pull from the bucket.

The key reframe

Your goal is mostly **already built** ? the archived DECOUPLED_COMPUTE project gave you `worker infer|train` over `gs://` and `deploy/gcp/` provisioning. The blocker was just *unpopulated data + no map*. Phase-1 closes the metadata gap; the local box is no longer needed for cloud **eval/training-data**.

A nice side-effect: the golden trajectory + label CSVs that a **separate session's golden-fixtures P3 thread** was hunting for are now sitting in GCS.

What's left (tracked in the doc, your call)

1. **Cleanup:** 7 stale pre-existing label objects (`*_proxy`, bare `GX010128`/`mahadevpura-1`) now coexist with the canonical names and would double-count in `worker train`'s glob ? I left them rather than blind-delete (different `_proxy` provenance). I can verify they're dupes and remove them.
2. **Phase 2 ? videos** (the only large migration): Drive ? bucket **from a cloud box**, proxies-first ? keeps your local machine out of the path entirely.
3. **Phase 3 ? flip the eval/litmus** to read `manifests/golden.json` by default so a fresh box/CI needs zero local corpus.

Want me to do the stale-label cleanup (#1) now, and/or set up the Phase-2 Drive?bucket video copy?

199. user (2026-06-22T10:12:40.021Z)

yes pleae continue with #1 and setup the Phase 2 Drive -> bucket video copy as well.. and bring this session to a great conclusion with a note about what to lookout for in the next sessio

200. assistant (2026-06-22T10:14:12.589Z)

The user wants both operational tasks done plus a strong session close. Note on approach: for the **destructive** label deletion, ultracode's "optimize for correctness" means a *deterministic byte/hash comparison* ? which is strictly more reliable than delegating byte-equality judgment to LLM agents ? so I verify those directly. I'll reserve a multi-agent **workflow for the adversarial completeness review** of the whole migration before I conclude, where independent perspectives genuinely raise confidence.

Let me start with two read-only investigations in parallel: hash-compare the 7 stale labels vs their canonical counterparts (to decide safe deletes), and check Drive accessibility + the golden folder structure (for Phase 2).

201. user (2026-06-22T10:14:45.887Z)

```
OLD object                           verdict                             (md5 old vs canonical)
GX010128.rallies.csv                 IDENTICAL -> safe delete   c4c9001c8fc4b12180b3f6c68874f1ee
/ c4c9001c8fc4b12180b3f6c68874f1ee
mahadevpura-1.rallies.csv            IDENTICAL -> safe delete   9e04ae52caca4408e25b145c0c4adb3a
/ 9e04ae52caca4408e25b145c0c4adb3a
GX020094_proxy.rallies.csv           IDENTICAL -> safe delete   6a7fa761040fe50ffe07d6dcb52f9573
/ 6a7fa761040fe50ffe07d6dcb52f9573
GX030094_proxy.rallies.csv           IDENTICAL -> safe delete   3a694b1ad18ed832fe5f1dd71a79af31
/ 3a694b1ad18ed832fe5f1dd71a79af31
Badminton BXH 2_proxy.rallies.csv    IDENTICAL -> safe delete   34140416abc713b757b9aba1384d9aff
/ 34140416abc713b757b9aba1384d9aff
Boxhill Doubles_proxy.rallies.csv    IDENTICAL -> safe delete   54968c8d2ba9413ec6d78ebacc26e358
/ 54968c8d2ba9413ec6d78ebacc26e358
mahadevpura-2_proxy.rallies.csv      IDENTICAL -> safe delete   357520405b1d8e94d81f87900360bb09
/ 357520405b1d8e94d81f87900360bb09
```

202. user (2026-06-22T10:14:49.471Z)

```
=== Drive mounted? ===
G:\ -> True
G:\My Drive -> True
golden folder -> True
--- contents ---
[Done] Video 10 - Badminton Multicourt
[Done] Video 12 - Badminton Multicourt
[Done] Video 13 - Badminton
[Done] Video 14 - Badminton
[Done] Video 3 - Badminton Multicourt
[Done] Video 4 - Badminton
[Done] Video 6 - Badminton
Video 1 - Badminton
Video 11 - Pickleball
Video 2 - Tabletennis
Video 5 - Pickleball
Video 7 - Badminton Multicourt
Video 8 - Table tennis
Video 9 - Badminton
7 ? 1 ? 9.txt
=== bucket videos/ today ===
gs://khelsutra-rally-corpus/videos/GX030094_proxy.mp4
gs://khelsutra-rally-corpus/videos/mahadevpura-1.MP4
gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4
=== F: proxies (KhelSutra_work) ===
F:\KhelSutra_work\19june_proxies ->
   GX010142_1080p.mp4  732MB
   GX010143_1080p.mp4  604MB
   GX010144_1080p.mp4  1,125MB
   GX010145_1080p.mp4  792MB
   GX010147_1080p.mp4  629MB
F:\KhelSutra_work\kushagra_june12_proxies ->
   GX010128.MP4  2,234MB
   GX010129.MP4  1,636MB
   GX020125.MP4  879MB
   mahadevpura-0.MP4  2,250MB
   mahadevpura-1.MP4  523MB
   mahadevpura-1.rallies.csv  0MB
   mahadevpura-2.MP4  1,853MB
```

203. assistant (2026-06-22T10:16:02.537Z)

```
Both investigations are conclusive:
- **All 7 stale labels are byte-identical (MD5) to their canonical counterparts** ? safe to
delete (content also preserved on C: and F:).
- **Google Drive is mounted** (`G:\My Drive`), and the golden folder has the `[Done] Video N`
structure ? the Phase-2 source.
```

Let me delete the 7 verified dupes (#1) and inspect the Drive golden folders to build the Phase-2 video?bucket mapping.

204. user (2026-06-22T10:16:44.826Z)

```
deleted: GX010128.rallies.csv
deleted: mahadevpura-1.rallies.csv
deleted: GX020094_proxy.rallies.csv
deleted: GX030094_proxy.rallies.csv
deleted: Badminton BXH 2_proxy.rallies.csv
deleted: Boxhill Doubles_proxy.rallies.csv
deleted: mahadevpura-2_proxy.rallies.csv

=== labels/ after cleanup (expect 15 canonical only) ===
15
```

205. user (2026-06-22T10:16:46.143Z)

```
=== [Done] Video 10 - Badminton Multicourt ===
   Badminton BXH 2_proxy.mp4          3,491 MB
   Badminton BXH 2_proxy.rallies.csv      0 MB
=== [Done] Video 12 - Badminton Multicourt ===
   Boxhill Doubles_proxy.mp4          1,482 MB
   Boxhill Doubles_proxy.rallies.csv      0 MB
=== [Done] Video 13 - Badminton ===
   GX010128.MP4                       2,234 MB
   GX010128.rallies.csv                   0 MB
=== [Done] Video 14 - Badminton ===
   mahadevpura-1.MP4                    523 MB
   mahadevpura-1.rallies.csv              0 MB
=== [Done] Video 3 - Badminton Multicourt ===
   mahadevpura-2_proxy.mp4            2,949 MB
   mahadevpura-2_proxy.rallies.csv        0 MB
=== [Done] Video 4 - Badminton ===
   GX020094_proxy.mp4                 1,482 MB
   GX020094_proxy.rallies.csv             0 MB
=== [Done] Video 6 - Badminton ===
   GX030094_proxy.mp4                 1,364 MB
   GX030094_proxy.rallies.csv             0 MB
=== Video 1 - Badminton ===
   GX010135_proxy.mp4                 3,682 MB
=== Video 11 - Pickleball ===
   Pickleball ELT 1_proxy.mp4         3,681 MB
=== Video 2 - Tabletennis ===
   GX010124_proxy.mp4                 7,327 MB
=== Video 5 - Pickleball ===
   GX020130_proxy.mp4                 3,285 MB
=== Video 7 - Badminton Multicourt ===
   GX010055_proxy.mp4                 6,525 MB
=== Video 8 - Table tennis ===
   GX010104_proxy.mp4                 3,367 MB
=== Video 9 - Badminton ===
   GX010132_proxy.mp4                 3,681 MB
```

206. assistant (2026-06-22T10:18:52.595Z)

#1 done ? 7 verified dupes deleted; `labels/` now holds exactly the 15 canonical objects.

For **Phase 2**, the Drive golden folder gives me the source mapping: 7 of the new clips have proxies in `[Done] Video N` folders (~13 GB total), GX010137/141 + the original-6 are large full MP4s on F:. Critically ? copying through this box's Drive mount uses the *local box*, which is exactly the blocker we're removing, so the **correct** Phase-2 is `rclone` on a cloud VM (Drive?GCS, no local box). I'll script + document both paths and demonstrate the mechanism with one small file.

Let me write the copy script with the full mapping.

207. assistant (2026-06-22T10:19:12.890Z)

Phase-2: copy golden-clip videos -> gs://khelsutra-rally-corpus/videos/<name>.mp4

Source = Google Drive golden folder (proxies, preferred) + F: archive (full MP4s).

Bucket object name = the eval/manifest `name` (consistent with trajectories/ + labels/)

#

Usage:

pwsh copy_videos_to_gcs.ps1 # DRY-RUN: print the plan + per-clip source

pwsh copy_videos_to_gcs.ps1 -Execute # copy everything missing (skip-existin

pwsh copy_videos_to_gcs.ps1 -Execute -Only mahadevpura_1 # copy ONE clip

#

NOTE: this runs from the OWNER box (Drive mounted G:, F: attached) and streams Dr

The decoupled path (no local box) = rclone on a cloud VM: `rclone copy gdrive:"<path

```
param([switch]$Execute, [string]$Only)

$BUCKET = "gs://khelsutra-rally-corpus/videos"
$DRIVE  = 'G:\My Drive\Sharing\Annotation Videos\Golden Label Videos Request - June 15'
$FROOT  = 'F:\[Khelsutra] GoPro Hero Black 12\KhelSutra'
```

name -> source path. proxy where available; * = large full MP4 (proxy-first later).

```
$src = [ordered]@{
  # --- Drive golden proxies (preferred) ---
  'mahadevpura_2'       = "$DRIVE\[Done] Video 3 - Badminton Multicourt\mahadevpura-2_proxy.mp4"
  'GX020094'            = "$DRIVE\[Done] Video 4 - Badminton\GX020094_proxy.mp4"
  'GX030094'            = "$DRIVE\[Done] Video 6 - Badminton\GX030094_proxy.mp4"
  'Badminton_BXH_2'     = "$DRIVE\[Done] Video 10 - Badminton Multicourt\Badminton BXH
2_proxy.mp4"
  'Boxhill_Doubles'     = "$DRIVE\[Done] Video 12 - Badminton Multicourt\Boxhill
Doubles_proxy.mp4"
  'GX010128'            = "$DRIVE\[Done] Video 13 - Badminton\GX010128.MP4"
  'mahadevpura_1'       = "$DRIVE\[Done] Video 14 - Badminton\mahadevpura-1.MP4"
  # --- F: full MP4s (large; consider proxy-first before upload) ---
  'GX010137'            = "$FROOT\Badminton - Boxhill - 17 June\GX010137.MP4"          # *6.4GB
  'GX010141'            = "$FROOT\Badminton - Boxhill - 17 June\GX010141.MP4"          # *5.5GB
  'adarsh_avi_singles'  = "$FROOT\Training\Golden Labelled\Adarsh and Avi.MP4"          #
*10.9GB
  'kushagra_singles'    = "$FROOT\Training\Golden Labelled\KushagraYashAviFirstVideo.MP4" #
*7.7GB
  'largetest_doubles'   = "$FROOT\Training\Golden Labelled\LargeTestVideo.MP4"          # *11GB
  'gbaaddy'             = "$FROOT\Training\Golden Labelled\gBaaddy20May2026.MP4"        # *6.7GB
  'testlarge_short'     = "$FROOT\Training\Golden Labelled\TestLargeVideo.MP4"          # *1.4GB
  'mahadevpura_singles' = "$FROOT\Training\Golden Labelled\MahadevpuraSingles.MP4"      # 187MB
}
```

which dests already exist in the bucket (skip those)

```
$have = @{}
gcloud storage ls "$BUCKET/" 2>$null | ForEach-Object { $have[[IO.Path]::GetFileName($_)] =
$true }

$copied = 0; $skipped = 0; $missing = 0
foreach ($name in $src.Keys) {
  if ($Only -and $name -ne $Only) { continue }
  $s = $src[$name]; $dst = "$BUCKET/$name.mp4"; $destName = "$name.mp4"
  $exists = Test-Path -LiteralPath $s
  $sizeMB = if ($exists) { [math]::Round((Get-Item -LiteralPath $s).Length/1MB) } else { 0 }
```

```
  if (-not $exists) { "MISSING SRC  {0,-20} {1}" -f $name, $s; $missing++; continue }
  if ($have[$destName]) { "skip(exists) {0,-20} ({1} MB)" -f $name, $sizeMB; $skipped++;
continue }
  if (-not $Execute) { "WOULD COPY   {0,-20} ({1} MB)  <- {2}" -f $name, $sizeMB, $s; continue }
  "COPYING      {0,-20} ({1} MB) ..." -f $name, $sizeMB
  gcloud storage cp $s $dst 2>$null
  if ($LASTEXITCODE -eq 0) { "  OK -> $dst"; $copied++ } else { "  FAIL -> $dst" }
}
"`nsummary: copied=$copied skipped=$skipped missing_src=$missing  (Execute=$Execute
Only='$Only')"
```

208. user (2026-06-22T10:19:12.916Z)

```
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-
indexer\scratch\copy_videos_to_gcs.ps1 (file state is current in your context ? no need to Read
it back)
```

209. assistant (2026-06-22T10:19:17.726Z)

```
Let me dry-run the mapping to confirm every source resolves, then kick off the demo copy
(smallest Drive proxy) in the background.
```

210. user (2026-06-22T10:19:22.001Z)

```
WOULD COPY   mahadevpura_2        (2949 MB)  <- G:\My Drive\Sharing\Annotation Videos\Golden
Label Videos Request - June 15\[Done] Video 3 - Badminton Multicourt\mahadevpura-2_proxy.mp4
WOULD COPY   GX020094             (1482 MB)  <- G:\My Drive\Sharing\Annotation Videos\Golden
Label Videos Request - June 15\[Done] Video 4 - Badminton\GX020094_proxy.mp4
WOULD COPY   GX030094             (1364 MB)  <- G:\My Drive\Sharing\Annotation Videos\Golden
Label Videos Request - June 15\[Done] Video 6 - Badminton\GX030094_proxy.mp4
WOULD COPY   Badminton_BXH_2      (3491 MB)  <- G:\My Drive\Sharing\Annotation Videos\Golden
Label Videos Request - June 15\[Done] Video 10 - Badminton Multicourt\Badminton BXH 2_proxy.mp4
WOULD COPY   Boxhill_Doubles      (1482 MB)  <- G:\My Drive\Sharing\Annotation Videos\Golden
Label Videos Request - June 15\[Done] Video 12 - Badminton Multicourt\Boxhill Doubles_proxy.mp4
WOULD COPY   GX010128             (2234 MB)  <- G:\My Drive\Sharing\Annotation Videos\Golden
Label Videos Request - June 15\[Done] Video 13 - Badminton\GX010128.MP4
WOULD COPY   mahadevpura_1        (523 MB)  <- G:\My Drive\Sharing\Annotation Videos\Golden
Label Videos Request - June 15\[Done] Video 14 - Badminton\mahadevpura-1.MP4
WOULD COPY   GX010137             (6449 MB)  <- F:\[Khelsutra] GoPro Hero Black
12\KhelSutra\Badminton - Boxhill - 17 June\GX010137.MP4
WOULD COPY   GX010141             (5505 MB)  <- F:\[Khelsutra] GoPro Hero Black
12\KhelSutra\Badminton - Boxhill - 17 June\GX010141.MP4
WOULD COPY   adarsh_avi_singles   (10915 MB)  <- F:\[Khelsutra] GoPro Hero Black
12\KhelSutra\Training\Golden Labelled\Adarsh and Avi.MP4
WOULD COPY   kushagra_singles     (7685 MB)  <- F:\[Khelsutra] GoPro Hero Black
12\KhelSutra\Training\Golden Labelled\KushagraYashAviFirstVideo.MP4
WOULD COPY   largetest_doubles    (10998 MB)  <- F:\[Khelsutra] GoPro Hero Black
12\KhelSutra\Training\Golden Labelled\LargeTestVideo.MP4
WOULD COPY   gbaaddy              (6747 MB)  <- F:\[Khelsutra] GoPro Hero Black
12\KhelSutra\Training\Golden Labelled\gBaaddy20May2026.MP4
WOULD COPY   testlarge_short      (1440 MB)  <- F:\[Khelsutra] GoPro Hero Black
12\KhelSutra\Training\Golden Labelled\TestLargeVideo.MP4
WOULD COPY   mahadevpura_singles  (187 MB)  <- F:\[Khelsutra] GoPro Hero Black
12\KhelSutra\Training\Golden Labelled\MahadevpuraSingles.MP4

summary: copied=0 skipped=0 missing_src=0  (Execute=False Only='')
```

211. assistant (2026-06-22T10:19:29.907Z)

```
All 15 sources resolve (~70 GB total: 7 Drive proxies ?13.5 GB + 8 large F: originals ?50 GB).
Let me kick off the demo copy (smallest Drive proxy, 523 MB) in the background to prove the
Drive?bucket mechanism + canonical naming, then document Phase 2 while it runs.
```

212. user (2026-06-22T10:19:31.795Z)

```
Command running in background with ID: b8q4kcp1v. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\78dabe12-41fd-46c2-ae90-ab74bdaec3c6\tasks\b8q4kcp1v.output. You will be notified when
it completes. To check interim output, use Read on that file path.
```

213. assistant (2026-06-22T10:19:45.288Z)

```
Demo copy is running in the background. Now let me document Phase 2 fully (mapping + both run
paths) and record the #1 cleanup in the doc.
```

214. user (2026-06-22T10:19:57.819Z)

```
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DATA_IN_GCS.md has been
updated successfully. (file state is current in your context ? no need to Read it back)
```

215. assistant (2026-06-22T10:20:02.450Z)

```
The demo copy completed. Let me confirm it landed correctly in the bucket.
```

216. user (2026-06-22T10:20:04.043Z)

```
COPYING      mahadevpura_1        (523 MB) ...
  FAIL -> gs://khelsutra-rally-corpus/videos/mahadevpura_1.mp4

summary: copied=0 skipped=0 missing_src=0  (Execute=True Only='mahadevpura_1')
```

217. user (2026-06-22T10:20:08.205Z)

```
=== videos/ now ===
1429944539  2026-06-20T14:27:35Z  gs://khelsutra-rally-corpus/videos/GX030094_proxy.mp4
 548827049  2026-06-20T14:26:23Z  gs://khelsutra-rally-corpus/videos/mahadevpura-1.MP4
  90194321  2026-06-20T14:20:32Z  gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4
TOTAL: 3 objects, 2068965909 bytes (1.93GiB)
```

218. assistant (2026-06-22T10:20:20.992Z)

```
The demo copy **failed** (the script suppressed the error). Let me re-run that one copy with the
error visible to diagnose.
```

219. user (2026-06-22T10:20:29.546Z)

```
Exit code 1
src exists (LiteralPath): True
src size: 523 MB
--- attempting copy (stderr visible) ---

...........
ERROR: (gcloud.storage.cp) The following URLs matched no objects or files:
G:\My Drive\Sharing\Annotation Videos\Golden Label Videos Request - June 15\[Done] Video
14 - Badminton\mahadevpura-1.MP4
```

220. assistant (2026-06-22T10:21:07.864Z)

```
Found it ? `gcloud storage cp` treats `[...]` as a **glob/character-class**, and every source
path has brackets (`[Done] Video N` on Drive, `[Khelsutra]` on F:), so gcloud can't match them.
This is yet another reason the cloud-native `rclone` path is the right Phase-2 answer. For the
interim owner-box script, I'll stage bracketed sources through a temp file. Let me fix the
script and re-run the demo.
```

221. user (2026-06-22T10:21:14.787Z)

```
The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\copy_videos_to_gcs.ps1 has
been updated successfully. (file state is current in your context ? no need to Read it back)
```

222. user (2026-06-22T10:21:20.449Z)

Command running in background with ID: begcvcrr9. Output is being written to: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\78dabe12-41fd-46c2-ae90-ab74bdaec3c6\tasks\begcvcrr9.output. You will be notified when it completes. To check interim output, use Read on that file path.

223. assistant (2026-06-22T10:21:32.468Z)

Re-running with the fix in the background. While it copies, let me record the #1 cleanup and the Phase-2 status in the doc.

224. user (2026-06-22T10:21:37.831Z)

<tool_use_error>String to replace not found in file.
String: > ? **Cleanup pending:** 7 pre-existing inconsistent label objects (`*_proxy.rallies.csv`, bare
> `GX010128.rallies.csv`, `mahadevpura-1.rallies.csv`) still sit in `labels/` alongside the new
> canonical names ? they're superseded and would double-count in `worker train`'s `labels/` glob.
> Verify they're dupes, then delete.
(note: Edit also tried swapping \uXXXX escapes and their characters; neither form matched, so the mismatch is likely elsewhere in old_string. Re-read the file and copy the exact surrounding text.)</tool_use_error>

225. assistant (2026-06-22T10:21:42.549Z)

Let me re-read the §7 area to get the exact current text.

226. user (2026-06-22T10:21:43.253Z)

100
101 A bare/relative path resolves under a configured `input_root`/`input_backend`
102 (`backend/storage/ref.py::resolve_input_uri`), so workflows can be written bucket-relative.
103
104 ---
105
106 ## 5. The gap ? what's still local-only (why the box is still a blocker)
107
108 All of this is **tiny except the originals** ? the metadata that blocks cloud eval is < 25 MB total:
109
110 | Data | Where it is now | Size | In bucket? | Needed for |
111 |---|---|---|---|---|
112 | 20 WASB trajectory CSVs | `C:\?\Annotation Setup\Collect\Trajectories\` | **17.5 MB** | ? none | eval/litmus (cached), train (`--trajectory-source cached`) |
113 | golden manifest | `output/human_lovo_manifest.json` (C:-absolute paths) | 4.5 KB | ? | eval corpus definition |
114 | original-6 labels | `F:\[Khelsutra]?\Training\Golden Labelled\*.rallies.csv` | ~12 KB | ? | train + eval |
115 | 7 new-clip labels | repo `output/*.rallies.csv` (gitignored) **+** F: archive (backed up 2026-06-22) | ~12 KB | ? partial/inconsistent | train + eval |
116 | source videos / proxies | F: archive + Google Drive golden folder | **100s of GB** | ? 3 only | infer + train (real WASB extraction) |
117
118 > The corpus *metadata* (trajectories + labels + manifest, ~25 MB) is what actually blocks GPU-free
119 > cloud eval and the served Gate-A litmus ? and it's trivially small to migrate. The **videos** are the
120 > only large migration, and they already exist on **Google Drive** (the golden-corpus folder), so they
121 > can go **Drive ? bucket from a cloud box**, keeping the local machine out of the path entirely.

```
---

## 6. Migration plan (close the gap)

**Phase 1 ? corpus metadata (?25 MB, do first, removes the eval blocker).** Stage all trajectories,
all labels, and a **bucket-relative eval manifest** into `gs://khelsutra-rally-corpus`. Pending the §7
naming decision. Sketch:
```

227. user (2026-06-22T10:21:48.000Z)

```
  the shared folder. This is the path that actually removes the local box.
- **Interim (today, owner box ? streams Drive/F: ? local ? GCS):** `pwsh scratch/copy_videos_to_gcs.ps1
  -Execute` (dry-run without `-Execute`; `-Only <name>` for one; skip-existing). Uses local bandwidth.

**Priority:** the 7 Drive **proxies** first (already 1080p, sufficient for detection). The 8 large F:
full MP4s should be **proxied to 1080p first** (cheaper storage + faster cloud decode) before upload.
? Two pre-existing non-canonical video objects (`videos/GX030094_proxy.mp4`, `videos/mahadevpura-1.MP4`)
are superseded once `videos/GX030094.mp4` / `videos/mahadevpura_1.mp4` land ? verify-dupe-then-delete
(same as the label cleanup). Track per-clip in §8.

**Phase 3 ? flip eval/training to read GCS by default.** Point the eval manifest + litmus at the
bucket (env override or `input_root`) so a fresh box/CI needs no local corpus. Re-verify the suite +
the served Gate-A litmus read from `gcs://` (and re-measure if the corpus set changes ? see PR #236).

---

## 7. Decisions

1. **`labels/` naming ? LOCKED (owner, 2026-06-22):** `labels/<YYYY-MM-DD>_<name>.rallies.csv`, where
   `<name>` is the eval/manifest clip name and `<YYYY-MM-DD>` is the label's **authoring date** =
   the *earliest* mtime across all known copies (C: working copy + F: archive ? robust to copy-resets;
   a restored C: copy shows today, the F: archive keeps the real date). Fixes the `_proxy` inconsistency
   and disambiguates recycled GoPro names by date. *(Future, when videos are in the bucket: optionally
   add a `video_id`/MD5 sidecar for the strongest keying ? deferred, needs hashing each video.)*
2. **Cache `trajectories/` ? YES, done 2026-06-22.** `trajectories/<name>.csv` cached so cloud/CI eval +
   the served Gate-A litmus run **GPU-free** (training can still regenerate via `--trajectory-source detector`).
3. **Eval manifest in GCS ? done 2026-06-22.** `gs://khelsutra-rally-corpus/manifests/golden.json`
   (15 clips, `gs://` trajectory + label paths). The **local** litmus keeps its local-path manifest;
```

228. user (2026-06-22T10:21:52.796Z)

```
   flipping the litmus to read the GCS manifest is Phase 3.
4. **Originals migration scope ? OPEN** ? proxies-only vs full 4K, and how much of the
```

15-clip golden set vs the wider archive. Source = Google Drive ? bucket from a cloud box (not the local box).

> Phase-1 was run via a dry-run planner (resolve every clip's trajectory + label, stamp the label's
> authoring date, build the `gs://` manifest) ? `gcloud storage cp` (planner kept in gitignored
> `scratch/`; promote it to `scripts/` if corpus-sync recurs).
> ? **Cleanup pending:** 7 pre-existing inconsistent label objects (`*_proxy.rallies.csv`, bare
> `GX010128.rallies.csv`, `mahadevpura-1.rallies.csv`) still sit in `labels/` alongside the new
> canonical names ? they're superseded and would double-count in `worker train`'s `labels/` glob.
> Verify they're dupes of the canonical ones, then delete.

---

8. Progress tracker

Item	Status
Bucket + SA + provisioning (deploy/gcp)	? built
Cloud worker `infer`/`train` by URI	? built (`backend/worker`)

229. user (2026-06-22T10:22:03.748Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DATA_IN_GCS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

230. assistant (2026-06-22T10:22:08.182Z)

The re-run completed. Let me verify the demo video landed.

231. user (2026-06-22T10:22:10.364Z)

```
COPYING      mahadevpura_1        (523 MB) ...
  OK -> gs://khelsutra-rally-corpus/videos/mahadevpura_1.mp4

summary: copied=1 skipped=0 missing_src=0  (Execute=True Only='mahadevpura_1')
```

232. user (2026-06-22T10:22:14.042Z)

```
 548827049  2026-06-22T10:21:57Z  gs://khelsutra-rally-corpus/videos/mahadevpura_1.mp4
TOTAL: 1 objects, 548827049 bytes (523.40MiB)
```

233. assistant (2026-06-22T10:22:24.207Z)

Demo confirmed ? `videos/mahadevpura_1.mp4` (523 MB) is in the bucket, proving the Drive?bucket path end-to-end. Let me finalize the tracker, then commit + push to the PR.

234. user (2026-06-22T10:22:28.521Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DATA_IN_GCS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

235. assistant (2026-06-22T10:22:32.756Z)

Committing the Phase-2 setup + cleanup record and pushing to update PR #316.

236. user (2026-06-22T10:22:42.619Z)

```
work/golden-corpus-hygiene
warning: in the working copy of 'docs/DATA_IN_GCS.md', LF will be replaced by CRLF the next time
Git touches it
[work/golden-corpus-hygiene 53467c9] docs(data): record stale-label cleanup + set up Phase-2
video copy
 1 file changed, 32 insertions(+), 9 deletions(-)
To https://github.com/Khelsutra/badminton-highlight-indexer.git
   6ae3503..53467c9  work/golden-corpus-hygiene -> work/golden-corpus-hygiene
=== branch vs master ===
53467c9 docs(data): record stale-label cleanup + set up Phase-2 video copy
6ae3503 docs(data): record Phase-1 GCS migration (trajectories + labels + manifest)
8884c9f docs(data): add DATA_IN_GCS.md ? the corpus source-of-truth map (GCS)
```

237. assistant (2026-06-22T10:23:24.002Z)

Pushed ? PR #316 now has 3 commits. Before concluding, let me verify the most safety-critical thing deterministically: that **every `gs://` path in the cloud manifest actually resolves** (a dangling manifest would silently break cloud eval).

238. user (2026-06-22T10:24:24.033Z)

```
manifest entries: 15
checked 30 gs:// refs across 15 entries -> dangling=0
OK: every trajectory + label path in the manifest resolves
```

239. assistant (2026-06-22T10:24:50.961Z)

Manifest integrity verified: **15 entries, 30 `gs://` refs, 0 dangling** ? the cloud eval manifest is sound.

Now, per the exhaustive-review bar, let me run a multi-agent adversarial review over the migration + doc before I conclude ? independent lenses for accuracy, completeness, risk, and next-session foresight (judgment work where independent perspectives genuinely raise confidence).

240. user (2026-06-22T10:25:19.382Z)

```
Workflow launched in background. Task ID: wxvt345s0
Summary: Adversarial review of the GCS corpus migration + DATA_IN_GCS.md before session close
Transcript dir: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\78dabe12-41fd-46c2-ae90-ab74bdaec3c6\subagents\workflows\wf_18dcd7ff-5ae
Script file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\78dabe12-41fd-46c2-ae90-ab74bdaec3c6\workflows\scripts\gcs-migration-review-
wf_18dcd7ff-5ae.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath:
"C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\78dabe12-41fd-46c2-ae90-ab74bdaec3c6\workflows\scripts\gcs-migration-review-
wf_18dcd7ff-5ae.js"} to iterate without resending the script.)
Run ID: wf_18dcd7ff-5ae
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\.claude\projects\C--
Users-avidu-Projects-badminton-highlight-
indexer\78dabe12-41fd-46c2-ae90-ab74bdaec3c6\workflows\scripts\gcs-migration-review-
wf_18dcd7ff-5ae.js", resumeFromRunId: "wf_18dcd7ff-5ae"}) ? completed agents return cached
results.

You will be notified when it completes. Use /workflows to watch live progress.
```

241. assistant (2026-06-22T10:25:45.009Z)

Both your tasks are done and verified; I've kicked off a final adversarial review (4 independent lenses: doc-accuracy, completeness, risk, next-session foresight) which is running in the background, and I'll fold its findings into the session wrap-up the moment it lands.

Status so far:

#1 ? stale-label cleanup ? ? the 7 old inconsistent label objects were MD5-verified byte-identical to the canonical names (content also safe on C:/F:) and deleted. `labels/` now holds exactly the 15 canonical objects.

Phase-2 Drive?bucket ? set up + demonstrated ? ? full per-clip source mapping (7 Drive proxies + 8 large F: originals), a reusable owner-box script, and the decoupled `rclone`-on-cloud path, all documented. Proved end-to-end: `videos/mahadevpura_1.mp4` (523 MB) copied Drive?bucket. Hit and fixed a real gotcha ? `gcloud storage cp` globs `[...]`, so the `[Done] Video N` / `[Khelsutra]` paths need temp-staging (another point in favor of the cloud `rclone` path).

Verified deterministically: the cloud manifest `manifests/golden.json` has 15 entries with **0 dangling `gs://` refs** ? cloud eval is wired correctly.

All committed to **PR [#316](https://github.com/Khelsutra/badminton-highlight-indexer/pull/316)** (now 3 commits). The full session conclusion + "what to watch next session" note will follow once the review panel reports back.

242. user (2026-06-22T10:28:04.141Z)

<task-notification>
<task-id>wxvt345s0</task-id>
<tool-use-id>toolu_01Cyxsu8U7FoPAaEzRgxUiiq</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\78dabe12-41fd-46c2-ae90-ab74bdaec3c6\tasks\wxvt345s0.output</output-file>
<status>completed</status>
<summary>Dynamic workflow "Adversarial review of the GCS corpus migration + DATA_IN_GCS.md before session close" completed</summary>
<result>[{"lens":"DOC ACCURACY ? docs/DATA_IN_GCS.md vs repo code + verified bucket state","summary":"Most load-bearing claims in docs/DATA_IN_GCS.md are accurate against the code: the worker `infer`/`train` subcommands and every documented flag exist (backend/worker/__main__.py:457-490), the URI-scheme/backend claims hold (backend/storage/ref.py:21-68; gcs/s3/local/gdrive backends + list_uris/fetch/put all present), the config snippet matches deploy/gcp/README.md:199, and the central \"train builds the manifest on-box, does NOT read manifests/golden.json\" claim is CORRECT (cmd_train reads labels/+videos/ and writes manifest.json to scratch; golden.json is never opened; a manifest.json appears only as a weights/<version>/ bundle OUTPUT ? training/gen0/harness.py:241). However there is one real BLOCKER: the §7.1-LOCKED label naming (`<date>_<name>.rallies.csv`) is incompatible with both the worker's video?label stem-join and the doc's own Phase-2 video naming, so `--trajectory-source detector` train-from-bucket cannot actually run as documented. Plus one internal contradiction (§2 calls trajectories/ \"PROPOSED\" while the rest of the doc + bucket state say DONE) and a couple of minor imprecisions.","findings":[{"severity":"blocker","issue":"The §7.1-LOCKED label naming `labels/<YYYY-MM-DD>_<name>.rallies.csv` breaks the worker's video?label join, so the documented `--trajectory-source detector` train-from-bucket flow cannot run. cmd_train derives the join key by stripping ONLY `.rallies.csv`/`.csv` from the label filename, so `2026-06-22_GX020094.rallies.csv` -> name=`2026-06-22_GX020094`, then looks up the video as `video_by_stem.get(name)`. But the doc's Phase-2 (§6) stages videos as `videos/<name>.mp4` with the BARE name (e.g. `GX020094.mp4`, `mahadevpura_1.mp4`) ? stems will never match, so every clip hits the `no video for ... skipping` branch and train aborts with <2 specs. The doc presents both the dated labels and the bare-named videos as the working contract and never flags the mismatch.","evidence":"backend/worker/__main__.py:384 (`name = fname.replace('.rallies.csv','').replace('.csv','')`) + :387 (`vuri = video_by_stem.get(name)`) + :378 (video_by_stem keyed by `os.path.splitext(vname)[0]`) vs docs/DATA_IN_GCS.md §7.1 (dated labels, LOCKED) and §6 Phase-2 (`videos/<name>.mp4`, name=bare clip name)"},{"severity":"major","issue":"Internal contradiction on trajectories/ status. The §2 contract table tags `trajectories/` as `(PROPOSED ? see §7.2)`, but §3's Phase-1 banner, §7.2, and §8 all state trajectories/ caching is DONE (2026-06-22), and the VERIFIED BUCKET STATE has 15 trajectory objects present. The §2 'PROPOSED' label is stale and contradicts the doc's own later sections and ground truth.","evidence":"docs/DATA_IN_GCS.md §2 row `trajectories/` = '(PROPOSED ? see §7.2)' vs §3 banner / §7.2 ('YES, done 2026-06-22') / §8 ('? 2026-06-22') and verified bucket (15 objects)"},{"severity":"minor","issue":"§2 contract table implies `models/wasb_badminton_best.pth.tar` in the bucket is read directly as 'input to the WASB detector'. The detector does NOT read the weight from gs://; it reads a LOCAL path (`~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar` or `$WASB_WEIGHTS`), which

must be STAGED from the bucket first. Minor overstatement of the bucket's role for weights.","evidence":"backend/config/models.py:65 + backend/pipeline/detectors/wasb_runner.py:37,58 (local weights_path / WASB_WEIGHTS env) and deploy/gcp/README.md:183-185 ('must be present at ~/models/...; stage from Drive/bucket') vs docs/DATA_IN_GCS.md §2 ('input to the WASB detector')"},{"severity":"nit","issue":"§2 note says train extracts trajectories 'via the native detector (--trajectory-source detector)' as if that were the train mode, but the worker's default `--trajectory-source` is `synth` (CPU-mock); `detector` is opt-in. The note is technically describing the detector path it names, but a reader could infer detector is the default. Worth a one-word 'when --trajectory-source detector'.","evidence":"backend/worker/__main__.py:477 (`--trajectory-source choices=['synth','detector'], default='synth'`) vs docs/DATA_IN_GCS.md §2 note"},{"severity":"nit","issue":"§3 'pre-Phase-1 snapshot' lists labels/ as 7 files but the prose/tracker elsewhere reference an 'original-6 + GX010137/GX010141' set; the snapshot and the §5 gap table ('original-6 labels' F:-only, '7 new-clip labels' partial) are consistent with each other and with the VERIFIED BUCKET STATE (15 canonical after reconcile), so no data error ? but the multiple count framings (7 stale / 15 canonical / original-6 / 7-new) are hard to reconcile on a single read. Presentation clarity only; numbers check out.","evidence":"docs/DATA_IN_GCS.md §3 table, §5 gap table, §8 tracker vs VERIFIED BUCKET STATE (labels/ = 15 canonical, 7 stale deleted)"}],"recommendations":["Resolve the BLOCKER before the doc is treated as the working contract: either (a) document that for `--trajectory-source detector` the videos MUST be named with the SAME dated stem as the labels (`videos/<date>_<name>.mp4`) ? and change Phase-2 accordingly ? or (b) note that detector-train requires the worker to strip a date prefix / match on the bare `<name>` (a code change), or (c) state plainly that the dated label scheme is EVAL/manifest-only and bucket train uses bare-named labels. Right now §7.1 and §6 silently disagree with backend/worker/__main__.py:384-390.","Fix the §2 trajectories/ row: change '(PROPOSED ? see §7.2)' to DONE/? to match §3/§7.2/§8 and the bucket (15 objects).","Clarify §2 that `models/` weights are STAGED to a local path for the detector (not read from gs:// in place), mirroring deploy/gcp/README.md:183-185.","Add 'when --trajectory-source detector' to the §2 train note so readers don't assume detector is the default (default is synth).","Optional: add a one-line reconciliation of the label counts (7 stale deleted -> 15 canonical = original-6 + GX010137/GX010141 + 7 new) so §3/§5/§8 read consistently."]},{"lens":"Completeness critic ? what is MISSING to actually run ALL train+infer from the cloud (remove the local Windows box + external drives as a blocker), beyond what docs/DATA_IN_GCS.md + SETUP_NEW_MACHINE.md + deploy/gcp/README.md already cover or flag.","summary":"Phase-1 (metadata: trajectories + canonical labels + GCS manifest) is genuinely done and self-consistent, but the stated goal ? \"a fresh GPU box or CI runs infer, eval, and Gen-0 training reading ONLY from gs://khelsutra-rally-corpus\" ? is NOT reachable yet, and two of the gaps are real code bugs the doc doesn't flag. (1) The new locked label naming `<date>_<name>.rallies.csv` BREAKS the cloud train path: `cmd_train` derives the join key as the full filename stem (`2026-06-14_GX010128`) and matches it against date-free video stems (`GX010128`), so `worker train --trajectory-source detector` ? the exact command in deploy/gcp/README §5 ? silently skips every video and aborts with \"<2 videos\". (2) Phase-3 \"flip eval/litmus to GCS\" is not a config flip: the eval code (`served_gate_a.evaluate_video`, `calibrate_wasb.load_golden_gts`, `TrackNetRunner.parse_trajectory_csv`) opens `spec[\"trajectory\"]`/`spec[\"labels\"]` with plain `open()` and gates on `os.path.isfile()` ? it has NO storage-layer fetch, so a `gs://` manifest cannot be consumed and the litmus skips on any box without local files. The doc honestly lists the org/infra gaps (14 un-uploaded videos, the 8 unproxied F: originals, GPU quota, who runs Phase 2/3) but understates them as a checklist when each is an unowned, unscheduled, owner-only step. Net: the box is still a hard blocker for cloud train+eval until the naming join, the eval fetch path, and the video upload are all done.","findings":[{"severity":"blocker","issue":"The LOCKED label naming `labels/<YYYY-MM-DD>_<name>.rallies.csv` breaks the cloud trai
... (truncated 27484 chars, full result in C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\78dabe12-41fd-46c2-ae90-ab74bdaec3c6\tasks\wxvt345s0.output)</result>
<usage><agent_count>4</agent_count><subagent_tokens>333546</subagent_tokens><tool_uses>59</tool_uses><duration_ms>164734</duration_ms></usage>
</task-notification>

243. assistant (2026-06-22T10:28:56.706Z)

The review surfaced a **real, verified-against-code blocker** I need to confirm and act on: my locked date-prefixed label naming may break `worker train`'s label?video stem-join. Per my own rule (verify engine claims at `file:line`, never relay unverified), let me read the actual join code before acting.

244. user (2026-06-22T10:28:57.515Z)

```
355         os.makedirs(labels_dir, exist_ok=True)
356         os.makedirs(traj_dir, exist_ok=True)
357
358         corpus = args.corpus_uri.rstrip("/")
359         label_uris = sorted(u for u in storage.list_uris(f"{corpus}/labels/", config=storage_cfg)
360                             if u.lower().endswith(".csv"))
361         if len(label_uris) < 2:
362             print(f"[worker] need >=2 labeled videos for leave-one-video-out; found {len(label_uris)} "
363                   f"under {corpus}/labels/")
364             return 2
365
366         # Real-detector source: resolve the runner once + index the bucket's videos/ by stem.
367         runner = None
368         video_by_stem: dict = {}
369         if args.trajectory_source == "detector":
370             os.makedirs(videos_dir, exist_ok=True)
371             runner = _resolve_train_detector(args, config)
372             if runner is None:
373                 return 2
374             for vuri in storage.list_uris(f"{corpus}/videos/", config=storage_cfg):
375                 vname = storage.parse_uri(vuri).name
376                 if not vname.lower().endswith(_VIDEO_EXTS):
377                     continue  # skip sidecars (.srt/.json/thumbnails) so they can't shadow the video
378                 video_by_stem[os.path.splitext(vname)[0]] = vuri
379
380         print(f"[worker] trajectory source: {args.trajectory_source}")
381         specs: List[dict] = []
382         for uri in label_uris:
383             fname = storage.parse_uri(uri).name                    # e.g. GX020094_proxy.rallies.csv
384             name = fname.replace(".rallies.csv", "").replace(".csv", "")
385             local_label = storage.fetch(uri, os.path.join(labels_dir, fname), config=storage_cfg)
386             if args.trajectory_source == "detector":
387                 vuri = video_by_stem.get(name)
388                 if not vuri:
389                     print(f"[worker] no video for {name!r} under {corpus}/videos/ ? skipping.")
390                     continue
391                 local_video = storage.fetch(vuri, os.path.join(videos_dir, storage.parse_uri(vuri).name),
392                                             config=storage_cfg)
393                 video_id = compute_video_id(local_video)
394                 # Real fps/width drive the trajectory frame->time mapping; PROBE (don't guess) ? a
395                 # wrong fps silently mis-aligns every label, so skip the video if ffprobe can't read it.
396                 meta = _probe_video_fps_width(local_video)
397                 if meta is None:
398                     print(f"[worker] could not probe fps/width for {name!r} (ffprobe) ? skipping (won't guess).")
399                     continue
```

245. assistant (2026-06-22T10:29:34.773Z)

Blocker **confirmed** at backend/worker/__main__.py:384 (`name` keeps the date prefix) vs [:378](backend/worker/__main__.py:378)/[:387](backend/worker/__main__.py:387) (videos keyed by bare stem) ? `worker train --trajectory-source detector` would skip every clip. Let me confirm the `--trajectory-source` choices (my doc mentioned `cached`) so I correct the doc accurately too.

246. user (2026-06-22T10:29:36.716Z)

```
    pt = sub.add_parser("train", help="Gen-0 train-from-bucket (labels [+videos] -> trajectories -> harness)")
    pt.add_argument("--corpus-uri", required=True, help="prefix containing labels/ (+videos/ for "
                    "--trajectory-source detector), e.g. s3://bucket or gcs://bucket")
    pt.add_argument("--out-uri", required=True, help="output prefix (weights/<version>/ is appended)")
    pt.add_argument("--version", required=True, help="semver for the artifact bundle")
    pt.add_argument("--trajectory-source", choices=["synth", "detector"], default="synth",
                    help="synth = CPU-mock label-consistent trajectories (default, no GPU); "
                         "detector = REAL shuttle trajectories via the resolved DetectorRunner "
                         "(needs videos/ in the corpus; honours detector_impl).")
    pt.add_argument("--segmenter", default="wasb_hybrid",
                    help="trajectory hybrid whose detector is used for --trajectory-source detector")
    pt.add_argument("--floor", default=None, help="floor baseline JSON (local path or storage URI)")
    pt.add_argument("--config", default=None)
    pt.add_argument("--scratch", default=None)
    pt.add_argument("--fps", type=float, default=30.0, help="synth trajectory fps (detector probes the video)")
    pt.add_argument("--frame-width", type=float, default=1280.0,
                    help="synth trajectory frame width (detector probes the video)")
    pt.add_argument("--set", action="append", default=[], metavar="k=v")
    pt.set_defaults(func=cmd_train)
    return p
```

247. assistant (2026-06-22T10:29:57.157Z)

Confirmed: `--trajectory-source` is only `synth|detector` (no `cached`) ? so my doc's §5 `--trajectory-source cached` mention is wrong, and the uploaded `trajectories/` cache serves **eval**, not train. Let me read the full review output (it was truncated) to fold in the completeness/risk/foresight findings before correcting the doc.

248. user (2026-06-22T10:30:01.889Z)

```
Traceback (most recent call last):
  File "<string>", line 1, in <module>
    import sys,json; d=json.load(sys.stdin); [print('\n===== '+r['lens'][:70]+'
=====\n'+r['summary']+'\n--findings--\n'+'\n'.join('['+f['severity']+'] '+f['issue'][:300] for f
in r['findings'])+'\n--recs--\n'+'\n'.join('- '+x for x in r['recommendations']) ) for r in d]
                                                               ~^^^^^^^^^
TypeError: string indices must be integers, not 'str'
```

249. user (2026-06-22T10:30:09.644Z)


```
{
  "summary": "Adversarial review of the GCS corpus migration + DATA_IN_GCS.md before session close",
  "agentCount": 4,
  "logs": [],
  "result": [
    {
      "lens": "DOC ACCURACY ? docs/DATA_IN_GCS.md vs repo code + verified bucket state",
```

"summary": "Most load-bearing claims in docs/DATA_IN_GCS.md are accurate against the code: the worker `infer`/`train` subcommands and every documented flag exist (backend/worker/__main__.py:457-490), the URI-scheme/backend claims hold (backend/storage/ref.py:21-68; gcs/s3/local/gdrive backends + list_uris/fetch/put all present), the config snippet matches deploy/gcp/README.md:199, and the central \"train builds the manifest on-box, does NOT read manifests/golden.json\" claim is CORRECT (cmd_train reads labels/+videos/ and writes manifest.json to scratch; golden.json is never opened; a manifest.json appears only as a weights/<version>/ bundle OUTPUT ? training/gen0/harness.py:241). However there is one real BLOCKER: the §7.1-LOCKED label naming (`<date>_<name>.rallies.csv`) is incompatible with both the worker's video?label stem-join and the doc's own Phase-2 video naming, so `--trajectory-source detector` train-from-bucket cannot actually run as documented. Plus one internal contradiction (§2 calls trajectories/ \"PROPOSED\" while the rest of the doc + bucket state say DONE) and a couple of minor imprecisions.",
"findings": [
{

250. user (2026-06-22T10:30:17.407Z)

===== DOC ACCURACY ? docs/DATA_IN_GCS.md vs repo code + verified bucket state =====
[BLOCKER] The §7.1-LOCKED label naming `labels/<YYYY-MM-DD>_<name>.rallies.csv` breaks the worker's video?label join, so the documented `--trajectory-source detector` train-from-bucket flow cannot run. cmd_train derives the join key by stripping ONLY `.rallies.csv`/`.cs
[MAJOR] Internal contradiction on trajectories/ status. The §2 contract table tags `trajectories/` as `(PROPOSED ? see §7.2)`, but §3's Phase-1 banner, §7.2, and §8 all state trajectories/ caching is DONE (2026-06-22), and the VERIFIED BUCKET STATE has 15 trajectory o
[MINOR] §2 contract table implies `models/wasb_badminton_best.pth.tar` in the bucket is read directly as 'input to the WASB detector'. The detector does NOT read the weight from gs://; it reads a LOCAL path (`~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.p
[NIT] §2 note says train extracts trajectories 'via the native detector (--trajectory-source detector)' as if that were the train mode, but the worker's default `--trajectory-source` is `synth` (CPU-mock); `detector` is opt-in. The note is technically describing the
[NIT] §3 'pre-Phase-1 snapshot' lists labels/ as 7 files but the prose/tracker elsewhere reference an 'original-6 + GX010137/GX010141' set; the snapshot and the §5 gap table ('original-6 labels' F:-only, '7 new-clip labels' partial) are consistent with each other an
-- recs --
  * Resolve the BLOCKER before the doc is treated as the working contract: either (a) document that for `--trajectory-source detector` the videos MUST be named with the SAME dated stem as the labels (`vid
  * Fix the §2 trajectories/ row: change '(PROPOSED ? see §7.2)' to DONE/? to match §3/§7.2/§8 and the bucket (15 objects).
  * Clarify §2 that `models/` weights are STAGED to a local path for the detector (not read from gs:// in place), mirroring deploy/gcp/README.md:183-185.
  * Add 'when --trajectory-source detector' to the §2 train note so readers don't assume detector is the default (default is synth).
  * Optional: add a one-line reconciliation of the label counts (7 stale deleted -> 15 canonical = original-6 + GX010137/GX010141 + 7 new) so §3/§5/§8 read consistently.

===== Completeness critic ? what is MISSING to actually run ALL train+infer from =====
[BLOCKER] The LOCKED label naming `labels/<YYYY-MM-DD>_<name>.rallies.csv` breaks the cloud train join. `cmd_train` computes the per-clip key as `name = fname.replace('.rallies.csv','').replace('.csv','')` ? `2026-06-14_GX010128`, then matches `video_by_stem.get(name)`
[BLOCKER] Phase-3 'flip eval/litmus to GCS' is not config-only ? the eval/litmus code physically cannot read gs://. `served_gate_a.evaluate_video` does `traj, labels = str(spec['trajectory']), str(spec['labels'])` then `TrackNetRunner.parse_trajectory_csv(traj)` / `load
[MAJOR] The 14 of 15 golden videos NOT yet in videos/ are the actual blocker for cloud TRAIN and INFER on real footage, and the migration has no owner/date/run-path committed. Only mahadevpura_1 is uploaded (a 1-of-15 demo). Without these, `--trajectory-source detecto
[MAJOR] The 8 large F: originals require a proxy-generation step that exists only as a TODO checkbox ? no tool, no command, no owner. The doc says 'proxy to 1080p first' but there is no referenced proxy script (unlike the copy script), so 'proxy the 8 large F: full-MP
[MAJOR] GPU quota is still 0 ? the GPU VM has NEVER been created, so no real cloud train/infer can run at all today. deploy/gcp/README §0 lists the VM as 'NOT created ? blocked on GPUS_ALL_REGIONS quota'; §2 is an explicit owner-only console action (request 1). DATA_I
[MAJOR] Secrets/creds bootstrap on the cloud box is owner-only and not yet done for a persistent

setup. The cloud box needs: GCS SA JSON staged to the VM + GOOGLE_APPLICATION_CREDENTIALS exported, a rotated GEMINI_API_KEY for real (non-stub) inference, and WASB weight
[MINOR] Doc-vs-code drift: DATA_IN_GCS §5 gap table cites a `--trajectory-source cached` train mode that does not exist. `cmd_train` only offers choices `synth` and `detector`. So 'cached trajectories for train' is not actually a supported path; cached trajectories on
[MINOR] Two split, non-converged manifest consumers and no single source of truth. `manifests/golden.json` (gs:// paths, built by the planner) is read ONLY by eval ? and eval can't fetch gs:// (above). `worker train` ignores that manifest entirely and rebuilds its own
[MINOR] The pre-existing non-canonical video objects (videos/GX030094_proxy.mp4, videos/mahadevpura-1.MP4) will collide/shadow once canonical names land, and cleanup is a manual verify-dupe-then-delete with no owner. While `_VIDEO_EXTS` filtering prevents sidecar shad
[MINOR] No CI wiring exists to actually exercise the gs:// corpus, so 'CI can run reading only from gs://' in the DoD has no implementation target. The regression test is designed to SKIP when local files are absent (which is the CI condition), and there is no CI job
-- recs --

* Fix the train join before any cloud train run: in cmd_train, strip the locked `<YYYY-MM-DD>_` date prefix when deriving the clip `name` (or match on the suffix), so date-prefixed labels join date-free
* Build the storage-fetch path into eval before claiming Phase-3: have served_gate_a.evaluate_video (and the regression test's presence gate) resolve `spec['trajectory']`/`spec['labels']` through storag
* Reconcile or unify the two manifest consumers: either make `worker train` honor manifests/golden.json as the corpus definition, or document explicitly that train's label-listing and eval's manifest mu
* Promote scratch/copy_videos_to_gcs.ps1 to scripts/ and write the missing proxy-generation step as a real, referenced tool/command; then assign + schedule the owner steps with a clear sequence: (a) GPU
* Make the cloud-box secrets/creds reproducible: script staging of the SA JSON + GEMINI_API_KEY + WASB weights (e.g. pull weights from gs://.../models/) into the VM startup/port procedure so the cattle-
* Correct the doc-vs-code drift: remove/adjust the `--trajectory-source cached` reference in DATA_IN_GCS §5 (no such flag), and add GPU-quota + secrets as explicit gating dependencies in §8 so the DoD (
* Add a real CI target that authenticates to GCS (workload-identity or a CI SA), points the litmus/eval at the gs:// manifest, and runs it ? so 'CI reads only from gs://' is verifiable rather than silen

===== RISK / SAFETY AUDIT =====
[MAJOR] The migration is irreproducible: the scripts that performed it (migrate_golden_to_gcs.py, stage_corpus_to_gcs.sh, make_cloud_manifest.py), the produced manifest (golden.gcs.json), and the source-of-truth manifest (human_lovo_manifest.json) ALL live in gitignor
[MAJOR] Two competing, incompatible migration paths exist. scratch/migrate_golden_to_gcs.py renames to canonical trajectories/<name>.csv + labels/<date>_<name>.rallies.csv and writes gs://-pathed golden.gcs.json (the path actually used). But scratch/stage_corpus_to_gc
[MAJOR] The bucket manifest manifests/golden.json is consumed by ZERO code paths, so its format/keys/path-resolution are untested by any CI gate. Every eval/litmus consumer defaults to the local C:-absolute output/human_lovo_manifest.json; the served Gate-A regression
[MINOR] Wrong-fps corruption risk is latent in the manifest. served_gate_a/calibrate map trajectory frame->time via the manifest's fps; a 60fps clip mislabeled 30 doubles every timestamp. The bucket manifest carries mixed fps (59.94 and 29.97) copied verbatim from hum
[MINOR] The labels/ rename is a one-way lossy transform whose ONLY inverse is the manifest. Canonical labels/<date>_<name>.rallies.csv discard the original authored filename (e.g. 'Adarsh and Avi.rallies.csv' -> 2026-06-08_adarsh_avi_singles.rallies.csv). If golden.gc
[MINOR] The 7-object deletion was safe ONLY because of off-bucket copies, which the doc does not guarantee persist. Deletion recoverability rests on (a) MD5 byte-equality to the new canonical objects (true) AND (b) the source labels still existing on C:/F:/output/. Th
[NIT] make_cloud_manifest.py's own docstring documents an invocation path that doesn't exist (`python deploy/gcp/make_cloud_manifest.py`), but the file lives in scratch/. Anyone following the docstring fails. Symptom of the scripts never being promoted out of scratc
[NIT] Bracket-glob gotcha is documented as a Phase-2 concern but is a live trap for any future operator. gcloud storage cp treats [...] as a glob, so Drive '[Done] Video N/' and F:'[Khelsutra] ...' paths fail with 'matched no objects'. The mitigation (bracket-free
-- recs --

* Promote the migration scripts out of gitignored scratch/ into scripts/ (or deploy/gcp/) and

include them in the PR: migrate_golden_to_gcs.py is the keeper; DELETE or clearly deprecate stage_corpus_to_
  * Commit the produced bucket manifest (golden.gcs.json) into the repo as the auditable source-of-truth (rally_seg_eval.py:46 already notes the local manifest is gitignored and a committed set is 'the AU
  * Add a tiny CI/manifest-validator that loads manifests/golden.json (or the committed copy) and asserts every entry has name+trajectory+labels+fps+frame_width and that names are unique ? so the bucket m
  * Before Phase-3 flips any consumer to GCS, re-derive (probe) fps/frame_width from the actual bucket videos rather than trusting the copied human_lovo_manifest.json values, given the wrong-fps-silently-
  * Do NOT re-run stage_corpus_to_gcs.sh against the bucket ? it would re-introduce the *_traj.csv / spaced-filename drift Phase-1 just cleaned. Add a loud comment/guard to that effect if the script is ke
  * For Phase-2 (~70GB videos), confirm region co-location (bucket asia-south1 + GPU VMs asia-south1, per the doc) so train/infer reads incur no cross-region egress, and prefer the rclone-on-cloud-VM path

===== NEXT-SESSION FORESIGHT ? what to watch for / pick up next after the GCS cor =====
[MAJOR] Phase-3 'flip eval/litmus to GCS' is net-new plumbing, NOT an env override ? the next session must scope it as real code work or it will spiral. The litmus and eval drivers are local-FS only.
[MAJOR] DoD is unmet and the doc says so honestly ? Phase-2 (14 of 15 golden videos not in bucket) is the gating blocker for real WASB extraction / detector-source training / true cloud-only operation. Do this before claiming the box is no longer a blocker.
[MAJOR] #236 served-Gate-A litmus is pinned to local paths AND its measurement was taken on a smaller corpus ? flipping to the GCS 15-clip manifest changes the eval set, which per the Launch-Report convention requires a re-measure + dual-set no-regression check, not a
[MINOR] Sibling golden-fixtures P3 thread needs the same golden traj+label CSVs ? these are now ALREADY in GCS (Phase-1), so that thread can be unblocked by pointing it at gs://khelsutra-rally-corpus/{trajectories,labels}/ instead of re-staging. Watch for accidental r
[MINOR] Path-fragility / shared-checkout hazards for the next session: the corpus-sync planner + copy_videos_to_gcs.ps1 live in gitignored scratch/ (not reproducible from the repo), the golden manifest (output/human_lovo_manifest.json) holds C:-absolute paths, and the
[MINOR] gcloud [...] glob footgun will bite anyone running Phase-2 from a path containing brackets ([Done] Video N/, [Khelsutra]?) ? the rclone-on-cloud path avoids it; the owner-box script works around it via a bracket-free temp file. Next session should prefer rclon
[NIT] Two non-canonical video objects (videos/GX030094_proxy.mp4, videos/mahadevpura-1.MP4) are superseded once canonical names land ? verify-dupe-then-delete (mirror the Phase-1 label cleanup) so videos/ ends clean at exactly the canonical set.
-- recs --
  * 1 (unblocks the most) ? Land PR #316 first: it is docs-only (verified 3 docs(data) commits, no code) so review/merge is low-risk; merging makes DATA_IN_GCS.md the authoritative cross-session data map
  * 2 ? Run Phase-2 videos via rclone on a cloud VM (preferred, keeps the local box out), proxies-first: the 7 Drive proxies (~13.5GB) first, then proxy the 8 large F: full-MP4s to 1080p before upload. Th
  * 3 ? Treat Phase-3 as real code, not a flag: thread resolve_input_uri / an input_root through served_gate_a.py, golden_real_fixtures.py, and run_phase3_eval.py so they can read the gs:// manifest + CSV
  * 4 ? Re-measure #236 served Gate-A on the new GCS 15-clip manifest as part of Phase-3, and ship a Launch Report with dual-set (RAW + golden) no-regression ? do NOT silently swap the eval set. This is a
  * 5 ? Unblock the sibling golden-fixtures P3 thread cheaply by pointing it at the already-migrated gs://khelsutra-rally-corpus/{trajectories,labels,manifests}/ rather than re-staging; reconcile against
  * 6 ? Harden path-fragility: capture/commit a reproducible corpus-sync script, and replace the C:-absolute output/human_lovo_manifest.json with the bucket-relative manifests/golden.json everywhere the l
  * 7 ? Housekeeping during Phase-2: verify-dupe-then-delete the 2 non-canonical video objects so videos/ converges to exactly the 15 canonical names, mirroring the Phase-1 label cleanup.

251. assistant (2026-06-22T10:31:43.386Z)

The review is sharp and mostly verified. Two clarifications before I act: I **confirmed both blockers against code** (the date-prefix train-join at `__main__.py:384/378/387`, and `--trajectory-source` having no `cached` option). But the risk-auditor **hallucinated two scripts** (`stage_corpus_to_gcs.sh`, `make_cloud_manifest.py`) that don't exist ? I'll disregard those specific claims (only `migrate_golden_to_gcs.py` + `copy_videos_to_gcs.ps1` exist, both in `scratch/`). The doc must be corrected for honesty before #316 is mergeable. Let me read §2 to fix it precisely.

252. user (2026-06-22T10:31:44.623Z)

```
71      | `outputs/` | one `<video_id>/`, `parity/` | ? dry-run outputs |
72      | `highlights/`, `promo/` | reels + promo | ? |
73
74      **Drift summary:** the bucket has extra prefixes (`highlights/`, `promo/`, `models/`
alongside
75      `weights/`) and **inconsistent `labels/` naming**, and is **missing the bulk of the
corpus inputs**
76      (`videos/`, all `trajectories/`, most `labels/`). It reflects past *dry runs*, not a
complete corpus.
77
78      ---
79
80      ## 4. How the cloud consumes it
81
82      Provisioning + the verified commands live in
[`deploy/gcp/README.md`](../deploy/gcp/README.md) §3?§5.
83      The data-facing entry points (`backend/worker/__main__.py`):
84
85      ```bash
86      # config (no secrets in config; auth via GOOGLE_APPLICATION_CREDENTIALS):
87      #   { "storage": { "backend": "local", "gcs": { "project": "gen-lang-client-0306026826"
} } }
88
89      # INFERENCE on one cloud video -> outputs/<video_id>/ in the bucket
90      python -m backend.worker infer \
91        --video-uri gcs://khelsutra-rally-corpus/videos/<clip>.mp4 \
92        --out-uri   gcs://khelsutra-rally-corpus --segmenter wasb_hybrid --config config.json
93
94      # TRAIN from the bucket (reads labels/ [+ videos/], extracts trajectories, builds
manifest on-box,
95      # runs training.gen0.harness, pushes weights/<version>/):
96      RALLY_DETECTOR_IMPL=native python -m backend.worker train \
97        --corpus-uri gcs://khelsutra-rally-corpus --out-uri gcs://khelsutra-rally-corpus \
98        --version <v> --trajectory-source detector --segmenter wasb_hybrid --config
config.json
99      ```
```

253. user (2026-06-22T10:31:57.606Z)

```
54      > (§7.3).
55
56      ---
57
58      ## 3. What's actually in the bucket today (2026-06-22)
59
60      > **Phase-1 landed 2026-06-22:** added `trajectories/` (15), 15 canonical
`labels/<date>_<name>.rallies.csv`,
61      > and `manifests/golden.json`. The table below is the **pre-Phase-1** snapshot + the
*remaining* gaps
62      > (videos, the 7 stale label objects to delete) ? see §7/§8.
63
64      | Prefix | Present | Drift / notes |
65      |---|---|---|
66      | `labels/` | 7 files: `GX010128`, `mahadevpura-1`, + `Badminton BXH 2_proxy`, `Boxhill
Doubles_proxy`, `GX020094_proxy`, `GX030094_proxy`, `mahadevpura-2_proxy` | ? **inconsistent
naming** (`_proxy` on some, not others); ? **missing** the original-6 labels + `GX010137`,
```

`GX010141` |
67 | `videos/` | 3: `GX030094_proxy.mp4`, `mahadevpura-1.MP4`, `mahadevpura_smoke_180s.mp4` | ? corpus videos/proxies almost entirely absent |
68 | `trajectories/` | ? | ? **does not exist** (every trajectory is local-only) |
69 | `models/` | `wasb_badminton_best.pth.tar` | ? (? WASB-C3 weight-provenance still an open legal item before any external share) |
70 | `weights/` | `0.1.0-l4/` | ? first real L4 train bundle |
71 | `outputs/` | one `<video_id>/`, `parity/` | ? dry-run outputs |

254. assistant (2026-06-22T10:32:04.377Z)

Let me grep the exact strings I need to correct, rather than read-hunting.

255. user (2026-06-22T10:32:06.394Z)

44:| `trajectories/` | cached WASB trajectory CSVs | **(PROPOSED ? see §7.2)** GPU-free eval/litmus input |
46:| `models/` | upstream pretrained weights (`wasb_badminton_best.pth.tar`) | input to the WASB detector |
112:| 20 WASB trajectory CSVs | `C:\?\Annotation Setup\Collect\Trajectories\` | **17.5 MB** | ? none | eval/litmus (cached), train (`--trajectory-source cached`) |
160:**Phase 3 ? flip eval/training to read GCS by default.** Point the eval manifest + litmus at the

256. user (2026-06-22T10:32:21.099Z)

40 | Prefix | Holds | Who reads/writes it |
41 |---|---|---|
42 | `videos/` | source videos / proxies (`<clip>.mp4`) | **input** to `worker infer` & `worker train` |
43 | `labels/` | golden rally labels (`<clip>.rallies.csv`) | **input** ground truth to `worker train` + all eval |
44 | `trajectories/` | cached WASB trajectory CSVs | **(PROPOSED ? see §7.2)** GPU-free eval/litmus input |
45 | `weights/<version>/` | trained Gen-0 bundles (`weights.json` + a run `manifest.json`) | **output** of `worker train` |
46 | `models/` | upstream pretrained weights (`wasb_badminton_best.pth.tar`) | input to the WASB detector |
47 | `outputs/<video_id>/` | per-video inference outputs (intervals, etc.) | **output** of `worker infer` |
48 | `highlights/`, `promo/` | rendered reels + promo artifacts | output / demo |
49
50   > **Note on the manifest.** `worker train` does **not** read a pre-staged corpus `manifest.json` ? it
51   > *builds* one on-box at train time from `labels/` (+`videos/`), extracting trajectories via the native
52   > detector (`--trajectory-source detector`). A `manifest.json` only appears as an **output** inside each
53   > `weights/<version>/` bundle. The **eval/litmus** path is what wants a stable, path-resolvable manifest
54   > (§7.3).

257. assistant (2026-06-22T10:32:42.229Z)

Now the targeted doc corrections (folding the verified review findings):

258. user (2026-06-22T10:32:45.759Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DATA_IN_GCS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

259. user (2026-06-22T10:32:54.931Z)

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260. user (2026-06-22T10:33:00.901Z)

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261. user (2026-06-22T10:33:13.632Z)

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262. assistant (2026-06-22T10:33:24.731Z)

Now let me add a "Known blockers" section capturing the verified must-fix items:

263. user (2026-06-22T10:33:37.023Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DATA_IN_GCS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

264. user (2026-06-22T10:33:47.318Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DATA_IN_GCS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

265. assistant (2026-06-22T10:33:52.056Z)

Now commit the accuracy corrections to PR #316.

266. user (2026-06-22T10:34:03.272Z)

```
work/golden-corpus-hygiene
warning: in the working copy of 'docs/DATA_IN_GCS.md', LF will be replaced by CRLF the next time
Git touches it
[work/golden-corpus-hygiene 1b33f55] docs(data): correct DATA_IN_GCS per adversarial review
(blockers + accuracy)
 1 file changed, 41 insertions(+), 12 deletions(-)
To https://github.com/Khelsutra/badminton-highlight-indexer.git
   53467c9..1b33f55  work/golden-corpus-hygiene -> work/golden-corpus-hygiene
=== #316 commits ===
1b33f55 docs(data): correct DATA_IN_GCS per adversarial review (blockers + accuracy)
53467c9 docs(data): record stale-label cleanup + set up Phase-2 video copy
6ae3503 docs(data): record Phase-1 GCS migration (trajectories + labels + manifest)
8884c9f docs(data): add DATA_IN_GCS.md ? the corpus source-of-truth map (GCS)
```

267. assistant (2026-06-22T10:34:26.047Z)

PR #316 now carries 4 clean docs(data) commits with the honest blocker section. Let me capture the one concrete engine fix the review surfaced as a spinnable task so it isn't lost, then finalize the handoff.

268. user (2026-06-22T10:34:37.458Z)

Noted (position 1, task_id: task_53b5e190). A chip is showing for the user ? they can start it in a fresh worktree with one click, or dismiss it. If this suggestion becomes stale or superseded, call dismiss_task with this task_id. Currently pending: task_53b5e190 "Fix worker train join for date-prefixed labels". Continue your current work.

269. assistant (2026-06-22T10:34:44.864Z)

Let me update the current-state checkpoint to the final state (re-reading since it's shared).

270. user (2026-06-22T10:34:45.921Z)

14   **Checkpoint:** 2026-06-22 (GOLDEN-FIXTURES **P3 ? #317 MERGED**; the golden-fixtures project is now IMPLEMENTATION-COMPLETE). master `7a40434`. Owner cleared ALL owned golden videos (owned+minors-free) + re-confirmed **option (a)** ? minted **6 curated, trimmed (?12k-frame ? few-min), de-attributed REAL Tier-0 fixtures `fx_r01`?`fx_r06`** (single / multi-court×2 / doubles×2 / continuously-active blob; both 29.97/59.94 fps; **~1.9 MB total**) via a NEW `freeze_real_fixture --max-frames` trim. De-attributed (opaque slug, time-origin reset, scenario-only labels, **NO** source name/path/video_id ? grep-verified + leak-scan clean); the characterization + 3-OS cross-OS gates now replay REAL data (F1 0.0?0.59 = real CONTROL-path, **behaviour-locking NOT field quality**; hard multi-court/blob collapse to a mega-window F1=0 = the known over-merge, now frozen). DATA found on this box: `C:\Users\avidu\Projects\Annotation Setup\Collect\{Trajectories,Golden Labelled}\` + repo `output\*.rallies.csv` (manifest `output/human_lovo_manifest.json` = 15 videos); F: has the source GoPro videos (not needed). **Surrogate?source map kept OFF-git** (given to owner in chat: fx_r01=mahadevpura_single, r02=kushagra, r03=mahadevpura_1, r04=largetest_doubles, r05=GX010128, r06=Boxhill_Doubles). Suite **1390?/7s**, cov **85.27%**, all CI green incl. leak-scan + 3-OS. **Golden-fixtures tracker now M1/M2/M3/MX/P1/P2/DOC/P0/P3 ALL ?** (P0=safe-scope; the FUNCTIONAL video-name rename DEFERRED to a coordinated pre-open-source effort; M4 quality-floor + O1 protobuf + O2 Tier-1-key-gated remain OPTIONAL). Track PRs = #186/#189/#191/#192/#193/#194 + #314 + #317, all merged+audited. [[golden-set-vendoring]] [[gotcha-shared-checkout-branch-collisions]]

15

16   **Checkpoint:** 2026-06-22 (**khelsutra brand-repo HYGIENE + DOC-FRESHNESS SWEEP ? 2 PRs merged, clean slate**). Session = "sync to remote head + low-hanging fruit + quick doc sweep." khelsutra was at master `e2570e4` (#93). ? **[#94] hygiene** ? `.gitignore` now covers `*.db` (runtime `site/access.db`) + `.claude/` (machine-specific; both were untracked ? committable) + corrected a stale NEXT_STEPS PR count; **caught my own `.gitignore` inline-comment bug via `git check-ignore` BEFORE commit** (`.gitignore` has no trailing-comment syntax ? comments must be own-line). ? **[#95] doc-freshness sweep** ? a **15-agent AUDIT** (one verifier/doc, every falsifiable claim checked at file:line incl. cross-repo vs engine `origin/master`) found **44 stale findings** (the pattern: docs froze at their authoring-PR while the SPA / engine API / localization / alpha-deploy shipped past them) ? a **12-agent FIXER fan-out** applied **63 surgical edits across 13 docs**; I **independently re-verified every load-bearing engine claim** on engine `origin/master` before merging (`/api/{health:308,capabilities:340,videos:374}` + compile `download_url:768` + `native_wasb_runner.py` on master + `auth.py::CF_ACCESS_HEADER` + `_allowed_hosts_from_env` REPLACE-semantics ? all real; brittle engine line-anchors ? symbol refs like `main.py::compile_highlights`). Honesty held (P6 stays `?` engine-done/SPA-pending, native-WASB "not production-verified", footage shots owner-gated, historical changelogs preserved; the 2 `fresh` docs untouched). Also pruned **6 stale merged local branches**. **All CI green (site+web+layout); 0 open PRs; only `master` locally; khelsutra master now `7dc7de5`.** *(Brand/site track ? distinct from the serving/packaging/UX/golden threads below.)* [[lesson-verify-engine-surface-before-scoping]] [[working-agreement-merging]] [[doc-status-register]]

17

18   **Checkpoint:** 2026-06-22 (GOLDEN-FIXTURES **P0 ? #314 MERGED**, master `5c2eec8`; a SEPARATE thread from packaging/UX). Owner: "all owned golden videos are owned+minor-free; pick (a); make P0 now." On scouting, P0 was bigger/riskier than the audit's "4 files": the video names are FUNCTIONAL (eval_baselines keys ? training floor + the gitignored `output/human_lovo_manifest.json` + data files; `MULTICOURT_CLIPS`; hot in QUALITY_ITERATIONS/the quality track) ? surfaced it ? owner picked **SAFE SCOPE**. Shipped: scrubbed personal `C:\Users\avidu` paths from 3 code docstrings + **`tools/leak_scan.py`** (structural PII guard: 32-hex MD5 + personal paths over backend/training/scripts/tools/eval_baselines; tests/ excluded; optional gitignored `.leakscan_private` for local name scanning) + CI `leak-scan` job + 10 tests. Clean scan; suite **1365?/7s**, cov **85.27%**, all CI green. **DEFERRED (safe-scope):** the functional video-name rename + git-history rewrite (need manifest/data/training-floor coordination + quality-track pause; do before open-sourcing). ?? **NEXT = P3 real Tier-0 fixtures** ? owner APPROVED (all owned cleared + **(a)** commit de-attributed per-frame CSV); needs the golden **trajectory CSVs + label CSVs** (`Annotation Setup/Collect/Trajectories/` + `Golden Labelled/` / OneDrive `golden-data/`) ? I locate them or owner points me, then `--freeze-real` per video. Golden-fixtures track = #186/#189/#191/#192/#193/#194 + #314, all merged+audited. [[golden-set-vendoring]]

19

20   **Checkpoint:** 2026-06-22 (? **GCS = corpus source-of-truth ? `docs/DATA_IN_GCS.md` + Phase-1 metadata MIGRATED ? PR [#316](https://github.com/Khelsutra/badminton-highlight-indexer/pull/316) OPEN; F: label backup done**). Owner goal: run ALL train+infer from the cloud

with ALL data in `gs://khelsutra-rally-corpus` ? remove the local box as a blocker; document where data lives so other sessions pick up from the bucket. [[gcs-data-source-of-truth]]
21   - ? **Authoritative map `docs/DATA_IN_GCS.md`** (indexed in docs/README.md; branch `work/golden-corpus-hygiene`, commit `8884c9f`): bucket layout · current contents + **drift** · how `backend.worker infer|train` consume `gs://` · the gap · phased migration plan + DoD. The **cloud engine already exists** (archived DECOUPLED_COMPUTE: `worker infer --video-uri` / `worker train --corpus-uri`, URI backends in `backend/storage/ref.py`, `deploy/gcp/`) ? the doc records WHERE bytes are, it does NOT redesign.
22   - ? **Gap measured:** corpus *metadata* is **~25 MB & entirely local** ? 20 trajectory CSVs (17.5 MB at `C:\?\Collect\Trajectories\`), `output/human_lovo_manifest.json` (C:-absolute paths), most labels. Bucket has only 3 videos, **no `trajectories/`**, inconsistent `_proxy` label naming. Only the VIDEOS are large ? migrate **Drive?bucket from a cloud box**, not via the local box.
23   - ? **F: backup done:** the 7 single-copy new-clip labels hash-verified into `F:\[Khelsutra] GoPro Hero Black 12\KhelSutra\Training\Golden Labelled\`. ? F: root was RENAMED `GoPro Hero Black 12`?`[Khelsutra] GoPro Hero Black 12` between turns ? live proof of the path-fragility GCS fixes.
24   - ? **Phase-1 DONE (uploaded to the bucket):** `trajectories/<name>.csv` (15), `labels/<YYYY-MM-DD>_<name>.rallies.csv` (15), `manifests/golden.json` (gs://-pathed, 15 clips). **Naming LOCKED (owner):** manifest name + authoring-date prefix = earliest mtime across C:/F: copies (robust to copy-resets). ? **These golden traj+label CSVs are exactly what the GOLDEN-FIXTURES P3 thread (#314 checkpoint above) was hunting for ? now in GCS.**
25   - ? **Owner decisions RESOLVED (DATA_IN_GCS §7):** naming=date-prefixed manifest name; cache trajectories=yes; GCS eval manifest=yes. **Still OPEN:** (4) originals migration scope (proxies-vs-4K; Drive?bucket from a cloud box, not local). **Cleanup pending:** 7 stale pre-existing `_proxy`/bare label objects in `labels/` (superseded; would double-count in `worker train`'s glob) ? verify-dupes-then-delete.

271. user (2026-06-22T10:34:55.071Z)

25   - ? **Owner decisions RESOLVED (DATA_IN_GCS §7):** naming=date-prefixed manifest name; cache trajectories=yes; GCS eval manifest=yes. **Still OPEN:** (4) originals migration scope (proxies-vs-4K; Drive?bucket from a cloud box, not local). **Cleanup pending:** 7 stale pre-existing `_proxy`/bare label objects in `labels/` (superseded; would double-count in `worker train`'s glob) ? verify-dupes-then-delete.
26   - ? Branch `work/golden-corpus-hygiene` off origin/master (`689c738`), 2 commits, **pushed ? PR #316 OPEN for review** (no self-merge). Serving WIP `6fdd1d4` + `scratch/PACKAGING_PLAN.serving-wip.md` + the gitignored `scratch/migrate_golden_to_gcs.py` planner preserved. Prior #236 served-Gate-A re-grounding MERGED 2026-06-19; original-6 labels turned out **recoverable from F:** (not lost) ? optional follow-up: re-measure litmus on the full 15-clip corpus (now feasible ? manifest in GCS). [[compute-stack-gcp-only]] [[gotcha-shared-checkout-branch-collisions]]
27

272. user (2026-06-22T10:35:18.213Z)

<tool_use_error>File has been modified since read, either by the user or by a linter. Read it again before attempting to write it.</tool_use_error>

273. assistant (2026-06-22T10:35:29.836Z)

The shared file changed under me (sibling session). Let me re-locate the GCS checkpoint and re-apply.

274. user (2026-06-22T10:35:31.798Z)

29:- ? **Owner decisions RESOLVED (DATA_IN_GCS §7):** naming=date-prefixed manifest name; cache trajectories=yes; GCS eval manifest=yes. **Still OPEN:** (4) originals migration scope (proxies-vs-4K; Drive?bucket from a cloud box, not local). **Cleanup pending:** 7 stale pre-existing `_proxy`/bare label objects in `labels/` (superseded; would double-count in `worker train`'s glob) ? verify-dupes-then-delete.
30:[Omitted long matching line]

275. user (2026-06-22T10:35:37.833Z)

29   - ? **Owner decisions RESOLVED (DATA_IN_GCS §7):** naming=date-prefixed manifest name; cache trajectories=yes; GCS eval manifest=yes. **Still OPEN:** (4) originals migration scope (proxies-vs-4K; Drive?bucket from a cloud box, not local). **Cleanup pending:** 7 stale pre-existing `_proxy`/bare label objects in `labels/` (superseded; would double-count in `worker train`'s glob) ? verify-dupes-then-delete.
30   - ? Branch `work/golden-corpus-hygiene` off origin/master (`689c738`), 2 commits, **pushed ? PR #316 OPEN for review** (no self-merge). Serving WIP `6fdd1d4` + `scratch/PACKAGING_PLAN.serving-wip.md` + the gitignored `scratch/migrate_golden_to_gcs.py` planner preserved. Prior #236 served-Gate-A re-grounding MERGED 2026-06-19; original-6 labels turned out **recoverable from F:** (not lost) ? optional follow-up: re-measure litmus on the full 15-clip corpus (now feasible ? manifest in GCS). [[compute-stack-gcp-only]] [[gotcha-shared-checkout-branch-collisions]]
31

276. user (2026-06-22T10:35:59.157Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

277. assistant (2026-06-22T10:36:41.976Z)

Everything's done and recorded. Here's the session wrap-up.

Session conclusion ? GCS corpus migration

Both requested tasks are complete and verified:

#1 ? stale-label cleanup ? All 7 old inconsistent label objects were MD5-verified **byte-identical** to the canonical names (content also safe on C:/F:) and deleted. `labels/` now holds exactly the **15 canonical** objects.

Phase-2 Drive?bucket ? set up + demonstrated ? Full per-clip mapping (7 Drive proxies + 8 large F: originals), a reusable owner-box script, and the decoupled `rclone`-on-cloud path ? all documented. Proved end-to-end: `videos/mahadevpura_1.mp4` (523 MB) copied Drive?bucket. Found + worked around a real gotcha (`gcloud storage cp` globs `[...]`, so `[Done] Video N` / `[Khelsutra]` paths need temp-staging ? another reason `rclone`-on-cloud is the right Phase-2 path).

Bucket state now (deterministically verified): `trajectories/` = 15 · `labels/` = 15 canonical · `manifests/golden.json` = 15 entries, **0 dangling `gs://` refs** · `videos/` = 1/15 (demo).

An honest catch from the adversarial review: a 4-lens panel (which I then verified against code) found a **real blocker** ? the date-prefixed label names you chose **break `worker train --trajectory-source detector`** (it joins the full label stem `2026-06-22_GX020094` against bare video stems at [`__main__.py:384`](backend/worker/__main__.py:384)). The eval path and synth-train are unaffected. The fix is ~1 line (strip the date prefix in `cmd_train`); I've **filed it as a chip (`task_53b5e190`)** and documented it in the doc's new "Known blockers" section rather than rush an untested engine change into the close. (The risk-agent also hallucinated two nonexistent scripts ? I disregarded those.)

PR [#316](https://github.com/Khelsutra/badminton-highlight-indexer/pull/316) ? 4 docs-only commits, open for your review (no self-merge). docs/DATA_IN_GCS.md is now honest about what works, what's dormant, and what's blocked.

? What to look out for / pick up next session (in order)

1. **Merge #316** ? docs-only, low-risk; it makes `DATA_IN_GCS.md` the authoritative cross-session data map.
2. **Worker date-strip fix** (chip `task_53b5e190`) ? unblocks cloud detector-training with your label naming. Own PR + test + full gate.
3. **Phase-2 videos** ? run `rclone` on a cloud VM (keeps the local box out), **proxies-first**: the 7 Drive proxies (~13.5 GB), then proxy the 8 large F: originals to 1080p before upload.

Needs a one-time owner `rclone config` (Drive OAuth).
4. **Phase-3 is real code, not a config flip** ? eval/litmus are local-FS only; thread `storage.fetch` through `served_gate_a` etc. to read the `gs://` manifest, **re-probe fps from the bucket videos**, then **re-measure the #236 litmus on the 15-clip set as a Launch-Report dual-set event** (don't silently swap the eval set).
5. **Free cross-thread win** ? the sibling golden-fixtures thread was hunting these same golden CSVs; they're now in `gs://khelsutra-rally-corpus/{trajectories,labels}/` ? point it there instead of re-staging.
6. **Owner-gated infra** ? GPU `GPUS_ALL_REGIONS` quota is still **0** (no VM ever created) and cloud-box secrets aren't set up; no real cloud run happens until those. Also promote the two `scratch/` migration scripts into `scripts/` for reproducibility.

Watch-outs: this checkout is a **shared worktree** (branches/files shifted under me 3× this session ? the F: archive folder was even renamed mid-session), so always branch from `origin/master` + re-verify HEAD; and absolute local paths keep drifting ? which is exactly the fragility GCS-as-source-of-truth removes.

All state is captured in [[gcs-data-source-of-truth]] + the current-state checkpoint, with the next-session order baked in. Good place to stop. ?